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Digital High-Frequency Chip – Linear Low-Dropout Voltage Regulator

Project Number: 3097

Project Report

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Project Carried Out At: University

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LDO for a DPLL chip

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Abstract:

LDOs are critical components in system-on-chips because of their small layout footprint and their ability to provide a precise voltage that is independent of supply voltage and load current variations. This work presents the design, simulation, layout and verification stages of a low-dropout (LDO) voltage regulator tailored for a digitally controlled phase-locked loop (DPLL) in TSMC 28nm CMOS technology. The LDO regulates a 0.9V output from a 1.8V supply and, through two enable signals, can also deliver 0V or pass through 1.8V. It is designed to meet the demanding requirements of a noise-sensitive, timing-critical DPLL, where voltage precision is critical for minimizing jitter and ensuring accurate frequency synthesis. As a result, tight line and load regulation, along with high power supply rejection, are essential design goals. The project was carried out under a strict timeline in collaboration with peer teams developing complementary DPLL blocks, with the full system scheduled for tape-out in August 2025.

The post-layout LDO delivers up to 50 mW of load power while consuming only 80 μA of quiescent current across its analog control components, including a bandgap reference (BGR), feedback network, and two operational transconductance amplifiers (OTAs). The regulation loop is driven by a two-stage Miller-compensated OTA, which features a differential input pair and common-source output stage. This OTA achieves a DC gain of 77.6 dB, unity gain bandwidth of approximately 5.6 MHz, and a phase margin of 74°, using a nulling resistor and compensation capacitor to ensure frequency stability.

The BGR employs a current-summing topology in which PTAT and CTAT currents, with closely matched temperature coefficients, are combined in a third branch and scaled by a resistor to generate a 0.72 V reference voltage. This design achieves excellent line regulation and minimal temperature drift without relying on second-order temperature compensation techniques or advanced current mirrors.

The LDO exhibits robust regulation characteristics with post-layout simulation results showing a line regulation of 0.0009 V/V, load regulation of 0.033 mV/mA, and temperature coefficient (TC) of 6.4 ppm/°C, verified across multiple corners. Power supply rejection ratio (PSRR) at 1 kHz is approximately −61 dB. The transient response was validated by applying a 10 ps-wide disturbance on the supply voltage and load current, with the output stabilizing within $1 - 10\mu S$ at worst case, which demonstrates the regulator's ability to suppress fast digital switching noise. The full LDO feedback loop maintains a phase margin of 58°, which ensures overall stability across operating conditions.

The layout stage was implemented following industry best practices, including usage of short and straight wires for current branches, common-centroid placement for OTA differential pairs and BJTs in the BGR, symmetry-based matching of devices and wires, and metal stacking to minimize IR drop in the LDO. All in all, the project successfully demonstrates the integration of an LDO with excellent TC and line and load regulation suitable for the DPLL architecture.

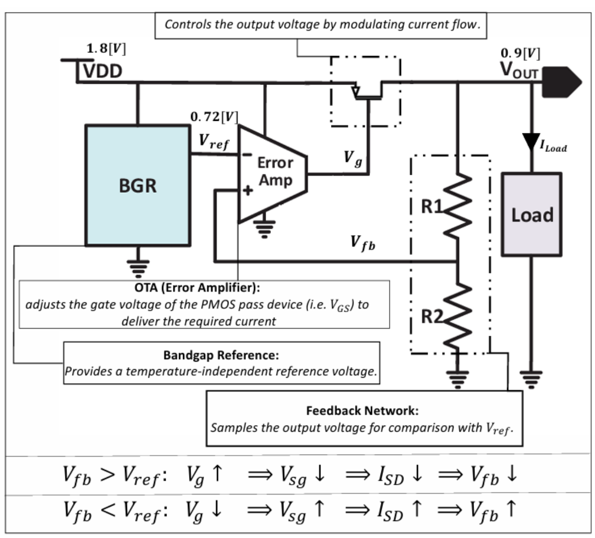

Figure 1. Block Diagram of the Designed LDO with Integrated OTA and BGR

1 Introduction

1.1 Goals of the Project

The main goal of this project was to fully design, simulate, implement layout and validate from scratch a linear low-dropout (LDO) voltage regulator in TSMC 28nm technology. The LDO should supply stable and accurate power to three critical digital blocks of a phase-locked loop (DPLL) system, including a DCO and TDC. These blocks, developed by other collaborating teams, require a clean and tightly regulated 0.9V supply derived from a 1.8V input. Since the DPLL is highly sensitive to voltage fluctuations, especially in its digitally controlled oscillator, it is essential to have tight line regulation, low temperature drift, and excellent noise immunity to minimize jitter and maintain signal integrity. The LDO is designed to deliver a constant power of 50mW, corresponding to a load current of approximately 55.5mA.

Unlike power supplies intended for highly dynamic loads, the DPLL presents a relatively steady DC load. As such, the LDO design emphasizes voltage accuracy, thermal robustness, and fast response to supply disturbances, rather than prioritizing load regulation. Transient response remains relevant due to fast digital switching in nearby blocks that may introduce noise on the power rails. The LDO must suppress these disturbances to maintain regulation integrity.

To meet the stringent performance demands of the system, the following design goals were set at the beginning:

- **Output voltage of BGR:** 0.72V from a 1.8V supply.
- **Output voltage of LDO:** 0.9V from a 1.8V supply.
- **Load current drawn from LDO:** 55.5mA (50mW).
- **Line regulation:** ≤ 0.01V/V (-40dB).
- **Load regulation:** ≤ 0.1mV/mA from 0–60mA
- **Temperature coefficient (TC):** ≤ 60ppm/°C across -40°C to 125°C.
- **PSRR:** ≤ -40dB at 1kHz.
- **Quiescent current:** ≤ 1mA (shared across both OTAs, the BGR, and feedback network).
- **OTA performance:** ≥ 70dB DC gain and ≥5MHz GBW for feedback response accuracy and speed.

In addition to circuit design, careful attention was given to the layout to ensure the LDO performs well after fabrication. Common-centroid placement was used for matched transistors in both the OTA and the BGR to reduce mismatch caused by systematic gradients and to improve robustness against process, voltage, and temperature (PVT) variations. In the OTA, dummy devices were added to improve symmetry and matching. Symmetrical metal routing was applied at sensitive nodes to balance parasitics and improve performance. For the LDO current path, metal stacking and symmetrical layout were used to reduce IR drop and ensure stable output voltage under varying conditions. The layout was planned not only to match post-layout simulation goals, but also to maintain performance after fabrication.

1.2 Motivation

Digitally controlled Phase-Locked Loops (DPLLs) are critical components in modern mixed-signal systems, often used for clock generation, data recovery, and frequency synthesis in communication and digital processing chips. These systems are highly sensitive to fluctuations in supply voltage, which can introduce jitter and degrade timing accuracy. As such, the power supply for a DPLL must offer exceptional stability, low noise, and minimal variation across temperature, line and load changes to ensure high signal integrity and frequency precision.

A Linear Low-Dropout (LDO) voltage regulator is particularly well-suited for this application due to its key advantages in noise performance, simplicity, and integration. Unlike switching regulators such as DC-DC buck converters or charge-pump-based solutions, LDOs do not rely on inductive or capacitive energy storage elements that typically require large passive components. This makes LDOs significantly more compact in layout—especially important in advanced CMOS nodes like TSMC 28nm, where silicon area is costly

Furthermore, the aforementioned switching regulators inherently generate high-frequency noise due to their switching operation, which can couple into sensitive analog nodes and degrade the performance of noise-sensitive circuits such as the digitally-controlled oscillator (DCO) and phase detector in the DPLL. In contrast, LDOs operate in a continuous, linear fashion, offering low output ripple and minimal high-frequency noise, which is essential for clean clock generation and low phase noise in the DPLL. However, because LDOs operate using linear (resistive) regulation, they have low power efficiency when the input voltage V_{DD} is much higher than the output voltage V_{out}, and the excess power is dissipated as heat. For more see [6].

Charge-pump regulators, while sometimes used in DPLL architectures themselves, are unsuitable as power supplies due to their reliance on large capacitors, limited drive capability, and sensitivity to leakage and switching mismatches. They also struggle with load regulation over wide ranges, making them incompatible with the constant power demand of multiple DPLL blocks.

In our system, the LDO is responsible for powering three internal digital and mixed-signal blocks of the DPLL. These blocks have a relatively constant and well-defined current draw of approximately 55.5 mA (50 mW at 0.9 V), allowing the LDO to be optimized for a narrow operating point. High power supply rejection ratio (PSRR) and excellent line regulation are prioritized to prevent external supply ripple from propagating into the DPLL's timing-critical circuitry.

Overall, since the LDO requires minimal external components and no inductors, they are ideal for on-chip integration with a DPLL because they offer the optimal balance of low noise, compact layout, fast transient response, and design simplicity.

1.3 The Approach to Solving the Problem

The project began with a detailed study of low-dropout (LDO) regulators, bandgap references (BGRs), and operational amplifier design. After reviewing the theory, we selected a BGR topology that initially used voltage summation of PTAT and CTAT voltages to generate a reference of around 1.2V with a small temperature coefficient. For the LDO's error amplifier, we chose a two-stage Miller-compensated OTA to meet our gain and stability requirements.

We first implemented the BGR using a current mirror configuration since it is simpler, but the TC and PSRR was poor due to voltage mismatch between branches. To overcome this, we replaced the current mirror with an ideal op-amp model provided in the simulation environment. This improved the performance significantly, allowing the BGR to reach a TC-stable output around 1.2V.

Next, we designed the two-stage Miller OTA, tuning its gain and compensation components to achieve a GBW of 5–10MHz and a phase margin of over 60°. After verifying its performance independently, we integrated it with the BGR to confirm compatibility. Once the system worked, we modified the BGR topology to generate a final reference voltage of 0.72V by summing PTAT and CTAT currents instead of voltages, while maintaining low temperature drift.

Before moving to layout, we confirmed in simulation that the complete LDO design functioned as intended, with proper regulation and stability. Then, we moved on to layout implementation. We started with the OTA, and went through several iterations to improve device matching and symmetry. Dummy devices and common-centroid placement were used to enhance performance. Once it passed LVS and DRC, and post-layout simulations confirmed proper function, we proceeded to the BGR.

The BGR layout also went through multiple refinements. Careful placement of BJTs in common-centroid configuration and attention to routing symmetry were used to minimize mismatch.

Finally, we constructed the full LDO layout, and integrated the OTA, BGR, and pass device. Emphasis was placed on minimizing parasitic resistance in the current path and achieving good symmetry to reduce IR drop. After additional design tweaks, the LDO met its key targets including line and load regulation, PSRR, and stability. Each layout stage passed layout-versus-schematic (LVS), design rule checks (DRC), and antenna effect checks.

The entire project workflow ensured that each block was verified before integration through rigorous simulation cycles, and the final layout checks validated the design's manufacturability and functional correctness.

1.4 Comparison Against Existing Implementations

Existing LDO regulator implementations found in the literature or commercially are often optimized for general-purpose use and are not tailored to meet the specific requirements of powering a DPLL system-on-chip. These designs typically have large layout footprints due to off-chip or non-area-constrained applications, making them unsuitable for integration in compact SoC environments. Moreover, their output voltage (V_{OUT}), supply voltage range (V_{DD}), and supported load current (I_{LOAD}) are frequently much higher and mismatched with the demands of this DPLL system.

Another significant limitation lies in the technology compatibility. Many of these regulators are implemented in older or incompatible process nodes, making direct integration with a 28 nm TSMC-based DPLL chip impractical or inefficient. Additionally, many LDO implementations often exhibit suboptimal line regulation and degraded performance over wide temperature ranges, particularly from $-40\,^{\circ}\mathrm{C}$ to $125\,^{\circ}\mathrm{C}$, which can be critical in precision timing applications.

In this work, a custom LDO designed in 28nm is presented to address the aforementioned shortcomings, with an emphasis on minimizing layout area and ensuring robust operation across process, voltage, and temperature (PVT) variations. The design also features specially tailored enable signals, aligned with the system-level requirements of the DPLL, for reliable control purposes. A detailed quantitative comparison between the proposed LDO and existing implementations, including key performance metrics such as PSRR, load/line regulation, and temperature stability, will be presented in the *Results and Analysis* section.

2 Theoretical Background

2.1 Low-Dropout Voltage Regulator

LDO Design Principles, Tradeoffs and Stability Considerations

The typical analog LDO regulator (shown in Figure 1) consists of a bandgap reference circuit, an error amplifier (EA), a resistive feedback network, and a PMOS pass transistor. The bandgap reference generates a stable voltage that is independent of supply variations and temperature changes. The EA compares this reference voltage to a scaled version of the output voltage through the feedback network. The EA responds to any voltage difference by adjusting the gate voltage of the PMOS pass transistor to regulate the output voltage by precisely controlling the current flow. Assuming an ideal EA with infinite input impedance and a virtual short between its inputs, the output voltage can be expressed as:

$$V_{\text{out}} = V_{\text{ref}}\left(1 + \frac{R_{\text{fb1}}}{R_{\text{fb2}}}\right)$$

Equation 1: LDO Output Voltage

An analog LDO with an error amplifier is preferred for a DPLL chip over a digital LDO which uses a comparator and doesn't use resistors because of design simplicity, better PSRR, and lower output noise (no ripple). However, a digital LDO has better process portability and lower power dissipation. The error amplifier is usually implemented with an OTA topology and not an op-amp because of its ability to drive capacitive loads like the gate of the PMOS pass transistor. It is clear that the gain of the OTA must be high enough to get a virtual short between its inputs to ensure precise regulation. The exact value of the gain is not important, but it should be high ($> 70dB$) for effective line and load regulation.

Although LDOs can use a pass transistor in triode mode to decrease the dropout voltage ($V_{DO} = V_{DD} - V_{out}$ and improve the power efficiency, usually saturation mode is preferred due to higher achievable loop gain which improves line and load regulation, and a better PSRR due to a larger r_{ds} in saturation mode. Additionally, the pass element is usually a PMOS because an NMOS requires an additional charge pump to generate a gate voltage higher than the supply voltage. On the other hand, the PMOS can safely operate below the input voltage and allows for higher loop-gain, with the cost of slightly increasing stability complexity.

In addition to achieving strict static performance targets such as line and load regulation, it is equally critical to meet dynamic metrics including PSRR, output noise, and transient response. Furthermore, a key design challenge lies in ensuring the LDO remains stable across the entire load current range. Typically, the dominant pole is located at the gate of the pass transistor. However, the output pole is load-dependent, shifting with variations in output impedance and load current. Consequently, the relative order of the poles can change under different load conditions, complicating frequency compensation. To address this, an external or on-chip output capacitor, often combined with a series damping resistor, is used to stabilize the feedback loop and maintain adequate phase margin throughout the entire operating range. This capacitor also improves high-frequency PSRR and improves load regulation. however, it comes at the cost of increased layout area and reduced bandwidth, which limits the LDO's transient response speed.

LDO Key Performance Metrics

- **Temperature Coefficient (TC)** — quantifies the relative change in the output voltage in μV per degree Celsius over the specified temperature range. It is typically expressed in parts per million per degree Celsius (ppm/°C) and can be minimized by designing an effective bandgap reference that provides a stable, temperature-independent voltage to the LDO.

$$\text{TC} = \frac{V_{\text{OUT}_{\text{max}}} - V_{\text{OUT}_{\text{min}}}}{V_{\text{OUT}_{\text{avg}}} \times \Delta T} \times 10^6 \quad \left[\frac{\text{ppm}}{°C}\right]$$

Equation 2: Temperature Coefficient Metric

where $V_{\text{OUT}_{\text{max}}}$ and $V_{\text{OUT}_{\text{min}}}$ are the maximum and minimum output voltages over the temperature range, $V_{\text{OUT}_{\text{avg}}}$ is the average output voltage, and ΔT is the temperature range in °C.
- **Line Regulation** — indicates how well a circuit maintains its specified output voltage in response to fluctuations in the supply voltage (I_{LOAD} is constant). It can be improved by increasing the DC open-loop voltage gain of the error amplifier.

$$\text{Line Regulation} = \frac{\Delta V_{\text{OUT}}}{\Delta V_{\text{DD}}} \quad \left[\frac{\text{V}}{\text{V}}\right]$$

Equation 3: Line Regulation Metric

- **Load Regulation** — indicates how well a circuit maintains its specified output voltage in response to variations in the load current (V_{DD} is constant). It can also be improved by increasing the DC open-loop voltage gain of the error amplifier.

$$\text{Load Regulation} = \frac{\Delta V_{\text{OUT}}}{\Delta I_{\text{LOAD}}} \quad \left[\frac{\text{mV}}{\text{mA}}\right]$$

Equation 4: Load Regulation Metric

- **Power Supply Rejection Ratio (PSRR)** — indicates how effectively the circuit suppresses ripple and noise from the supply voltage in dB. For example, a PSRR better than −60 dB at 1 kHz means the LDO attenuates a 1 mV ripple at that frequency to less than 1 μV at the output. Typically, an LDO provides excellent rejection at low frequencies because of the high loop gain and PSRR of the reference voltage, however, PSRR typically degrades at higher frequencies. This is generally acceptable since the LDO's output usually does not experience switching noise. PSRR is analogous to line regulation but measured across different frequencies—therefore, improving line regulation through high loop-gain also improves PSRR.

$$\text{PSRR} = 20\log_{10}\left(\frac{\Delta v_{out}}{\Delta v_{\text{DD}}}\right) \quad \text{(dB)}$$

Equation 5: PSRR Metric

Commercial bandgap-based LDOs, examples of which are summarized in Table 1, typically exhibit a temperature coefficient (TC) of 20–100 [ppm/°C], line regulation in the range of 0.0001–0.0005 [V/V], and load regulation between 0.02–0.1 [mV/mA].

Parameter	**NCP4681**	**ADP121**	**LD39015**
Temperature Coefficient (ppm/°C)	100	10	30
Load Regulation (mV/mA) $V_{out} = 1V$	0.268	0.01	0.02
Line Regulation (V/V) $I_{LOAD} = 1mA$	0.0002-0.001	0.0003	0.0001
PSRR (dB)	@1kHz - 25	@10kHz - 66	@1kHz - 65

Table 1
Comparison of Typical Parameters for Commercial LDOs, Taken from [30] [31] [32]

- **Quiescent Current** I_Q — indicates the minimal current necessary to operate the LDO's periphery circuitry, such as its error amplifier, the bandgap reference and the feedback network. Essentially this is the difference between the input current and the load current. It can be minimized by using large resistors in the feedback network of the LDO and minimizing current usage in the OTA and BGR.

$$I_Q = I_{\text{Total}} - I_{\text{Load}}$$

Equation 6: Quiescent Current Metric

- **Dropout Voltage** V_{DO} — represents the minimum difference between the supply voltage V_{DD} and the output voltage V_{OUT} at full load current for which the LDO can still maintain effective regulation. In an LDO, this corresponds roughly to the drain-to-source voltage V_{DS} of the pass transistor and can get as low as 50 mV to 200 mV. Reducing the dropout voltage can be achieved by increasing the width-to-length ratio (W/L) of the pass transistor. However, this also significantly increases the transistor's gate parasitic capacitance, often to hundreds of fF up to the low pF range, which may introduce stability challenges in the regulator's control loop.

$$V_{\mathrm{DO}} = V_{\mathrm{DD}} - V_{\mathrm{OUT}} = I_{\mathrm{Load}} \cdot R_{\mathrm{DS(on)}} = V_{\mathrm{DS}} \quad \text{(of the PMOS pass transistor)}$$

Equation 7: Dropout Voltage Metric

- **Power Efficiency** — indicates the ratio of power delivered to the load relative to the total power consumed by the system. The remaining power is dissipated as heat in the pass element. Power efficiency can be improved by minimizing both the quiescent current and the dropout voltage.

$$\eta = \frac{V_{\mathrm{OUT}} \times I_{\mathrm{Load}}}{V_{DD} \times I_{\mathrm{Total}}} = \frac{V_{\mathrm{OUT}} \times I_{\mathrm{Load}}}{V_{DD} \times (I_{\mathrm{Load}} + I_Q)} = \frac{(V_{DD} - V_{DO}) \times I_{\mathrm{Load}}}{V_{DD} \times (I_{\mathrm{Load}} + I_Q)}$$

Equation 8: Power Efficiency Metric

- **Input/Output Range** — defines the range of V_{DD}, V_{out}, and I_{Load} over which the LDO maintains effective output regulation. In particular, the minimum V_{DD} and maximum I_{Load} at which V_{out} remains regulated are critical design considerations.
- **Transient Response** — indicates the LDO's response for an impulse/step change in the load current or the supply voltage. It is highly dependent on the OTA loop gain and the output filter network, which influence key aspects such as response time, settling behavior, and overshoot.
- **Loop Gain and Stability** — This represents the behavior of the LDO across frequency in terms of gain and phase. While the total DC loop gain directly influences load and line regulation, the gain-bandwidth product (GBW) and phase response are critical for determining the LDO's transient behavior and overall stability.
  To assess and ensure stability, the following frequency-domain metrics are commonly used:
  - **Phase Margin (PM)** – The phase difference between the loop gain phase at the unity-gain frequency (0 dB) and $-180°$. A phase margin in the range of $45°$ to $90°$ is typically considered stable. Low or negative phase margin indicates the risk of oscillation, as the negative feedback may turn into positive feedback.
  - **Gain Margin (GM)** – The difference in gain (in dB) between the point where the phase crosses $-180°$ and 0 dB. A gain margin above 10–20 dB is generally targeted to ensure robust stability.
  - **Gain-Bandwidth Product (GBW)** – Defined as the product of the open-loop DC gain and the frequency at which the gain drops by 3 dB (bandwidth):

    $$GBW = |A_{DC,OL}| \cdot f_{-3dB}$$

    For a single-pole system, the GBW is equal to the unity-gain bandwidth (UGB). In well-compensated two or three pole systems, the GBW should be designed to closely match the UGB, effectively obtaining a single-pole system and ensuring high phase margin and predictable transient response.
- **Output Noise** — LDOs generate less output noise than switching regulators but are still influenced by thermal, flicker, and BJT shot noise. At low frequencies, OTA flicker noise tends to dominate [11] and it is given by:

$$\overline{V^2_{\mathrm{noise,output}}} = \left(1 + \frac{R_1}{R_2}\right)^2 \cdot \overline{V^2_{\mathrm{noise,OTA}}}$$

Equation 9: Output Noise Metric

where $\overline{V^2_{\mathrm{noise,OTA}}}$ denotes the input-referred noise of the OTA. Since this noise is amplified by the closed-loop gain of the regulator, designing a low-noise OTA to minimize the overall output noise of the LDO is favorable.

- **Figure of Merit (FOM)** — To allow reliable comparison of LDO performance, various FOM definitions have been proposed in the literature. While no single FOM captures all aspects of LDO behavior, the most widely adopted FOM is given by:
  Here, C_{out} is the total output capacitance, ΔV_{out} is the output voltage deviation between maximum and minimum load current, I_Q is the quiescent current, and $I_{\mathrm{Load,max}}$ is the maximum load current.

$$FOM = \frac{C_{\text{out}} \cdot \Delta V_{\text{out}} \cdot I_Q}{I^2_{\text{Load,max}}}$$

Equation 10: Standard LDO Figure of Merit

This FOM captures how efficiently the LDO regulates output voltage under dynamic loading conditions. Lower FOM values indicate better transient performance, minimal output deviation, and more efficient regulation with minimal quiescent current and output capacitance.

Line Regulation Analysis

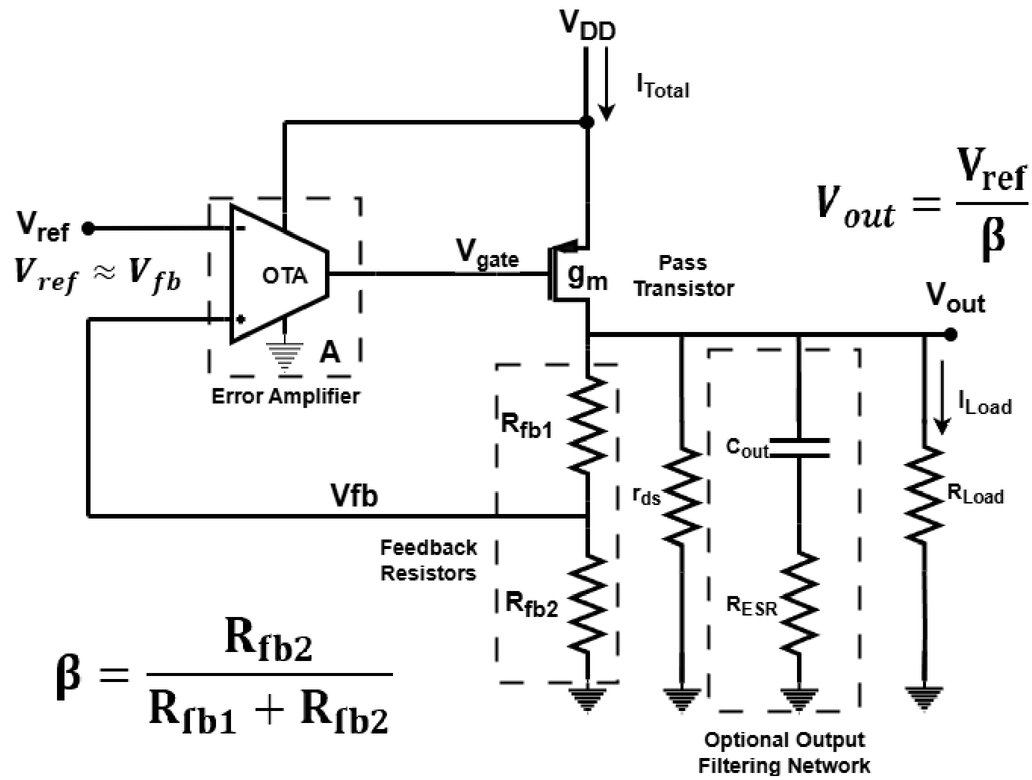

Figure 2. LDO Circuit for Line/Load Regulation and Stability Analysis

The next analysis is based on Figure 2. The OTA regulates the gate-source voltage (V_{SG}) of the pass transistor. This relationship is described by:

$$V_{SG} = V_{DD} - A\beta \cdot (V_{OUT} - V_{REF})$$

For small-signal analysis, we take the differential form:

$$\Delta V_{SG} = \Delta V_{DD} - A\beta \cdot \Delta V_{OUT}$$

Here, V_{REF} is treated as constant under small supply variations due to the high line regulation of the bandgap reference circuit.

This gate-source voltage perturbation is manifested at the output through the PMOS current source stage. The resulting expression that describes the line regulation behavior is:

$$(\Delta V_{DD} - \beta A \cdot \Delta V_{OUT})\, g_m \left(R_{Load} \parallel r_{DS} \parallel (R_{fb1} + R_{fb2})\right) = \Delta V_{OUT}$$

In this expression, ΔV_{DD} is a small change in the supply voltage, ΔV_{OUT} is the resulting change in output voltage, $\beta = \frac{R_{fb2}}{R_{fb1}+R_{fb2}}$ is the feedback factor from the resistor divider, A is the DC gain of the error amplifier, g_m is the transconductance of the PMOS pass transistor, R_L is the load resistance, r_{DS} is the output resistance of the PMOS, and $(R_{fb1} + R_{fb2})$ is the total resistance of the feedback network.

Solving for $\frac{\Delta V_{\text{OUT}}}{\Delta V_{\text{DD}}}$ gives:

$$\frac{\Delta V_{\text{OUT}}}{\Delta V_{\text{DD}}} = \frac{g_m \left(R_{\text{Load}} \parallel r_{DS} \parallel (R_{fb1} + R_{fb2})\right)}{1 + \beta A g_m \left(R_{\text{Load}} \parallel r_{DS} \parallel (R_{fb1} + R_{fb2})\right)} \approx \frac{1}{\beta A}$$

Equation 11: Line Regulation / PSRR as a Function of the Loop Gain

This expression clearly shows that the line regulation performance and PSRR of a Low-Dropout Regulator (LDO) improves with a higher error amplifier gain A, as the high open-loop gain of the OTA suppresses changes in V_{DD} from propagating to the output voltage V_{OUT}. Additionally, increasing the feedback factor β also enhances line regulation. However, this design choice forces V_{OUT} to be close to V_{REF}, which may not be appropriate in all applications. In our case, we take advantage of this to improve performance. Thus for $\beta = 0.8$, the OTA gain would need to be around 62dB to get a line regulation of 0.001 $[V/V]$.

Load Regulation Analysis

The following analysis is based on Figure 2. When the output voltage ΔV_{OUT} changes, the total variation in load current ΔI_{Load} consists of two components:

- A passive current flowing through the output impedance of the LDO, composed of the pass transistor's output resistance and the resistive feedback network.
- An active correction current through the feedback loop of the OTA and pass device to regulate the output voltage.

Hence, the load regulation behavior of the LDO can be described by the following expression:

$$\Delta I_{\text{Load}} = \frac{\Delta V_{\text{OUT}}}{r_{\text{DS}} \parallel (R_{\text{fb1}} + R_{\text{fb2}}) \parallel R_{\text{Load}}} + \Delta V_{\text{OUT}} \cdot \beta A g_m$$

Solving for $\frac{\Delta V_{\text{OUT}}}{\Delta I_{\text{Load}}}$ gives:

$$\frac{\Delta V_{\text{OUT}}}{\Delta I_{\text{Load}}} = \frac{r_{\text{DS}} \parallel (R_{\text{fb1}} + R_{\text{fb2}}) \parallel R_{\text{Load}}}{1 + \beta A g_m \left(r_{\text{DS}} \parallel (R_{\text{fb1}} + R_{\text{fb2}}) \parallel R_{\text{Load}}\right)} \approx \frac{1}{\beta A g_m}$$

Equation 12: Load Regulation as a Function of the Loop Gain

This approximation holds when the loop gain is large enough. A high open-loop gain A of the OTA improves the LDO's load regulation by minimizing deviations in V_{OUT} under varying I_{Load}. Additionally, a large pass transistor with high transconductance g_m further improves regulation performance. Since the transconductance g_m is unlikely to exceed 100 mS, it becomes evident that the load regulation cannot be better than the line regulation. Assuming a feedback factor $\beta = 0.8$ and an OTA gain of 62 dB, the best-case load regulation, based on a maximum achievable g_m of 100 mS, is approximately 0.01 V/A. To achieve significantly better load regulation, an open-loop gain closer to 80 dB is desirable. This provides sufficient margin, especially considering that post-layout parasitics may degrade performance beyond the allowed specification.

Loop Gain and Stability Analysis

The feedback loop gain of an LDO with a 2-stage OTA is shaped by three poles located at high-impedance nodes. The first pole is the output pole, given by:

$$\omega_{p1} = \frac{1}{R_{\text{out}} C_{\text{out}}} = \frac{1}{(r_{\text{ds}} \parallel R_{\text{Load}} \parallel (R_{fb1} + R_{fb2})) C_{\text{out}}}$$

Here, C_{out} can represent an on-chip decoupling capacitor placed near the load between the LDO output and ground to act as a local energy reservoir and suppress voltage spikes or dips during fast load transients. It can also be a dedicated on-chip or off-chip capacitor added specifically for the LDO to shape its frequency response and improve noise suppression.

The other two poles are related to the OTA. The first one is the internal pole of the first stage, which is usually the smallest one, hence it is the dominant pole in both the OTA and the LDO and dictates their open-loop bandwidth. The final pole appears at the output of the OTA (the gate of the pass transistor) and is given by:

$$\omega_{p2} = \frac{1}{R_{\text{out,OTA}} C_{G,\text{PMOS}}}$$

where $R_{\text{out,OTA}}$ is the output resistance of the OTA, and $C_{G,\text{PMOS}}$ is the parasitic gate capacitance of the PMOS pass transistor. This parasitic capacitance increases as the pass transistor width is increased to support higher load currents.

Depending on the relative values of the resistances and capacitances in the circuit, these poles can be positioned at varying frequencies. When the poles are close to each other, the phase margin is significantly reduced, which compromises loop stability. Therefore, it is crucial to space these poles as far apart as possible. A widely used guideline is to place the non-dominant poles beyond the unity-gain frequency. This frequency defines the LDO's bandwidth and thus determines the transient response speed. A wider bandwidth results in faster transient response.

The challenge of ensuring stability becomes more even more complex because the position of the output pole itself is heavily influenced by the load current, as expressed by:

$$\omega_{p1} = \frac{1}{R_{\text{out}} C_{\text{out}}} \approx \frac{1}{r_{\text{ds}} C_{\text{out}}} = \frac{I_{\text{Load}}}{V_A C_{\text{out}}}$$

where V_A is the Early voltage and C_{out} is the output capacitance, both assumed constant. Consequently, the dominant pole ω_{p1} increases linearly with the load current I_{Load}. This implies that for a large output capacitor (on the order of 1 μF), the output pole may dominate at low load currents but can shift to higher frequencies as the load increases. Such a shift can cause the output pole to cross other poles, and become non-dominant and potentially destabilize the loop. To maintain stability across the entire load range, a resistor is often introduced in series with the output capacitor. This creates a left-half-plane (LHP) zero, which improves phase margin and helps preserve loop stability.

Alternatively, poles can be spaced far apart so that the output pole remains non-dominant under all load conditions. This approach sometimes leads to the design of capless LDOs that operate without an external output capacitor. However, omitting output capacitance entirely is not always practical, as it plays a critical role in damping transient voltage spikes and dips, suppressing output noise, and smooth the output response..

The loop gain LG of the feedback system is given by:

$$LG = A \cdot \beta \cdot g_m \cdot R_{\text{out}} = A \cdot \beta \cdot g_m \cdot \left(r_{ds} \parallel R_{\text{Load}} \parallel (R_{fb1} + R_{fb2}) \right)$$

Here, A is the gain of the error amplifier, $\beta = \frac{R_{fb2}}{R_{fb1}+R_{fb2}}$ is the feedback factor set by the resistor divider, g_m is the transconductance of the pass transistor, and R_{out} is the equivalent output resistance formed by the parallel combination of the transistor output resistance r_{ds}, the load resistance R_{Load}, and the total feedback resistance $R_{fb1} + R_{fb2}$.

Hence, the open-loop transfer function $T(s)$ of the feedback system is given by:

$$T(s) = \frac{V_{\text{fb}}}{V_+} = \frac{A \beta g_m \left(r_{ds} \parallel R_{\text{Load}} \parallel (R_{fb1} + R_{fb2}) \right) \left(1 + \frac{s}{\omega_z}\right)}{\left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \left(1 + \frac{s}{\omega_{p3}}\right)}$$

Equation 13: Open-loop Transfer Function of the LDO

where V_+ is the non-inverting input of the OTA, at which a small-signal source is connected, and V_{fb} is the voltage at the feedback node between the two feedback resistors. The additional LHP zero is the internal compensation zero in the OTA. This transfer function represents the open-loop response of the system, with the feedback loop temporarily broken between the OTA input and the resistive feedback network for the purpose of frequency-domain analysis.

Design Implications

In the case of the DPLL chip, the output capacitance is determined primarily by decoupling capacitors at the input of each DPLL block. These capacitors typically range from 20 fF to 60 fF, thus no more than 300 fF in total across all blocks. As a result, the output pole frequency is expected to be very high (around 1 GHz) and will not affect stability. Therefore, in this scenario, the LDO stability can be ensured by focusing solely on OTA stability. The small decoupling capacitors should be enough to dampen possible voltage dips and suppress noise, hence there is no need for a dedicated output capacitor and series resistor in the LDO itself. Consequently, the layout of our LDO will not include an output capacitor. To ensure stability and seamless integration, we must verify that the LDO remains stable across this range of output capacitance, and simulations must be performed to analyze the associated trade-offs between their exact sizes. The decision to include a series damping resistor with the output capacitor is deferred to the top-level design, depending on the overall system performance following full integration.

2.2 OTA as an Error Amplifier

Function in LDO and BGR Systems

The Operational Transconductance Amplifier (OTA) plays a central role in both the LDO regulator and the bandgap reference (BGR) used in this project. In both cases, it serves as a high-gain analog block that enables closed-loop regulation.

In the LDO, the OTA compares a scaled-down version of the output voltage to a reference voltage and adjusts the gate of the pass transistor accordingly. This enables the regulator to maintain a steady output despite variations in load current or supply voltage.

In the BGR, the OTA ensures equal voltages across two branches, facilitating the generation of a Proportional To Absolute Temperature (PTAT) current. This current is then combined with a Complementary-To-Absolute-Temperature (CTAT) component to produce a stable, temperature-independent reference voltage.

Why Use a Transconductance Amplifier?

Unlike traditional voltage amplifiers, an operational transconductance amplifier (OTA) produces an output current proportional to its differential input voltage. While operational amplifiers typically drive resistive loads and have low output impedance, OTAs drive capacitive loads and have high output impedance. Although the OTA is fundamentally a current-output device, in practical LDO and BGR applications, it effectively functions as part of a voltage amplification stage. This is because the output current charges or discharges capacitive nodes, which leads to a voltage swing. Ultimately, the overall loop behavior and regulation performance are mostly determined by the gain of the OTA.

General Structure and Need for High Gain

The OTA used in this project follows a two-stage architecture as in Figure 3:

- A differential input stage that subtracts the two input voltages and provides initial gain.
- A common-source gain stage that further boosts the gain and drives the capacitive load.

This structure provides high open-loop gain, typically exceeding 60 dB, which is essential for precise regulation:

- In the LDO, high gain improves both line and load regulation.
- In the BGR, it enables better voltage matching between nodes, which enhances reference voltage accuracy over temperature and PSRR.

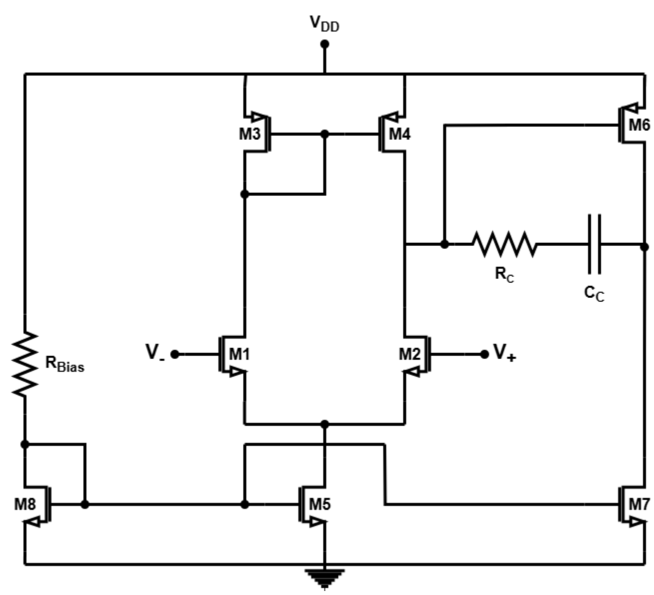

Figure 3. The 2-stage Miller OTA Topology used in the LDO and the BGR

Stability Concerns and Pole Splitting

In an error amplifier, ensuring stability is essential for reliable operation under all expected conditions. Instability can arise when a system intended to operate under negative feedback unintentionally becomes positive feedback,typically at high frequencies. This leads to oscillations at the output.

The two-stage OTA is naturally not stable and requires frequency compensation since it introduces two internal poles, which tend to lie close together in frequency due to the structure of the circuit. The output pole of the first stage is located at

$$\omega_{p1} = \frac{1}{(r_{ds2} \parallel r_{ds4})C_{\text{par}}},$$

where C_{par} is the total parasitic capacitance seen at the drain of the first stage. The second stage's output pole is similarly located at

$$\omega_{p2} = \frac{1}{(r_{ds6} \parallel r_{ds7})C_{\text{Load}}}.$$

Since transistors M_2, M_4, M_6, and M_7 are typically designed with similar channel lengths (L) and carry similar drain currents (I_{DS}), the resulting output resistances and capacitive parasitics are similar. Without any compensation, this causes the poles to lie close together, which degrades phase margin and lead to instability.

The gain of each stage can be expressed as:

$$A_{v1} = -g_{m2}(r_{ds2} \parallel r_{ds4}), \quad A_{v2} = -g_{m6}(r_{ds6} \parallel r_{ds7}).$$

If these poles are not properly split, the phase drops rapidly near their frequencies, which can lead to a low phase margin and increase the risk of oscillation.

To ensure stability, the poles must be split: the first stage pole (the dominant pole) should be shifted to a lower frequency, while the second stage pole (the non-dominant pole) should be moved to a higher frequency. This is typically achieved by adding a compensating capacitor between the output of the first stage and the input of the second stage. While this capacitor does not affect the DC gain, it introduces a high impedance path at low frequencies, while preserving high gain at low frequencies and attenuating the gain at higher frequencies. Hence, the first stage pole is shifted to lower frequencies.

Additionally, the non-dominant pole can be pushed to higher frequencies by increasing the transconductance g_{m6} of the second stage. This can be done by either increasing the transistor width (W) or increasing the bias current in that stage. As a result the poles are effectively separated and stability is improved by ensuring the gain drops sufficiently before excessive phase shift accumulates.

The Miller Effect and Trade-offs

Instead of adding a large capacitor directly to the input, the Miller effect can be used to achieve a large effective input capacitance using a relatively small compensation capacitor C_c. This effect occurs in amplifying stages with negative voltage gain, such as the common-source stage. Since the output inverts the input ($A_v < 0$), a capacitor connected between the input (gate) and output (drain) appears as a much larger capacitance at the input node of the second stage. The effective input capacitance is given by:

$$C_{\text{in,eff}} = C_c(1 - A_v) = C_c(1 + |A_v|)$$

Equation 14: Miller Input Capacitance

When the gain of the stage is high (e.g., $A_v = -100$), even a small C_c can significantly increase the effective input capacitance. As a result, the input pole shifts to a lower frequency:

$$p_1 = \frac{1}{R_1(C_{\text{par}} + C_c(1 + |A_v|))}.$$

This pole often becomes the dominant pole of the OTA and limits its bandwidth, thereby reducing its speed. Figure 4 highlights a fundamental trade-off between stability and bandwidth in this topology:

- With a small or no C_c, the amplifier retains a higher bandwidth but suffers from a low phase margin and potential instability.
- With a larger C_c, the phase margin improves due to better pole separation, but the dominant pole moves to lower frequencies, which reduces bandwidth.

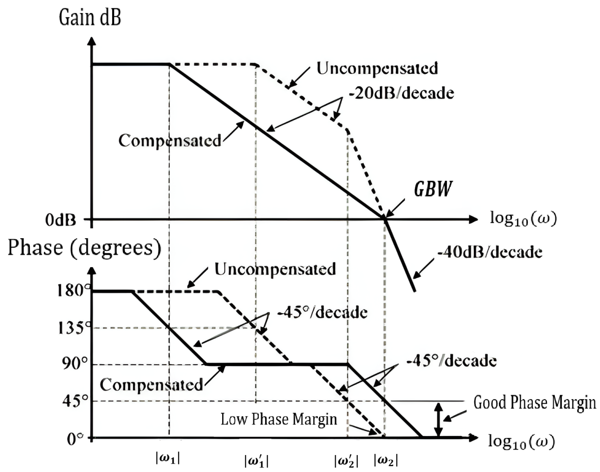

Figure 4. Bode Plot for a Compensated and Non-compensated OTA

At the output, the Miller capacitor isn't amplified, so its effect on the non-dominant pole is modest. This behavior is illustrated in Figure 5.

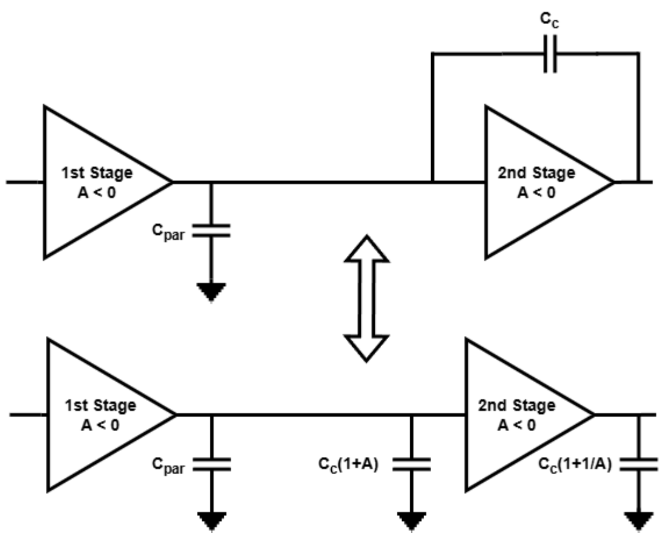

Figure 5. Miller Capacitor C_c modeled as Equivalent Input and Output Capacitors

However, Miller compensation introduces an undesired side-effect. It creates a feedback path that can lead to a right-half-plane (RHP) zero. This occurs since the compensation capacitor connects the output of the first stage to that of the second stage, which allows high-frequency current to feed directly back into the first stage. As frequency increases, the capacitor effectively bypasses the second stage, which leads to a reduction of its gain.

Unlike a left-half-plane zero, which adds positive phase, an RHP zero lowers phase margin. The larger the compensation capacitor, the lower the frequency at which this zero appears. This can bring it very close to the unity-gain frequency and exacerbate stability issues.

To mitigate this, a series resistor can be added in series with the compensation capacitor to shift the zero either into the left half-plane or to higher frequencies within the right half-plane. This simple and widely used compensation network stabilizes OTAs with low area and power usage.

Small Signal Analysis of the OTA

In Figure 6, R_1, R_2 are the output resistances of the first, and second stages, respectively. C_{par} is the total parasitic capacitance at the first stage's output node, which consists of C_{ds4}, C_{gs6} etc. It is much smaller than C_C. g_{m_1}, g_{m_2} are the trans-conductances of the first, and second stages, respectively.

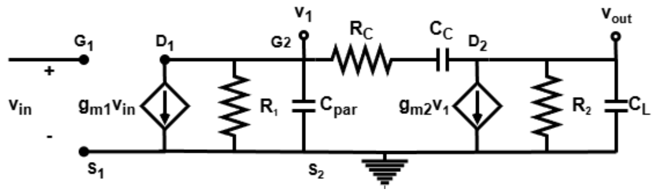

Figure 6. The Small Signal Model of the OTA

To derive the transfer function analytically, we simplify by taking $R_C = 0$ and splitting the transfer function into two separate transfer functions using KCL in both stages' output nodes:

$$\frac{v_{out}}{v_{in}} = \frac{v_1}{v_{in}}\frac{v_{out}}{v_1}$$

$$\frac{v_1}{\frac{1}{sC_{par}}} + \frac{v_1}{R_1} + g_{m_1}v_{in} + \frac{v_1 - v_{out}}{R_C + \frac{1}{sC_C}} = 0 \quad (KCL\ first\ stage)$$

$$v_1 = \frac{v_{\text{out}}sR_1C_C - v_{in}g_{m_1}R_1}{1 + sR_1(C_C + C_{par})} \tag{1}$$

$$\frac{v_{out}}{\frac{1}{sC_L}} + \frac{v_{out}}{R_2} + g_{m_2}v_1 + \frac{v_{out} - v_1}{R_C + \frac{1}{sC_C}} = 0 \quad (KCL\ second\ stage)$$

$$v_{\text{out}}\left(s(C_L + C_C) + \frac{1}{R_2}\right) = v_1\,(sC_C - g_{m2}) \tag{2}$$

Plugging (1) into (2) to obtain the full transfer function:

$$\frac{v_{\text{out}}}{v_{\text{in}}} = \frac{g_{m1}R_1g_{m2}R_2\left(1 - \frac{sC_C}{g_{m2}}\right)}{s^2R_1R_2(C_{\text{par}}C_C + C_{\text{par}}C_L + C_CC_L) + s\left[R_2(C_C + C_L) + R_1(C_C + C_{\text{par}}) + g_{m2}R_2R_1C_C\right] + 1}$$

Equation 15: Voltage Gain of the OTA

As expected, the transfer function is of the form:

$$\frac{v_{out}}{v_{in}} = \frac{A(1 - \frac{s}{z})}{(1 + \frac{s}{p_1})(1 + \frac{s}{p_2})} = \frac{A(1 - \frac{s}{z})}{s^2(\frac{1}{p_1p_2}) + s(\frac{1}{p_1} + \frac{1}{p_2}) + 1} \approx \frac{A(1 - \frac{s}{z})}{s^2(\frac{1}{p_1p_2}) + \frac{s}{p_1} + 1}$$

where A is the DC gain, and assuming $p_1 << p_2$ because of the pole splitting. Therefore, comparing the coefficients of both transfer functions gives:

$$A = g_{m1}R_1g_{m2}R_2$$

$$p_1 = \frac{1}{R_2(C_c + C_L) + R_1(C_c + C_{\text{par}}) + g_{m2}R_2C_CR_1} \approx \frac{1}{g_{m2}R_2C_CR_1}$$

Since $g_{m2}R_2$ is the gain of the second stage, hence this is the dominant term (Miller input pole). Therefore, the GBW is given by:

$$GBW = DC\ gain \times p_1 = \frac{g_{m1}}{C_c}$$

The second pole is located at:

$$p_1p_2 = \frac{1}{R_1R_2(C_{\text{par}}C_C + C_{\text{par}}C_L + C_CC_L)} \rightarrow p_2 = \frac{g_{m2}C_C}{C_{\text{par}}C_C + C_{\text{par}}C_L + C_CC_L} \approx \frac{g_{m2}}{C_L}$$

The zero is located at:

$$z = \frac{g_{m2}}{C_c}$$

Thus, for larger C_c, the zero shifts to lower frequencies, and worsens the stability problem. Without a resistor, the only way to mitigate this is by pushing the RHP zero beyond the unity-gain frequency, typically done by increasing the transconductance of the second stage (g_{m2}), which often requires wider transistors or bias current and results in higher area or power consumption.

Hence, introducing a series resistor to C_c offers a more elegant solution. It has minimal effect on pole locations but can either cancel the RHP zero or shift it into the left-half-plane (LHP). This improves phase margin by contributing up to 90° of positive phase shift.

The shifted zero is located at:

$$z = \frac{1}{C_c\left(\frac{1}{g_{m2}} - R_c\right)} \quad \text{LHP zero for: } R_c > \frac{1}{g_{m2}}.$$

Summary and Design Implications

The OTA in this project is designed to achieve high open-loop gain, adequate phase margin, and robust operation of both the LDO and BGR across all conditions. Miller compensation with a series resistor is used to maintain stability while minimizing power and area, which facilitates integration within a DPLL system.

2.3 Bandgap Reference Circuit

Motivation for Using a Bandgap Reference

Low-dropout regulators (LDOs) require a precise and stable voltage reference to ensure consistent output regulation across variations in temperature, supply voltage (V_{DD}), and manufacturing process parameters. Since no naturally occurring component provides a voltage with zero temperature dependence, various reference circuit designs have been developed, but most suffer from key drawbacks. Passive dividers lack both supply and temperature independence, while MOS-based and threshold voltage references are highly process-sensitive and exhibit significant temperature variation. Diode-based references and thermo-voltage references, though more predictable, still show substantial temperature coefficients.

In contrast, the Bandgap Reference (BGR) is widely regarded as the most suitable for precision applications due to its excellent temperature stability and insensitivity to supply and process variations. This makes it uniquely well-suited for robust and precise voltage referencing in analog and mixed-signal circuits. As a result, BGRs are commonly used in LDOs, ADCs, and DACs to provide a stable and accurate reference voltage.

Bandgap Reference Circuit Operation Principle

In most cases, Bandgap References (BGRs) operate in voltage mode (an example topology is shown in Figure 7), where temperature variation, typically spanning from −40 °C to 125 °C, is compensated by combining two opposing temperature-dependent voltages: a complementary-to-absolute-temperature (CTAT) voltage that decreases with temperature, and a proportional-to-absolute-temperature (PTAT) voltage that increases with temperature. When properly scaled, their sum yields a reference voltage with a near-zero temperature coefficient, i.e., $\frac{dV_{\text{ref}}}{dT} \approx 0$.

A naturally occurring CTAT voltage is the base-emitter voltage (V_{EB}) of a bipolar junction transistor (BJT) operating in forward-active mode. It is well known that V_{EB} decreases by approximately 1.5 mV/°C, although this dependence is nonlinear, especially over a wide temperature range. As such, second-order curvature compensation techniques can be employed to improve accuracy.

On the other hand, the thermal voltage $V_T = \frac{kT}{q}$ exhibits PTAT behavior, increasing by roughly 86.25 μV/°C, where k is Boltzmann's constant, T is the absolute temperature in kelvin, and q is the elementary charge. Since this PTAT slope is significantly weaker than that of the CTAT voltage, a BGR must carefully scale the PTAT component to balance the stronger CTAT effect. Proper design ensures that the resulting reference voltage remains stable across process and temperature variations.

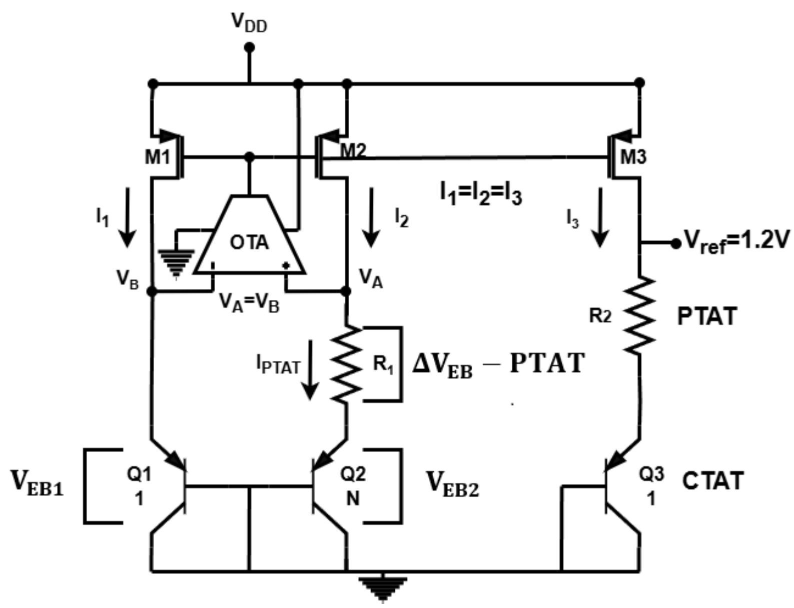

Figure 7. Voltage Mode Bandgap Reference Circuit Topology

A diode-connected BJT, where the base and collector are shorted, effectively eliminates the collector-base junction. In this configuration, the base-emitter junction behaves like a standard forward-biased diode. Consequently, the collector current in a diode-connected BJT operating in the forward-active region follows the exponential relationship:

$$I_C = I_S e^{\frac{V_{EB}}{V_T}} \ (for\ V_{EB} \gg V_T)$$

where I_S is the saturation current, V_{EB} is the base-emitter voltage, and $V_T = \frac{kT}{q}$ is the thermal voltage, which is $26mV$ at room temperature. Therefore, the base-emitter voltage is given by:

$$V_{EB} = V_T \ln\left(\frac{I_C}{I_S}\right)$$

Equation 16: Base-Emitter Voltage of a BJT

The PTAT term, given by the thermal voltage V_T, is relatively weak in comparison to the dominant CTAT component in the base-emitter voltage. Specifically,

$$\frac{dV_T}{dT} = \frac{k}{q} \approx 86.25\,\mu\text{V}/^\circ\text{C},$$

where k is Boltzmann's constant and q is the elementary charge. Assuming the collector current I_C is constant with respect to temperature (for this derivation), then the CTAT behavior originates from the exponential dependence of the base-emitter voltage on the saturation current I_S, which varies strongly with temperature. The saturation current of a BJT is given by:

$$I_S = A_E q D_n \frac{n_i^2}{N_A L_n},$$

where:

- A_E: the emitter area,
- q: the elementary charge (temperature-independent constant),
- D_n: the diffusion coefficient of electrons, which depends on temperature,
- n_i: the intrinsic carrier concentration, strongly temperature-dependent,
- N_A: the acceptor doping concentration,
- L_n: the diffusion length of electrons.

From Einstein's relation, the diffusion coefficient D_n is given by:

$$D_n = \mu \frac{kT}{q},$$

where μ is the carrier mobility, which typically varies with temperature as $\mu \propto T^{-1.5}$. Hence, the diffusion coefficient approximately follows

$$D_n \propto T^{-0.5}.$$

Additionally, the intrinsic carrier concentration depends on temperature as

$$n_i^2 \propto T^3 e^{\frac{-E_g}{kT}},$$

where E_g is the energy bandgap of silicon and k is Boltzmann's constant. Since A_E and q are temperature-independent, the strong temperature dependence of the saturation current I_S primarily arises from the temperature variation in D_n and n_i^2. This results in the exponential increase of I_S with temperature and leads to the complementary-to-absolute-temperature (CTAT) behavior of the base-emitter voltage V_{EB}, which typically has a temperature coefficient around $-1.5\,\text{mV}/^\circ\text{C}$.

Therefore, denoting constants as C, we obtain:

$$I_S = CT^{2.5} e^{\frac{-E_g}{kT}}$$

To get the TC of the saturation current, we derive it with respect to temperature:

$$\frac{dI_s}{dT} = C\left(T^{2.5} e^{\frac{-E_g}{kT}} \cdot \frac{E_g}{kT^2} + 2.5T^{1.5} e^{\frac{-E_g}{kT}}\right) = \underbrace{CT^{2.5} e^{\frac{-E_g}{kT}}}_{=I_s} \left(\frac{E_g}{kT^2} + \frac{2.5}{T}\right) = I_s \left(\frac{E_g}{kT^2} + \frac{2.5}{T}\right)$$

Given $I_s, V_T, \frac{dI_s}{dT}, \frac{dV_T}{dT}$, we can derive V_{EB} with respect to temperature:

$$\begin{aligned}
\frac{dV_{EB}}{dT} &= \frac{dV_T}{dT}\ln\left(\frac{I_C}{I_s}\right) + V_T\frac{d}{dT}\left(\ln\left(\frac{I_C}{I_s}\right)\right) \\
&= \frac{k}{q}\ln\left(\frac{I_C}{I_s}\right) - \frac{V_T}{I_s}\cdot\frac{dI_s}{dT} \\
&= \frac{k}{q}\ln\left(\frac{I_C}{I_s}\right) - V_T\left(\frac{E_g}{kT^2}+\frac{2.5}{T}\right) \\
&= \frac{V_T\ln\left(\frac{I_C}{I_s}\right)}{T} - \frac{2.5V_T}{T} - V_T\cdot\frac{E_g}{kT^2} \\
&= \underbrace{\frac{V_T\ln\left(\frac{I_C}{I_s}\right)}{T}}_{=\frac{V_{EB}}{T}} - \frac{2.5V_T}{T} - \frac{kT}{q}\cdot\frac{E_g}{kT^2} \\
&= \frac{V_{EB} - 2.5V_T - \frac{E_g}{q}}{T}
\end{aligned}$$

To find the voltage change across the base-emitter junction of the BJT wrt. temperature at $27^\circ C = 300^\circ K$, we plug in the values:

$$\frac{dV_{BE}}{dT}(T = 300^\circ K) = \frac{0.7 - 2.5\times 0.026 - 1.12}{300} = -1.616\frac{mV}{^\circ K}$$

The derived value is very close to values reported in the literature. However, the previous derivation assumes that the collector current I_C is temperature-independent, which is not accurate. In practice, the current in all three branches of the bandgap reference circuit shown in Figure 7 is set by the resistor R_1, if the following three conditions are satisfied:

1. **Equal Branch Currents:** The current through both PNP transistors must be equal to ensure they have the same collector current I_C. This can be achieved by using a current mirror composed of transistors M_1, M_2, and M_3 which accurately replicate the reference current. To minimize the impact of the Early effect and improve current matching, all transistors in the mirror are designed with long channel lengths and operate with the same drain-source voltage (V_{DS}).
2. **Equal Node Voltages:** The voltages at the two nodes V_A and V_B must be equal. While this could also be enforced with a current mirror, a high-gain operational transconductance amplifier (OTA) is preferred because its higher loop gain provides better voltage matching accuracy.
3. **Different Emitter Areas:** To generate a proportional-to-absolute-temperature (PTAT) voltage, two BJTs with different emitter areas must be used. One transistor must have a base-emitter junction area that is N times larger than the other, which is effectively equivalent to having N transistors in parallel. Due to this increased area, the saturation current I_s is also increased by a factor of N, resulting in a slightly lower V_{EB} for the larger device at the same current. Common choices for the emitter area ratio N include 8, 15, 24, or 80 ($N = integer^2 - 1$). These values facilitate better matching and symmetry in layout, which allow for common centroid structures. Increasing N improves the signal-to-noise ratio by generating a larger ΔV_{EB}, which improves robustness against noise and device mismatch that can lead to input offset in the OTA and thus to an error in V_{ref}. However, this comes at the cost of increased layout area and parasitic effects. As a result, a practical tradeoff is usually made, with $N = 8$ being a widely adopted compromise between precision and layout efficiency.

Having all these satisfied, the OTA, having a virtual short between its inputs, forces $V_A = V_B$, meaning the difference in V_{EB} of both BJTs with the same current but different areas (ΔV_{EB}) must appear across resistor R_1. This gives:

$$V_{R_1} = V_{EB1} - V_{EB2} = V_T\left[\ln\left(\frac{I_C}{I_s}\right) - \ln\left(\frac{I_C}{NI_s}\right)\right] = V_T\ln(N) = \frac{kT}{q}\ln(N)$$

where N is the emitter area ratio between the two BJTs.

Since this voltage is directly proportional to T, it is a PTAT voltage. The current through resistor R_1, which sets the collector current of the BJTs (and is mirrored by transistors M_1, M_2, M_3), is therefore:

$$I_{R_1} = I_C = \frac{V_T\ln(N)}{R_1}$$

This means the current can be tuned via the resistor size R_1. Moreover, power consumption can be minimized by choosing a larger resistance value, as long as the PNP transistors remain in the forward-active region, which is valid because of the diode connection of the BJTs.

The expression above clearly shows that I_C is a PTAT current, and thus I_C is not temperature-independent. The derivation below assumes that R_1 is temperature-independent. This is not accurate, however, by selecting a resistor with a very small temperature coefficient (TC) like a polysilicon-made resistor, its variation on the CTAT slope calculation becomes negligible. The temperature derivative of I_C is given by:

$$\frac{dI_C}{dT} = \frac{\ln(N)}{R_1} \cdot \frac{dV_T}{dT} = \frac{\ln(N)k}{R_1 q} = \frac{V_T \ln(N)}{R_1 T} = \frac{I_C}{T}$$

Thus, we can now derive V_{EB} again with respect to temperature, this time accounting for the PTAT behavior of I_C:

$$\begin{aligned}
\frac{dV_{EB}}{dT} &= \frac{dV_T}{dT} \ln\left(\frac{I_C}{I_s}\right) + V_T \cdot \frac{d}{dT}\left(\ln\left(\frac{I_C}{I_s}\right)\right) \\
&= \frac{V_T}{T} \ln\left(\frac{I_C}{I_s}\right) + V_T \left(\underbrace{\frac{1}{I_C} \cdot \frac{dI_C}{dT}}_{\text{new } I_C \text{ term}} - \frac{1}{I_s} \cdot \frac{dI_s}{dT} \right) \\
&= \frac{V_T}{T} \ln\left(\frac{I_C}{I_s}\right) + V_T \left(\underbrace{\frac{1}{T}}_{\text{new } I_C \text{ term}} - \frac{2.5}{T} - \frac{E_g}{kT^2} \right) \\
&= \frac{V_{EB}}{T} - \frac{1.5V_T}{T} - \frac{kT}{q} \cdot \frac{E_g}{kT^2} \\
&= \frac{V_{EB} - 1.5V_T - \frac{E_g}{q}}{T}
\end{aligned}$$

Thus, the only difference is that the term marked in red changed from $-2.5V_T$ to $-1.5V_T$, meaning the influence of I_C's temperature dependence is small. The more precise temperature coefficient of the base-emitter voltage at room temperature ($27°\text{C} = 300\,\text{K}$) is:

$$\left.\frac{dV_{EB}}{dT}\right|_{T=300\,\text{K}} = \frac{0.7 - 1.5 \times 0.026 - 1.12}{300} = -1.53\,\frac{\text{mV}}{°\text{K}}$$

Simulations in Virtuoso showed approximately a slope of $-1.65\,\text{mV}/°\text{C}$ for the CTAT component and around $100\,\mu\text{V}/°\text{C}$ for the PTAT component, which confirm the accuracy of the theoretical derivation.

To form a temperature-independent reference voltage V_{ref}, a third branch is added which includes a third BJT (with unit emitter area) and a second resistor. The reference voltage is then given by:

$$V_{\text{ref}} = \alpha \cdot \text{PTAT} + \beta \cdot \text{CTAT} = \frac{R_2}{R_1} \ln(N) V_T + \beta V_{EB}$$

where $\alpha = \frac{R_2}{R_1} \ln(N)$, and typically $\beta = 1$.

Since we require $\frac{dV_{\text{ref}}}{dT} = 0$ for temperature independence:

$$\alpha \cdot \frac{dV_T}{dT} + \frac{dV_{EB}}{dT} = 0$$

Substituting known values found previously:

$$\alpha \cdot 86.25\,\mu\text{V/K} - 1.53\,\text{mV/K} = 0$$

Solving for α:

$$\alpha = \frac{1.53\,\mathrm{mV/K}}{86.25\,\mu\mathrm{V/K}} \approx 17.74$$

This determines the approximate resistor ratio and emitter scaling needed to achieve a temperature-independent reference voltage. However, because there is only one degree of freedom, the value of V_{ref} is essentially fixed once the temperature coefficient is canceled:

$$V_{\mathrm{ref}} = \alpha V_T + V_{EB} \approx 17.74 \cdot V_T + 0.7 = 1.16\,\mathrm{V}$$

This value closely matches the bandgap voltage of Silicon where silicon-based BJTs exhibit near-zero temperature coefficient (see Figure 8).

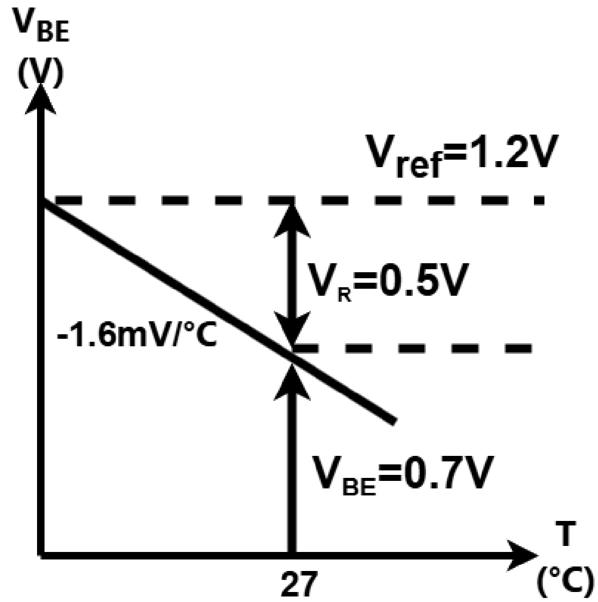

Figure 8. Temperature Compensation in a Bandgap Reference: Combining CTAT and PTAT Components to sum to a Temperature-Compensated Reference Voltage around $\sim 1.2\,\mathrm{V}$.

Feedback Loop Gain Analysis

As previously discussed, since this is a DC circuit, the OTA is not functioning as a typical linear signal amplifier. Instead, its high open-loop gain forces the input differential voltage toward zero, which enables the generation of a stable PTAT (Proportional To Absolute Temperature) current.

Because the OTA is connected across two branches, the circuit contains both negative and positive feedback paths. For correct and stable operation, it is essential that the negative feedback loop dominates. A common design guideline is to ensure that the gain of the negative feedback path is at least twice that of the positive feedback path. Fortunately, this condition is typically met by the inherent structure of the bandgap reference topology.

What remains is selecting the correct polarity for the OTA inputs to ensure proper functionality. Supposing node V_A is connected to the non-inverting (+) input and node V_B to the inverting (˜) input. If V_A increases and V_B stays constant, the OTA's input differential voltage $V_A - V_B$ increases, causing the OTA output voltage $V_{\mathrm{out}} = V_{G1,2}$ to rise (the OTA preserves phase). Since M_1 and M_2 are PMOS transistors, an increase in gate voltage reduces their gate–source voltage V_{SG}, thereby decreasing their drain currents. This leads to a drop in V_A, counteracting the initial difference. This confirms that the loop exhibits negative feedback and the chosen input polarity is correct.

In this configuration, the negative feedback loop dominates over the positive one. To analyze this, we break the loop between the OTA output and the gates of transistors M_1 and M_2. The gain of the negative feedback path is given by:

$$A_{\mathrm{neg}} = \frac{V_{\text{out, OTA}}}{V_{\mathrm{gate}}} = A_{\mathrm{OTA}} \cdot \left(-g_{m2}\left(R_1 + \frac{1}{g_{m,Q2}}\right)\right),$$

while the gain of the positive feedback path is:

$$A_{\text{pos}} = \frac{V_{\text{out, OTA}}}{V_{\text{gate}}} = A_{\text{OTA}} \cdot \left(-g_{m1} \cdot \frac{1}{g_{m,Q1}}\right).$$

Since transistors M_1 and M_2 are matched in size, they exhibit equal transconductance, $g_{m1} = g_{m2} = g_m$. Additionally, the transconductance of a BJT is independent of its base-emitter area, implying $g_{m,Q1} = g_{m,Q2}$. Typical values for the transconductance g_m of both MOS and BJT devices in this context are on the order of 0.1 mS. Selecting a resistor R_1 bigger than 10 kΩ ensures that the gain of the negative feedback path significantly exceeds that of the positive path, thereby guaranteeing stable loop behavior.

The approximate loop gain (LG) of the BGR can thus be expressed as:

$$A_{\text{Total}} = \frac{V_{\text{out, OTA}}}{V_{\text{gate}}} \approx A_{\text{neg}} - A_{\text{pos}} = -A_{\text{OTA}} \cdot g_m \cdot R_1$$

Since the OTA is phase-preserving and the current mirror stage formed by M_1 and M_2 introduces a common-source configuration, the overall loop exhibits a 180° phase shift at low frequencies.

Current-Mode Bandgap Reference Circuit

While the voltage-mode bandgap reference (BGR) topology is straightforward and can generate a nearly temperature-independent output near 1.2V, it provides few degrees of freedom for adjusting the output voltage V_{ref}.

To overcome this limitation, a slight modification to the BGR topology is introduced, as shown in Figure 9. It generates a temperature-independent current by combining a CTAT current with a PTAT current of equivalent but opposite temperature coefficient (TC). This current is then converted into a scalable, temperature-independent voltage reference through a resistor.

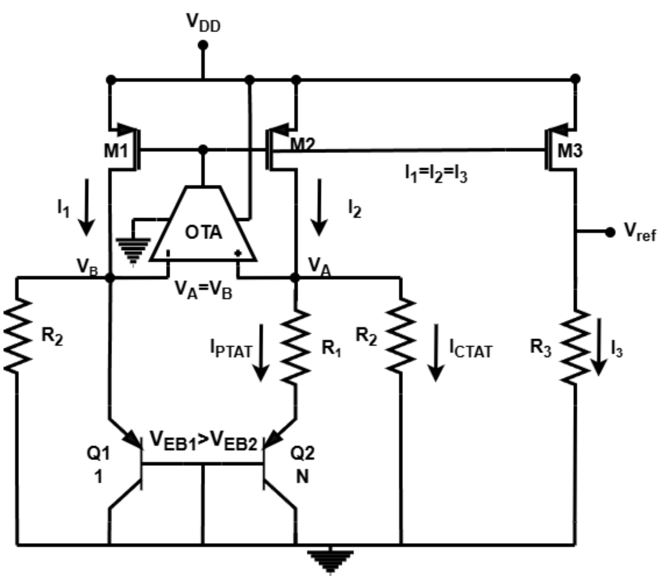

Figure 9. Current Mode Bandgap Reference Circuit Topology

In this topology, the total reference current consists of a PTAT component and a CTAT component. The PTAT current is generated through resistor R_1, similar to the previous topology. However, additional branches with resistors R_2 introduce a CTAT current due to the base-emitter voltage V_{EB}, which is a voltage with a negative temperature coefficient (CTAT). Since the high-gain OTA equalizes the voltages at nodes A and B, $V_A = V_B = V_{\text{EB1}}$, enforcing this CTAT behavior in these added branches. The combined current through the current mirror transistors, therefore, is a sum of PTAT and CTAT currents.

To achieve a temperature-independent current, R_1 is selected based on the power consumption requirements, and R_2 is then precisely sized such that the positive TC of the PTAT current and the negative TC of the CTAT current cancel each other out, such that the TC of the combined current is approximately:

$$\frac{dV_{\text{ref}}}{dT} \approx 0$$

In the third branch, the resistor R_3 is used to scale the current sum into a voltage reference:

$$V_{\text{ref}} = R_3(I_{\text{PTAT}} + I_{\text{CTAT}})$$

This enables the generation of a temperature-independent and scalable V_{ref}. Unlike the voltage-mode bandgap topology, which inherently fixes V_{ref} near 1.2 V due to its intrinsic zero-TC point, this current-mode approach introduces an additional degree of freedom due to resistor ratio, that allows V_{ref} to be tuned as needed, for example, to values lower than 1.2 V, which is necessary to meet the specifications of the DPLL system.

The derivation of the temperature-independent reference voltage is shown below:

$$I = I_1 = I_2 = I_3$$

$$I_{\text{PTAT}} = \frac{V_{\text{EB}_1} - V_{\text{EB}_2}}{R_1} = \frac{kT}{qR_1}\ln(N) \quad \text{(PTAT current)}$$

$$I_{\text{CTAT}} = \frac{V_{\text{EB}}}{R_2} \quad \text{(CTAT current)}$$

$$V_{\text{ref}} = R_3\left(I_{\text{PTAT}} + I_{\text{CTAT}}\right) = R_3\left(\frac{kT}{qR_1}\ln(N) + \frac{V_{\text{EB}}}{R_2}\right) = \underbrace{R_3\left(\frac{\ln(N)}{R_1}V_T + \frac{V_{\text{EB}}}{R_2}\right)}_{\text{Temperature independent if TCs cancel}}$$

Equation 17: V_{ref} Equation for BGR Current-Mode Topology

Since the exact temperature dependence of the resistors is difficult to characterize due to process variations and second-order effects, a precise analytical derivation of V_{ref} with respect to temperature is omitted. Instead, the simplified expression derived above can be used to estimate initial resistor values. These approximate values serve as a good starting point, and the final resistor ratios can then be fine-tuned through iterative simulations to minimize the temperature coefficient (TC) of V_{ref}.

2.4 Transistor Operation Considerations for the LDO

In analog building blocks such as LDOs, bandgap references (BGRs), and operational transconductance amplifiers (OTAs), MOSFETs typically operate in the saturation region, as it gives several critical benefits: it provides high intrinsic gain essential for OTAs, acts as a current source mostly independent of the drain-source voltage V_{DS} (a key property for current mirrors in BGRs and bias circuits), and ensures predictable, linear behavior across process variations within feedback systems such as LDOs. This relative independence from V_{DS} is especially important in LDO pass transistors, where the OTA feedback loop relies on a controlled output current to maintain precise voltage regulation despite variations in load current or supply voltage. Although in practice MOSFETs show some dependence on V_{DS} due to channel length modulation and other second-order effects, the saturation region remains the most stable and predictable operating point for analog design.

The saturation region itself can be subdivided into weak, moderate, and strong inversion sub-regions, depending on the gate-source voltage V_{GS} relative to the threshold voltage V_{TH} and the resulting charge distribution in the inversion layer. For simplicity, moderate inversion is grouped with either weak or strong inversion due to the gradual transition between these sub-regions. In general, MOSFETs are considered to be in strong inversion when $V_{GS} > V_{TH}$ and in weak inversion when $V_{GS} < V_{TH}$, given that the transistor is not cutoff. Moderate inversion typically corresponds to a gate voltage window approximately ± 100 mV around V_{TH}.

In strong inversion, the channel is pinched-off near the drain, which makes the drain current effectively independent of V_{DS} and primarily determined by device parameters and the gate overdrive voltage, as described by:

$$I_{DS} = \frac{\mu C_{ox}}{2} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$$

Equation 18: MOSFET Drain Current in Saturation — Strong Inversion with Early Effect

where μ is the carrier mobility, C_{ox} is the oxide capacitance per unit area, W and L are the transistor's width and length respectively, V_{GS} is the gate-to-source voltage, V_{TH} is the threshold voltage, V_{DS} is the drain-to-source voltage, and λ is the channel-length modulation parameter, which models the Early effect, and can be minimized by using long channels.

In contrast, in the weak inversion region, the dominant mechanism for current is no longer drift but diffusion, similar to a bipolar junction transistor, which results in an exponential dependence of the drain current on gate voltage:

$$I_{DS} = I_0 \frac{W}{L} e^{\frac{V_{GS} - V_{TH}}{n V_T}} \left(1 - e^{-\frac{V_{DS}}{V_T}}\right) \underset{\substack{\text{for } V_{DS} > 3V_T \\ \text{(weak inversion saturation)}}}{\approx} I_0 \frac{W}{L} e^{\frac{V_{GS} - V_{TH}}{n V_T}}$$

Equation 19: MOSFET Drain Current in Weak Inversion (Subthreshold)

where I_0 is a process-dependent constant, W and L are the transistor width and length, V_{GS} is the gate-to-source voltage, n is the sub-threshold slope factor, V_T is the thermal voltage, V_{DS} is the drain-to-source voltage.

For $V_{DS} > 3V_T$ (approximately 75 mV at room temperature), the exponential term $e^{-\frac{V_{DS}}{V_T}}$ becomes negligible, and the current no longer depends on V_{DS} which explains why the device operation is still considered saturation in weak inversion.

In strong inversion, the transconductance strongly depends on the gate voltage, given by:

$$g_{m,\mathrm{SI}} = \frac{\partial I_{DS}}{\partial V_{GS}} = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) = \frac{2 I_{DS}}{V_{GS} - V_{TH}}$$

On the other hand, in weak inversion, the transconductance is independent of the gate voltage, and is given by:

$$g_{m,\mathrm{WI}} = \frac{\partial I_{DS}}{\partial V_{GS}} = \frac{I_{DS}}{n V_T}$$

In weak inversion, the transconductance-to-current efficiency $\frac{g_m}{I_{DS}}$ is inherently higher than in strong inversion and is largely independent of device geometry (W/L). Furthermore, the small bias current due to the weak inversion layer results in very high output resistance $r_o = \frac{V_A}{I_{DS}}$, where V_A is the Early voltage. Together, these give a high intrinsic gain $A_v = g_m r_o$ with excellent transconductance efficiency. This allows transistors operating in weak inversion to achieve higher gain at lower bias currents, which makes them ideal for low-power designs such as the LDO in a DPLL chip.

However, the high r_o from the small bias current imposes a fundamental trade-off: increasing bandwidth requires reducing r_o, which directly reduces gain. As a result, circuits operating in weak inversion naturally exhibit lower bandwidth, typically below 1 kHz, which limits circuit speed. Thus, while weak inversion enables high gain and low power, it inherently sacrifices bandwidth.

Additionally, weak inversion devices are more sensitive to process, voltage, and temperature (PVT) variations, which may affect reliability. Therefore, as a trade-off, it is often best to operate near the boundary between weak and strong inversion ($V_{GS} \approx V_{TH}$) to maintain high gain while improving robustness across corners.

Despite these limitations, weak inversion operation is well suited for DPLL applications, where control loop bandwidths are typically around 1 kHz or lower. Although flicker and thermal noise are higher in weak inversion, the low bandwidth and inherent filtering of the LDO sufficiently suppress noise and ensure clean supply voltages for sensitive sub-blocks in the DPLL.

3 Implementation

3.1 Project Implementation Steps

The LDO was implemented following a step by step process, as outlined below:

1. **Initial BGR Topology Selection:** We began by selecting a Bandgap Reference (BGR) topology that generates approximately 1.2 V. To verify functionality quickly, we first used an ideal operational transconductance amplifier (OTA) in the feedback loop and confirmed that the BGR produced the expected output.
2. **Determining OTA Gain Requirements:** After validating basic operation, we investigated the minimum OTA gain required to maintain acceptable performance. Simulations showed that a gain of at least 60 dB was sufficient, but improvements were still noticeable up to 80 dB, beyond which the benefits diminished compared to design complexity.
3. **OTA Topology and Design:** A two-stage Miller-compensated OTA was chosen for implementation, with the goal of achieving wide swing and a gain of at least 60 dB. The compensation capacitor was sized based on estimated gate capacitance from initial pass transistor size in the LDO.
4. **Integration into BGR:** After verifying OTA stability and performance, we replaced the ideal OTA in the BGR with the designed OTA. The full feedback loop remained stable and operated as expected.
5. **Scaling for Low V_{REF}:** To achieve a reference voltage below 0.9 V, we replaced the original BGR topology with a current-mode variant. This allowed V_{REF} scaling and involved adding a current-scaling branch and adjusting resistor values. The updated BGR was verified to function correctly.
6. **LDO Implementation and PMOS Sizing:** We implemented the LDO using the verified BGR and OTA. Initially, it was tested under 10ṁA load current. After specifications were updated to drive a load current of 55.5 mA, we increased the width the PMOS pass transistor to handle the increased current. The increased gate capacitance required minor tuning of the compensation capacitor, slightly lowering the OTA bandwidth to maintain good phase margin.
7. **Output Capacitance and Stability Testing:** The LDO was tested over a range of output capacitance values and series resistances. It remained stable and performed well under varying load and filter configurations.
8. **Layout Process:**
   - **OTA Layout:** This was the most time-consuming step. Although the OTA is not high-speed (its bandwidth is only around 1 kHz), it functions as a precision error amplifier in the feedback loop and is therefore highly sensitive to mismatch, asymmetry, and layout-induced offsets. As such, we had to learn and apply many analog layout techniques that go beyond ideal simulation, focusing on symmetry, common-centroid placement, and current mirror matching to preserve DC gain and minimize offset. Multiple iterations were required to meet these goals, satisfy design rule checks, and get an approval from the layout advisor.
   - **BGR Layout:** After finalizing the OTA, we integrated it into the BGR layout. This stage was significantly simpler, as the BGR block is smaller and primarily operates in the DC domain. As a result, it is far less sensitive to layout parasitics and mismatch compared to the OTA, but best practices like common centroid for the BJTs were key to minimize PVT variations.
   - **LDO Layout:** Special attention was given to minimizing IR drop by carefully routing high-current paths. Additionally, to improve symmetry and create a clean rectangular block for easier integration at the top level, the OTA for the LDO was slightly resized and compacted. Once the parasitic resistance at the pass transistor current path was reduced to a satisfactory value, and IR drop effect was essentially eliminated, the full layout was approved by the project advisors.
9. **Enable Signal Integration:** In July 2025, a design update introduced two enable signals to enable control over the LDO output. These signals allow the output to be selectively configured to 0 V (disabled), 0.9 V (regulated), or 1.8 V (bypassing the LDO). This necessitated making minor changes to the LDO schematic and layout.
10. **Final Validation:** The final design was validated and found to be working reliably across a myriad of process, voltage, and temperature (PVT) corners.

3.2 BGR

Attempting a Design with a Current Mirror:

Initially, to enforce equal voltages at points A and B, the topology seen in Figure 10 using a current mirror was attempted. In this design, the gates of the MOSFETs were shorted together, so if the transistors are sized identically and carry the same current, their source voltages should ideally be equal, thereby forcing both nodes to the same voltage. However, in practice, even with transistors having very large channel lengths, variations in the drain voltage affect the source voltage, causing differences in the drain-to-source voltage (V_{DS}) between the devices. This mismatch led to poor voltage matching and resulted in low PSRR of about −17 dB.

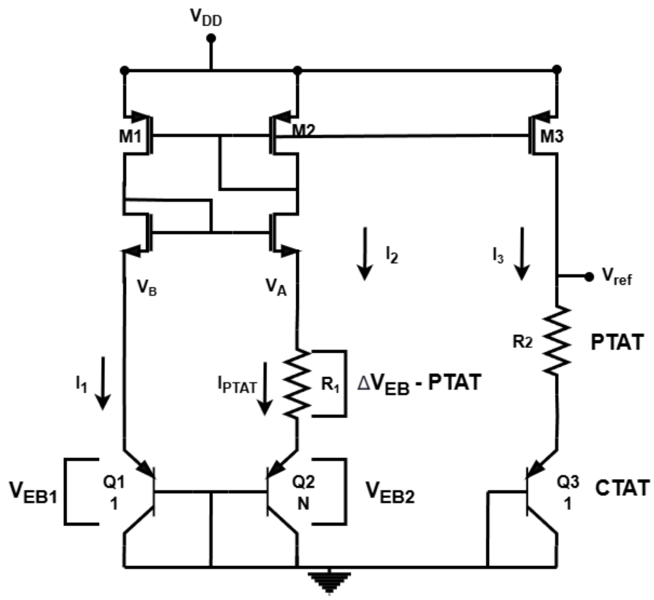

Figure 10. Current-Mirror Based BGR Topology

Attempting an Design with an OTA:

The following design process refers to the circuit in Figure 9. To overcome these limitations, the design transitioned to using an operational transconductance amplifier (OTA). While the OTA is significantly larger and requires a greater layout area, it offers improved performance. An ideal OTA model from the library with configurable gain was first used to evaluate the relationship between OTA gain, PSRR, and temperature coefficient (TC) at the reference voltage. At a gain of 60 dB, the circuit achieved a reasonable PSRR of −43 dB. Based on these results, it was decided to design an OTA with at least 60 dB gain. Gains in the range of 70–80 dB provided further improvements in PSRR and TC, while gains beyond this range exhibited diminishing returns in terms of circuit complexity.

Selecting the Base-Emitter Area Factor:

The ratio between the base-emitter areas of the BJTs was chosen to be 1:8. This ratio enables a common-centroid layout, where the larger transistors' base-emitters surround that of the smaller transistor. This configuration helps to minimize the impact of process gradients and temperature variations, since any changes affecting the central transistor are similarly experienced by the surrounding devices, which improves matching. Choosing $n = 8$ is a good balance between matching accuracy and layout area, as BJTs consume significant silicon area. Larger ratios, such as 15 or 24, would double or triple the layout size without providing much benefit in matching performance.

Resistor Type Selection for Temperature Stability:

Among the various resistor options available in the TSMC 28 nm technology, we selected the Rupoly resistor for our Bandgap Reference (BGR) design. This is an unsilicided polysilicon resistor, widely regarded as the standard precision resistor in most CMOS processes, and commonly used in BGR implementations. Rupoly resistors are especially well suited for this circuit due to their low temperature coefficient (TC). Their resistance remains more stable with temperature compared to diffusion-based or n-well resistors, which typically exhibit a higher TC. Selecting a resistor with a low TC is critical for achieving a stable output voltage V_{ref}. A resistor with a high TC would cause the current to decrease at high temperatures, thereby degrading the linearity of the PTAT current and negatively affecting overall temperature compensation.

Current Mirror and V_{ref} Considerations:

The current mirrors in the Bandgap Reference (BGR) replicate the current generated in the central branch to the side branches. Accurate current replication is critical to improving both the PSRR and the temperature coefficient (TC) of the reference voltage.

To enhance current mirror performance, the transistor channel length L is increased, as the Early voltage V_A scales approximately with L. A larger V_A increases the output resistance r_{out} of the current mirror transistors, which makes the current source more ideal and less sensitive to variations in the drain-to-source voltage (V_{DS}). This improved output impedance leads to a more constant current with supply voltage changes, thus PSRR is improved significantly. In this design, $L = 1\,\mu$m, which is six times the minimal size, was chosen as a practical compromise to reduce Early effect without excessive area. The transistor width was sized to ensure operation in strong inversion saturation, which offers better accuracy and reliability compared to weak inversion.

Furthermore, minimizing the mismatch in V_{DS} among the current mirror transistors is essential to reduce channel length modulation errors. Since the output current depends on V_{DS}:

$$I_{DS,\text{out}} = I_{\text{ideal}}\left(1 + \frac{V_{DS}}{V_A}\right),$$

keeping the drain voltages of all mirror transistors equal reduces this mismatch and improves matching. This is achieved by setting the reference voltage $V_{\text{ref}} \approx V_A = V_B$.

With this approach, the PSRR of the reference voltage improves significantly to -60 dB even with a simple current mirror topology. This avoids the complexity and larger layout area of cascoded or Wilson mirrors, which offer higher output resistance but are more challenging to match in layout.

The improved current replication also minimzes the TC of the reference voltage, while using only first-order compensation. Although V_{ref} is constrained to approximately 0.7 V (the typical base-emitter voltage), this fits well with the target output voltage of 0.9 V in the LDO design.

Operating Current Selection:

The operating current in each branch of the BGR was chosen to be approximately $5\,\mu$A. This current balances several design constraints: it is low enough to ensure high PSRR and low power consumption, yet sufficiently high to guarantee reliable startup and maintain thermal noise at acceptable levels. Lower current improves PSRR by increasing the output impedance of the current mirror transistors, since $r_{\text{ds}} = \frac{V_A}{I_{\text{DS}}}$, where V_A is the Early voltage. At the same time, very large resistors increase area, degrade matching due to temperature and process gradients, and lower operating currents, making the circuit more vulnerable to leakage.

Precision Resistor Sizing for Temperature Compensation:

After selecting an operating current of approximately $5\,\mu$A, the resistor R_1 was chosen to generate a PTAT current using the relation:

$$R_1 = \frac{V_T \ln(N)}{I_{\text{PTAT}}} = \frac{0.026 \times \ln(8)}{5\,\mu\text{A}} \approx 10.98\,k\Omega.$$

The resistor R_2 was then tuned to produce a CTAT current with an opposite temperature coefficient, such that the sum of the PTAT and CTAT currents is effectively temperature-independent. This yielded $R_2 = 121.6\,k\Omega$.

Subsequently, the reference voltage V_{ref} was scaled to approximately 0.72 V by adjusting R_3 according to:

$$R_3 = \frac{V_{\text{ref}}}{\frac{\ln(N) V_T}{R_1} + \frac{V_{BE}}{R_2}} = \frac{0.72}{\frac{\ln(8)\times 0.026}{10.98\,k\Omega} + \frac{0.715}{121.6\,k\Omega}} \approx 66.63\,k\Omega.$$

Multiple simulation iterations confirmed this configuration yields an optimal temperature coefficient, which proves the effectiveness of both the OTA regulation and current mirroring, as these calculations assume their proper operation.

Final Transistor and Resistor Dimensions for BGR Layout:

As a result of equalizing the V_{DS} voltages across all transistors in the current mirror, achieved by scaling the reference voltage from approximately 1.2 V down to 0.72 V, the need for large channel lengths was reduced. This allowed the transistor lengths to be decreased from $1\,\mu$m to 250 nm to save layout area after adding dummy devices for matching. The effect of this on line regulation and PSRR was negligible.

Table 2 displays the final sizes of all the components in the BGR:

Component	Type	Width (μm)	Length (μm)	Purpose
M0, M1, M2	PMOS (pch_18_mac)	1.8	0.25	Current mirror transistors
M3	PMOS (pch_18_mac)	0.9	0.25	Dummy for current mirror
M4	PMOS (pch_18_mac)	3.6	0.25	Dummy for current mirror
C1	PNP5 BJT	–	–	BGR diode (area: 2.5×10^{-11} m^2)
C2	PNP5 BJT	–	–	BGR diode (area: 2.0×10^{-10} m^2)
R3, R4	Resistor (rupolym_m)	0.43	90	CTAT resistors ($\sim$ 121.6 $k\Omega$)
R6	Resistor (rupolym_m)	0.4	46	Output scaling resistor ($\sim$ 66.63 $k\Omega$)
R7	Resistor (rupolym_m)	0.53	10	PTAT resistor ($\sim$ 10.98 $k\Omega$)

Table 2
Component Sizing Summary for BGR Circuit

3.3 OTA

Establishing Initial Design Specifications:

The design process refers to the OTA in Figure 3. The initial design of the OTA was performed early in the year, at a time when the precise circuit specifications were still unclear. To develop a baseline design with reasonable performance, we used a textbook methodology for LDO design as described in [15], using preliminary specifications as a starting point for future refinement. The initial design specifications included a DC gain of at least 60 dB, a gain-bandwidth product (GBW) in the range of 10–50 MHz, and a phase margin of approximately 60° to ensure loop stability. Since the OTA is used twice within the LDO, minimizing its power consumption was a key constraint, with a target of less than 200 μW.

Initially, the maximum load current was estimated at 8 mA, which dictated the sizing of the PMOS pass transistor. To estimate its parasitic gate capacitance, we performed a transient simulation (Figure 11) using a 0–1.8 V pulse source in series with a 500 kΩ resistor connected to the transistor's gate. The time constant τ was extracted at the point where the gate voltage reached 63.2% of its final value ($V = 0.632V_{\text{final}}$). From the relation:

$$\tau = RC \quad \Rightarrow \quad C = \frac{\tau}{R} = \frac{31.921\,\text{ns}}{500\,\text{k}\Omega} \approx 64\,\text{fF},$$

An estimate for the gate capacitance of a PMOS with dimensions $W = 60\,\mu$m, $L = 150$ nm was obtained, which was the pass transistor size. This helped determining the required size for the compensation capacitor.

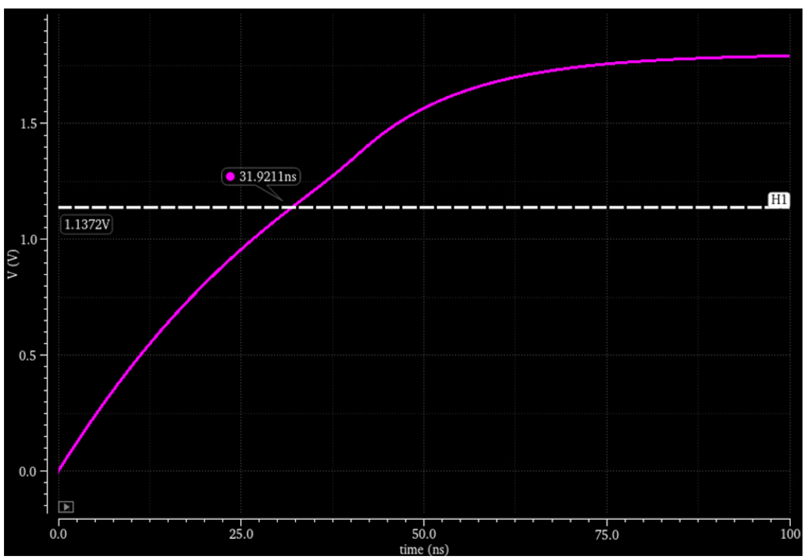

Figure 11. Finding the Parasitic Capacitance at the Gate of the Pass Transistor Through Simulation

To provide sufficient headroom, the load capacitor C_L was initially set to 100 fF. This value was sufficient not only for the OTA but also for the much smaller BGR current mirror transistors. However, mid-year, the maximum load current specification was increased to 55 mA, which required a significantly larger pass transistor. This increased the gate capacitance by roughly a factor of 10, to approximately 640 fF. Consequently, C_L was increased to 1 pF to maintain the desired frequency response. To preserve loop stability and maintain a good phase margin, the compensation capacitor was also increased, which in turn reduced the GBW by about half, but still within the acceptable design range.

The OTA's differential pair uses NMOS input transistors, and the expected common-mode input voltage was expected to be around $V_{\text{ref}} = 0.72V$. While the input voltage is not expected to fall significantly below this level, it was important to ensure that all transistors remain in saturation, including in weak inversion. Therefore, the input common-mode range (ICMR−) was carefully minimized. The upper limit of the ICMR is not a concern, as it can extend up to 1.6 V without issue.

Slew rate (SR), which is the maximum rate of change of the OTA output, was not considered, since the OTA is used in low-speed analog blocks such as the BGR and LDO, where fast transients are uncommon.

Table 3 summarizes the inital OTA performance goals and the achieved results:

Specification	Initial Goal	Initial Results	Final Results - Pre Layout
DC Gain (dB)	>60	69.8	77.6
GBW (MHz)	5–30	23	6.4
ICMR (-)	Minimize	0.67 V	0.69
V_{out} Dynamic Range	Rail-to-Rail	0.1 V–1.7 V	0.1-1.7
Phase Margin (deg)	>60	72	76.5
C_{Load} (pF)	0.1 → 1	0.1	1
C_C (pF)	>0.4	1	1.5
Power (μW)	<200	120	37

Table 3
OTA Design Specifications

OTA First Stage Design

The OTA consists of two stages. We begin by designing the first stage, which includes the biasing branch, tail transistor, differential input pair, and active load (current mirror) transistors. The second stage is initially modeled as a 1 pF capacitive load. The design targets for the first stage is to achieve approximately 40 dB gain, GBW of $30MHz$ and bias current of around $20\mu A$.

The transistor sizing for proper differential operation is:

$$M_1 = M_2 \quad \text{(input NMOS)}, \quad M_3 = M_4 \quad \text{(PMOS current mirror)}, \quad M_5 = M_8 \quad \text{(bias and tail NMOS)}$$

Active Load Transistor Sizing (M_3, M_4): The sizes of the PMOS current mirror transistors M_3, M_4 are determined based on the positive input common-mode range (ICMR^+). To keep M_3, M_4 in strong inversion and saturation, we require:

$$V_{\text{DS}} > V_{\text{GS}} - V_{\text{TH}}$$

Assuming the maximum input voltage is 1.2 V (arbitrary), the drain voltage of M_1 must be:

$$V_{D1} > V_{in} - V_{TH} = 1.2\,\text{V} - 0.5\,\text{V} = 0.7\,\text{V}$$

To provide margin, we aim for $V_{D1} \approx 1\,\text{V}$.

Simulating a diode-connected PMOS with $W = 10\,\mu\text{m}, L = 1\,\mu\text{m}$, and bias current $I = 10\,\mu\text{A}$, we estimate:

$$\mu_p C_{ox} \approx 100\,\mu\text{A/V}^2, \quad V_{TH} \approx 0.5\,\text{V}$$

For a current of $I_3 = 10\,\mu\text{A}$ and $V_{GS} \approx 0.8\,\text{V}$, the size ratio is:

$$\left(\frac{W}{L}\right)_{3,4} = \frac{2I_3}{\mu_p C_{ox}(V_{GS} - V_{th})^2} = \frac{20\,\mu}{100\,\mu \cdot (0.3)^2} \approx 2.2$$

We therefore choose:

$$\left(\frac{W}{L}\right)_{3,4} = \frac{6\,\mu}{2\,\mu} = 3$$

The relatively long channel length helps increase the output resistance and thus the stage gain.

Input Transistor Sizing (M_1, M_2): The sizing of the NMOS input pair is based on the GBW requirement. Assuming a single-pole first stage and DC gain of:

$$A_{v1} = g_{m1} \cdot (r_{o2} \parallel r_{o4})$$

The pole is located at:

$$\omega_p = \frac{1}{C_L(r_{o2} \parallel r_{o4})}$$

Thus, the gain-bandwidth product (GBW) becomes:

$$\text{GBW} = \frac{g_{m1}}{2\pi C_L}$$

For a target GBW of 30 MHz and $C_L = 1\,\text{pF}$:

$$g_{m1} = 2\pi \cdot 1\,\text{pF} \cdot 30\,\text{MHz} = 188.5\,\mu\text{S}$$

Extracting from simulation $\mu_n C_{ox} \approx 1\,\mathrm{mA/V^2}$, and $I_{D1} = 10\,\mu\mathrm{A}$, we solve:

$$\left(\frac{W}{L}\right)_{1,2} = \frac{g_{m1}^2}{2 I_{D1} \mu_n C_{ox}} = \frac{(188.5\,\mu)^2}{2 \cdot 10\,\mu \cdot 1\,\mathrm{m}} \approx 1.77$$

We round this to:

$$\left(\frac{W}{L}\right)_{1,2} = \frac{2.8\,\mu}{1.6\,\mu} \approx 1.75$$

This sizing ensures the OTA meets its bandwidth and stability targets with sufficient gain.

Tail Current Source and Biasing Branch Design (M_5, M_8): The sizing of transistors M_5 and M_8 is determined based on the negative input common-mode range (ICMR⁻) requirement. Since the input voltage is expected to be around $0.72V$, we must ensure that M_5 remains in saturation even at low input voltages. For proper operation of M_1, the condition is:

$$V_{\mathrm{in,min}} > V_{\mathrm{DSAT}_5} + V_{\mathrm{GS1}}.$$

To estimate V_{GS1}, we use the quadratic current relation in saturation:

$$(V_{\mathrm{GS1}} - V_{\mathrm{th}})^2 = \frac{2I_1}{\mu_n C_{\mathrm{ox}} \frac{W}{L}} = \frac{20\,\mu\mathrm{A}}{1\,\mathrm{mA/V^2} \times 1.75} \approx 0.0114,$$

$$\Rightarrow V_{\mathrm{GS1}} \approx V_{\mathrm{th}} + \sqrt{0.0114} \approx 0.5\,\mathrm{V} + 0.107 \approx 0.6\,\mathrm{V}.$$

If we target $V_{\mathrm{in,min}} = 0.7V$, then $V_{\mathrm{DSAT}_5} \le 0.093\,\mathrm{V}$, which is challenging at the original tail current of $10\mu A$. Hence we reduce the tail current slightly to $7.5\mu A$ and increase the size of M_5 accordingly.

Using the saturation current equation:

$$\left(\frac{W}{L}\right)_{5,8} = \frac{2I_5}{\mu_n C_{\mathrm{ox}} (V_{\mathrm{GS}} - V_{\mathrm{th}})^2},$$

To get an overdrive voltage $50mV$, we compute:

$$\left(\frac{W}{L}\right)_{5,8} = \frac{2 \cdot 15\,\mu\mathrm{A}}{1\,\mathrm{mA/V^2} \cdot (50\,\mathrm{mV})^2} = \frac{30\,\mu}{2.5\,\mu} = 12.$$

Using simulations, the final value chosen was:

$$\left(\frac{W}{L}\right)_{5,8} = \frac{8.4\,\mu\mathrm{m}}{0.48\,\mu\mathrm{m}} \approx 17.5.$$

Resistor R_1 Sizing for Bias Current: To generate the bias current $I_8 \approx 15\,\mu\mathrm{A}$, we select R_1 such that the gate voltage of M_8 provides the correct V_{GS}. Assuming the drain of M_8 is approximately $V_{\mathrm{DD}} = 1.3\,\mathrm{V}$ and its source is connected through R_1 to ground, we estimate:

$$V_{GS8} = V_D - V_S = V_{DD} - I_8 R_1 \approx 0.55V$$

Substituting into the saturation current equation:

$$I_8 = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (1.25 - I_8 R_1)^2 = 8.75m(1.25 - 15\mu \times R_1)^2 = 15\mu$$

Hence, we get a quadrastic equation on R_1, which gives us an initial estimation of $R_1 = 80k\Omega$. In simulation, a resistor of $70k\Omega$ gave us a current of around $15\mu A$ in the tail transistor.

After finalizing the transistor sizing, simulation confirmed that all devices remain in saturation across the expected input range. The input common-mode range (ICMR⁻) was verified to be around $0.67V$, while the upper bound exceeded $1V$, which is sufficient for the LDO. The first stage gain was measured to be approximately $36dB$, which is slightly below the design target of around $40dB$

OTA Second Stage Design

We proceeded to the second stage, which is a common-source (CS) amplifier designed to boost the overall gain. To achieve a large output swing and improve common-mode rejection, this stage is implemented using a PMOS transistor. The gain of the second stage is given by:

$$A_2 = g_{m6}(r_{ds6} \parallel r_{ds7})$$

The second stage introduces an additional pole located at:

$$p_2 = \frac{g_{m6}}{C_L}$$

where g_{m6} is the transconductance of the second-stage transistor, and C_L is the load capacitance, which at first was estimated to be $0.1pF$.

As a result of adding this stage, the circuit is no longer a single-pole system, and compensation is required to control the pole locations. Thus, we introduce a compensation capacitor, C_C. Its addition also introduces a right-half plane (RHP) zero at:

$$z_{\text{RHP}} = \frac{g_{m6}}{C_C}$$

As discussed in the theory, there's Miller effect on the compensating capacitor, which shifts the first-stage pole to a lower frequency:

$$p_1 = \frac{1}{(r_{ds1} \parallel r_{ds3})A_2 C_C}$$

This shift determines the gain-bandwidth product (GBW), which is now:

$$\text{GBW} = \text{DC gain} \times p_1 = \frac{g_{m1}}{C_C}$$

Frequency Compensation and Phase Margin Optimization: Due to the added left-half plane (LHP) pole and the RHP zero, the phase margin is expected to decrease. To improve it, both the non-dominant pole and the RHP zero must be pushed to higher frequencies. The zero is placed at a frequency one decade beyond the GBW:

$$z = 10 \times \text{GBW}$$

This implies:

$$\frac{g_{m6}}{C_C} = \frac{10 g_{m1}}{C_C} \quad \Rightarrow \quad g_{m6} = 10 g_{m1}$$

To determine where to place the non-dominant pole, we must consider the desired phase margin. The phase of the transfer function at the gain-bandwidth (GBW) frequency, which approximately corresponds to the 0 dB frequency in a well compensated design, is given by:

$$\angle \frac{v_{\text{out}}}{v_{\text{in}}}(\text{GBW}) = -\tan^{-1}\left(\frac{\text{GBW}}{z}\right) - \tan^{-1}\left(\underbrace{\frac{\text{GBW}}{p_1}}_{A_{\text{DC}}}\right) - \tan^{-1}\left(\frac{\text{GBW}}{p_2}\right)$$

Assuming the zero is placed at $z = 10 \times \text{GBW}$ and A_{DC} is very large ($\tan^{-1}(A_{\text{DC}}) \approx 90^\circ$), we can approximate:

$$\angle \frac{v_{\text{out}}}{v_{\text{in}}}(\text{GBW}) = -\tan^{-1}\left(\frac{1}{10}\right) - 90^\circ - \tan^{-1}\left(\frac{\text{GBW}}{p_2}\right)$$

$$= -5.71^\circ - 90^\circ - \tan^{-1}\left(\frac{\text{GBW}}{p_2}\right)$$

Since phase margin (PM) is defined as the difference between 180° and the phase at the unity-gain frequency (GBW), we have:

$$\text{PM} = 180^\circ - 5.71^\circ - 90^\circ - \tan^{-1}\left(\frac{\text{GBW}}{p_2}\right)$$

To achieve a phase margin of at least 60°, the following inequality must hold:

$$\tan^{-1}\left(\frac{\text{GBW}}{p_2}\right) \geq 180^\circ - 60^\circ - 90^\circ - 5.71^\circ = 24.29^\circ$$

Solving for p_2, we get:

$$p_2 \geq \frac{\text{GBW}}{\tan(24.29^\circ)} \approx 2.2 \times \text{GBW}$$

Since $p_2 = \frac{g_{m6}}{C_L}$, we can write:

$$\frac{g_{m6}}{C_L} \geq 2.2 \times \frac{g_{m1}}{C_C}$$

The condition of $g_{m6} = 10g_{m1}$ for the zero placement significantly raises the current consumption in the second stage, which is not power efficient. Therefore, we choose to add a series resistor to the compensation capacitor. This can shift the zero to the left-half plane (LHP). An LHP zero contributes positive phase shift, which allows the second (non-dominant) pole to be placed closer to the gain-bandwidth product (GBW) without degrading stablity. As a result, a value of $g_{m6} = 2$ to $3\,g_{m1}$ is sufficient. Therefore, we get:

$$\frac{3}{C_L} \geq \frac{2.2}{C_C} \quad \Rightarrow \quad C_C \geq 0.74\,C_L = 74\,\text{fF}$$

While this gives a phase margin close to 60°, we prefer a higher margin (70°) for increased headroom, especially considering post-layout parasitic capacitances may increase the load capacitance. To achieve a phase margin of 70°, we need:

$$\tan^{-1}\left(\frac{\text{GBW}}{p_2}\right) = 180^\circ - 70^\circ - 90^\circ - 5.71^\circ = 14.29^\circ$$

$$p_2 = \frac{\text{GBW}}{\tan(14.29^\circ)} \approx 3.93 \times \text{GBW}$$

This translates to:

$$C_C \geq \frac{g_{m1}}{g_{m6}} \cdot \frac{C_L}{3.93} \Rightarrow C_C \geq 1.3\,C_L = 130\,\text{fF}$$

However, increasing C_C reduces the GBW, which highlights the trade-off between bandwidth and phase margin. Despite the reduced GBW, we choose a higher phase margin (75°) to ensure the circuit maintains stability under various loading conditions.

Hence, we select:

$$C_C = 300\,\text{fF}$$

Sizing the CS Stage Transistor M6: We now proceed to size the transistor M_6, which is the common-source (CS) gain stage. Since its gate-source voltage V_{GS} is the same as that of transistors M_3 and M_4, we can use the following approximate relation:

$$\frac{I_6}{I_4} = \frac{(W/L)_6}{(W/L)_4} = \frac{g_{m6}}{g_{m4}}$$

From earlier design steps, we determined that with the compensation capacitor and the nulling resistor, the required transconductance is approximately:

$$g_{m6} \approx 3\,g_{m1} \approx 300\,\mu\text{S}$$

Given that:

$$g_{m4} = \sqrt{2I_4\mu_p C_{\text{ox}}\left(\frac{W}{L}\right)_4} = 67\,\mu\text{S}$$

We compute the required aspect ratio:

$$\left(\frac{W}{L}\right)_6 = \frac{g_{m6}}{g_{m4}} \cdot \left(\frac{W}{L}\right)_4 = \frac{300\,\mu\text{S}}{67\,\mu\text{S}} \times 3 \approx 13.4$$

In practice, the following aspect ratio gave us the best simulation results:

$$\left(\frac{W}{L}\right)_6 = \frac{6\,\mu\text{m}}{0.48\,\mu\text{m}} = 12.5$$

Sizing the CS Stage Load Transistor M7: Next, we size the load transistor M_7, which forms part of the current mirror along with M_5 and M_8. Its sizing follows a similar principle:

$$\left(\frac{W}{L}\right)_7 = \frac{I_7}{I_5} \cdot \left(\frac{W}{L}\right)_5 = \frac{4I_4}{I_5} \cdot \left(\frac{W}{L}\right)_5$$

Substituting known values:

$$\left(\frac{W}{L}\right)_7 = \frac{32\,\mu\text{A}}{15\,\mu\text{A}} \times \frac{8.4\,\mu\text{m}}{0.48\,\mu\text{m}} \approx 37$$

However, due to the short channel length of 0.48 μm, the second-stage gain was too low. To improve gain, we increased the channel length of M_7 to 13 μm. This also reduced the threshold voltage V_{th} of the device from 500 mV to 450 mV, which allowed us to reduce the required width. The final sizing that gave the best performance in simulation was:

$$\left(\frac{W}{L}\right)_7 = \frac{135\,\mu\text{m}}{13\,\mu\text{m}} \approx 10.4$$

After extensive simulation and tuning, the optimal values for the compensation capacitor and nulling resistor were found to be $C_C = 0.8\,\text{pF}$ and $R_z = 2.9\,\text{k}\Omega$, respectively. This configuration achieved a phase margin (PM) of 72°. The resulting open-loop gain was 69.8 dB, and the remaining performance metrics are summarized in Table 3. The DC operating point of this initial design is displayed in Figure 12.

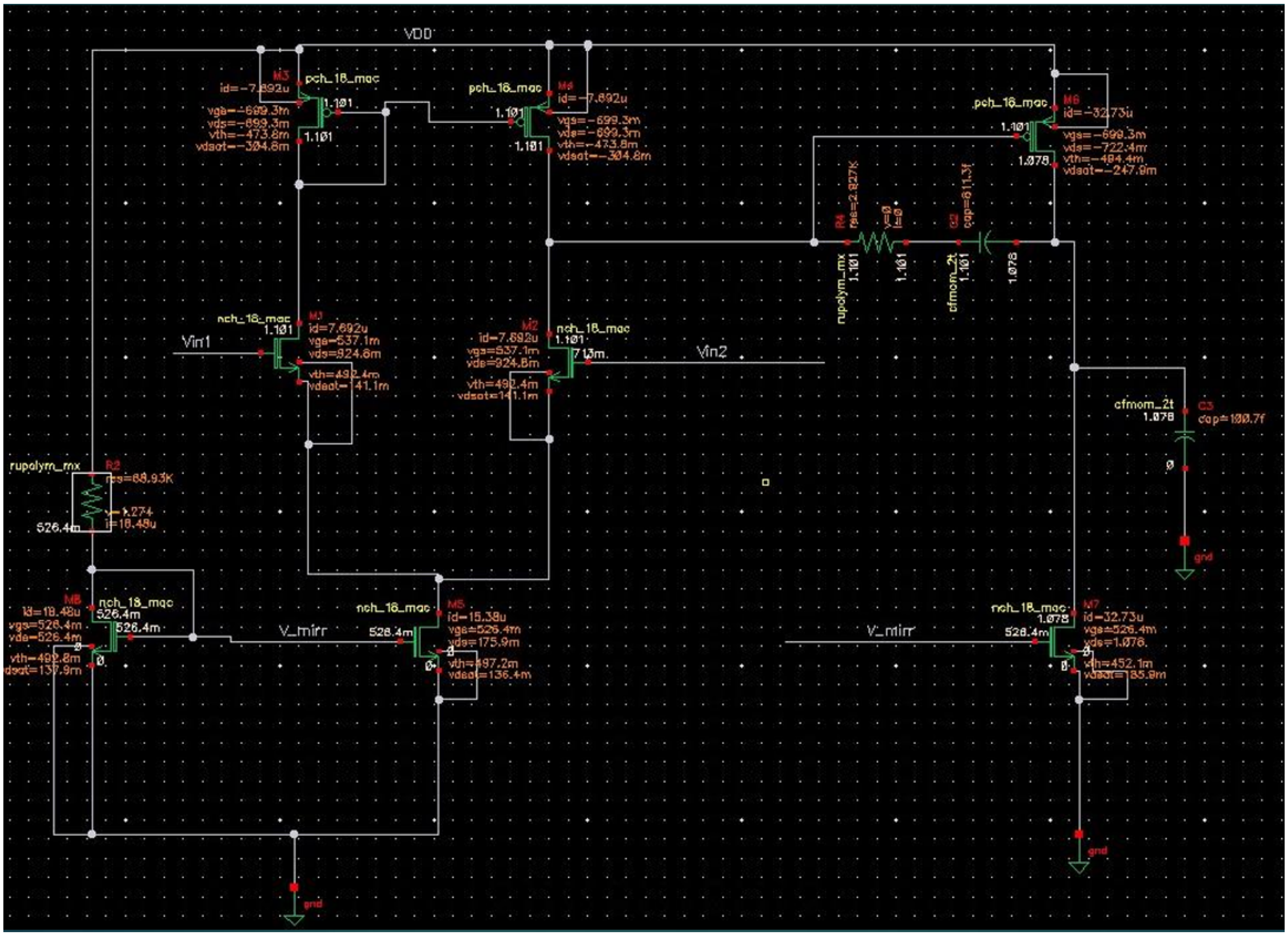

Figure 12. Initial OTA DC Simulation

Load Current Specification Update and OTA Re-Optimization

Later in the year, the load current specification for the LDO was increased to 55.5 mA. There, the pass transistor's size had to be increased, which in turn required a larger load capacitor, rising from 0.1 pF to approximately 0.8 pF. As a result, to maintain a phase margin (PM) of around 70°, the compensation capacitor was increased to 1.5 pF, and the series resistor was adjusted to 8.73 kΩ. These changes reduced the gain-bandwidth product (GBW) from 23 MHz to approximately 15 MHz.

During post-layout optimization, additional dummy devices were added to improve matching and layout symmetry. This significantly increased the OTA's area. To mitigate this, transistor M_7 was scaled down by a factor of ten while maintaining the same aspect ratio. Furthermore, the bias resistor was increased to 200 kΩ to reduce the bias current, resulting in lower branch currents throughout the circuit.

The input transistors M_1 and M_2 were re-biased to operate in the weak inversion (subthreshold) region. This mode offers higher transconductance efficiency (g_m/I_{DS}), leading to increased gain for lower power consumption. This is particularly beneficial for LDO regulation accuracy and voltage matching in the bandgap reference (BGR) circuit. Since both circuits are low-speed applications, the reduced bandwidth from the weak inversion operation is not a limitation.

To obtain the lower $V_{GS} - V_{th}$ required in weak inversion, the width of M_1 and M_2 was increased from 2.8 μm to 12 μm. Because the drain current in weak inversion is small, the output resistance increases, leading to a substantial gain improvement in the first stage—from 36 dB to 43 dB. As a result, the overall open-loop gain increased to 77.55 dB.

However, the reduced g_{m1} due to the lower bias current also led to a decrease in GBW, from 15 MHz to approximately 6 MHz, but this value remains within the design specifications. The remaining transistors retained their original sizing.

The complete sizing table for the OTA used in the BGR circuit is shown in Table 4.

Component	Type	Width (μm)	Length (μm)	Multiplier	Purpose
R2	Resistor (rupolym_m)	0.4	73.92	–	Bias resistor ($\sim$ 107.06 $k\Omega$)
R3	Resistor (rupolym_m)	0.4	69	–	Bias resistor ($\sim$ 99.94 $k\Omega$)
R4	Resistor (rupolym_m)	0.4	8	–	Compensation resistor ($\sim$ 8.73 $k\Omega$)
M5	NMOS (nch_18_mac)	4	1.4	2	Tail transistor of diff pair
M8	NMOS (nch_18_mac)	4	1.4	2	Bias transistor in current mirror
M7	NMOS (nch_18_mac)	4	1.4	4	2nd stage active load transistor
M1, M2	NMOS (nch_18_mac)	6	1.8	1	Differential input pair
M3, M4	PMOS (pch_18_mac)	6	2.0	1	Active load of diff pair (current mirror)
M6	PMOS (pch_18_mac)	8	0.58	1	Second stage common-source amplifier
C1	Capacitor (cfmom_2t)	–	–	–	Compensation capacitor (1 pF)
C2	Capacitor (cfmom_2t)	–	–	–	Compensation capacitor (132.58 fF)
C5	Capacitor (cfmom_2t)	–	–	–	Compensation capacitor (372.14 fF)
M10	NMOS (nch_18_mac)	6	1.4	1	Dummy transistor (lower current mirror)
M14	NMOS (nch_18_mac)	1	1.4	8	Dummy transistor (lower current mirror)
M15	NMOS (nch_18_mac)	6	1.4	1	Dummy transistor (lower current mirror)
M17	NMOS (nch_18_mac)	1	1.4	8	Dummy transistor (lower current mirror)
M11	PMOS (pch_18_mac)	8	2.0	2	Dummy transistor (upper current mirror)
M12	PMOS (pch_18_mac)	1	2.0	4	Dummy transistor (upper current mirror)
M9	NMOS (nch_18_mac)	1	1.8	4	Dummy transistor (common-centroid for diff pair)
M0	NMOS (nch_18_mac)	14	1.8	2	Dummy transistor (common-centroid for diff pair)

Table 4
Component Sizing Summary for OTA in BGR

The *Multiplier* column indicates the number of parallel unit devices used for each component. This allows the effective width to be scaled while maintaining small individual device dimensions, which improves matching in the layout. The splitting of the compensation capacitors and the bias resistor is done purely for layout purposes.

LDO OTA Adaptation for Layout and Bias Sharing

The performance of the OTA used in the LDO, specifically in terms of gain, bandwidth, and phase margin, is nearly identical to that of the OTA used in the BGR. The primary difference lies in the sizing of the transistors. While the aspect ratios of the transistors were preserved, their absolute dimensions were scaled down to make the design more compact. This allowed for a rectangular layout that integrates more easily into the DPLL chip. Additionally, the LDO OTA does not include a dedicated biasing branch. This is because the bias point can be shared with the OTA used in the BGR, which makes a separate biasing circuit redundant and saves area. The complete sizing table for the OTA used in the BGR circuit is shown in Table 5.

Component	Type	Width (μm)	Length (μm)	Multiplier	Purpose
M1, M2	NMOS (nch_18_mac)	2	1.8	2	Differential input pair
M5	NMOS (nch_18_mac)	2	1.4	2	Tail transistor
M3, M4	PMOS (pch_18_mac)	2	1.5	1	Active load (1st stage current mirror)
M6	PMOS (pch_18_mac)	4	0.56	1	Second stage common-source amplifier
M7	NMOS (nch_18_mac)	2	1.4	4	Active load (2nd stage)
R4	Resistor (rupolym_m)	0.4	7	–	Compensation resistor ($\sim$ 10.18 $k\Omega$)
C1	Capacitor (cfmom_2t)	–	–	–	Compensation capacitor (1.66 pF)
M10	NMOS (nch_18_mac)	4	1.4	1	Dummy transistor (lower current mirror)
M14	NMOS (nch_18_mac)	1	1.4	4	Dummy transistor (lower current mirror)
M15	NMOS (nch_18_mac)	4	1.4	1	Dummy transistor (lower current mirror)
M17	NMOS (nch_18_mac)	1	1.4	8	Dummy transistor (lower current mirror)
M12	PMOS (pch_18_mac)	1	1.5	4	Dummy transistor (upper current mirror)
M11	PMOS (pch_18_mac)	4	1.5	2	Dummy transistor (upper current mirror)
M9	NMOS (nch_18_mac)	1	1.8	4	Dummy transistor (common-centroid for diff pair)
M0	NMOS (nch_18_mac)	12	1.8	2	Dummy transistor (common-centroid for diff pair)

Table 5
Component Sizing Summary for OTA in LDO

3.4 LDO

Sizing the Pass Transistor:

In terms of low-frequency PSRR and output temperature coefficient (TC), the bandgap reference already ensures good performance. Line regulation is also well satisfied due to the high open-loop gain of the OTA, as shown in Equation 3. What remains as the primary concern is load regulation, which is strongly dependent on the transconductance g_m of the pass transistor, as described in Equation 4.

As discussed in the theoretical section, the PMOS pass transistor should operate in deep saturation for optimal regulation performance. To support the maximum load current of 55.5 mA, its aspect ratio $\frac{W}{L}$ must be large. We chose the minimum allowed channel length, $L = 150$ nm, to maximize current density and maintain compact layout.

The required width W must ensure that the transistor can handle the full load current across the 0–55.5 mA range. Because the drain current I_D in saturation is proportional to W, a width that is too small would require a large overdrive voltage $(V_{SG} - V_{th})^2$ to meet the current demand. This is problematic, as it may exceed the supply voltage V_{DD}. Therefore, there exists a minimum viable width, below which the pass device cannot reliably deliver the required load current. To minimize this width, the OTA output swing must be wide enough to fully control the pass transistor's gate voltage.

Although increasing W helps reduce dropout voltage (as $R_{\text{DS,ON}} \propto \frac{1}{W}$), this is not a major constraint in this design, as both V_{DD} and V_{OUT} are fixed by specification.

Higher W of the pass transistor increases the transconductance g_m, thereby improving load regulation. However, excessively increasing W introduces trade-offs:

- Area Overhead: Since the minimum required width is already substantial, further increases significantly increase silicon area.
- Gate Capacitance: Gate capacitance grows linearly with W, which introduces a lower-frequency pole in the LDO feedback loop. This can degrade phase margin and stability.

Therefore, the width of the pass transistor should be selected to:

- Support the maximum load current of 55.5 mA.
- Ensure adequate phase margin $(45 - -60^\circ)$ for closed-loop stability.
- Meet a reasonable area constraint.

Beyond this point, further increasing W results in diminishing returns and can negatively affect performance.

To obtain an initial estimate for the required width of the PMOS pass transistor, we use the saturation-region current equation:

$$W \geq \frac{2 I_{\text{Load}} L}{\mu_p C_{ox} (V_{SG} - V_{th})^2}$$

The parameters μ_p and C_{ox} can be extracted from a DC sweep of V_{SG} versus I_D on a diode-connected PMOS transistor. By plotting $\sqrt{I_D}$ versus V_{SG}, we can estimate $\mu_p C_{ox}$ from the slope of the linear region. For this process, this value was found to be approximately 60–70 μA/V^2.

Using this estimate, the required width range for the PMOS transistor was calculated to be between 550 μm and 700 μm to support a maximum load current of 55.5 mA with sufficient margin. After simulation-based tuning and stability analysis, a width of 600 μm was selected as the optimal value, offering a good trade-off between current drive capability, layout area, and gate capacitance.

Sizing the Feedback Network Resistors:

Regarding the feedback resistors, their ratio sets the output voltage according to:

$$\frac{V_{\text{out}}}{V_{\text{ref}}} = 1 + \frac{R_{fb1}}{R_{fb2}}$$

While their ratio defines the DC gain of the feedback network, the absolute values have important trade-offs:

- **Power consumption:** Lower resistor values draw more current from the output through the feedback network, which increases quiescent power consumption and reduces power efficiency.
- **Stability:** Higher resistor values, combined with the parasitic capacitance at the feedback node, form an RC low-pass filter. If the resistors are too large, this can shift the output pole to low frequencies that can degrade the phase margin and lead to oscillations in the transient response.

- **Increased thermal noise:** Larger resistors generate higher thermal noise, which can couple into the feedback path and degrade output voltage regulation,

Balancing these considerations, the feedback resistor values were chosen to be approximately $R_1 = 40\,k\Omega$ and $R_2 = 10\,k\Omega$. These provide the correct output voltage ratio and draw low current while maintaining low enough resistance to preserve phase margin and ensure stable operation across load conditions.

Finding Stable Range for the Output Filter Network:

The output capacitor and the optional series resistor were swept in simulation to determine the range of values that ensure stable LDO operation. Between $100fF$ to $10pF$ output capacitance, many combinations provided good regulation. Howver, they differed significantly in transient performance, particularly in settling time, which varied from approximately 1 μs to 100 μs, depending on the selected capacitor.

These simulations also confirmed that the LDO remains stable and functions correctly with minimal voltage undershoot and overshoot when the load capacitance is approximately 150–300 fF, which is the estimated total load due to the decoupling capacitors at the input of the DPLL blocks.

Adding Enable Signals to Extend LDO Functionality:

In mid-July 2025, the LDO design was updated to include two control signals:

- **EN** — Enables or disables the LDO. When `EN = 0`, the LDO is turned off and the output voltage is forced to 0 V.
- **BYPASS_EN** — Allows bypassing the LDO. When `BYPASS_EN = 1`, the output is directly connected to the supply voltage (1.8 V).

To implement this functionality, two transistors were added at the LDO output node to act as switches:

- An NMOS pulldown transistor forces the output to 0 V when `EN = 0`.
- A PMOS pullup transistor drives the output to 1.8 V when `BYPASS_EN = 1`.

However, in both of these states, it is critical to prevent unnecessary power consumption. To ensure the pass transistor does not source current (up to 55.5 mA) when it is not needed, a third PMOS pullup transistor was added at the gate of the pass device. This transistor shorts the gate to source voltage (V_{GS}) of the pass transistor whenever `EN = 0` or `BYPASS_EN = 1`, to disable it.

The nominal LDO operation—regulating the output at 0.9 V—is achieved when `EN = 1` and `BYPASS_EN = 0`. In this mode, all three control transistors remain off, allowing the OTA to regulate the output normally.

The state `EN = 0` and `BYPASS_EN = 1` results in undefined behavior due to contention between the pullup and pulldown paths at the output. This condition serves no functional purpose and should be avoided during normal operation.

A summary of the enable logic and the resulting output behavior is provided in Table 6.

EN	bypass_en	Vout	**gate_en** = EN · $\overline{\text{bypass\_en}}$	Description
0	0	0V	0	NMOS pulls down output to 0V, PMOS pull-up disabled. Pass transistor disabled.
0	1	forbidden	0	Pulldown competes with pull-up, output undefined. Pass transistor disabled.
1	0	0.9V	1	Nominal operation. Pulldown and pull-up transistors disabled. Pass transistor enabled.
1	1	1.8V	0	PMOS pulls up output to 1.8V, NMOS pulldown disabled. Pass transistor disabled.

Table 6
Logic for Enable Signals

Since the NMOS pulldown transistor must pull the output node to ground when `EN = 0`, its gate should be driven by the inverted enable signal, `EN_N`. Similarly, the PMOS pullup transistor must drive the output to the supply voltage when `BYPASS_EN = 1`, so its gate should be controlled by the inverted bypass signal, `BYPASS_EN_N`.

To ensure the pass transistor is only active during normal regulation (`EN = 1` and `BYPASS_EN = 0`), its gate is controlled by a signal called `GATE_EN`, which is given by:

$$\texttt{GATE\_EN} = \texttt{EN} \cdot \overline{\texttt{BYPASS\_EN}}$$

The logic implementation of these control signals is shown in Figure 13. These components can be fully integrated into the LDO layout. Both the logic gates and the three associated switching transistors occupy minimal area and have negligible impact on the overall layout size.

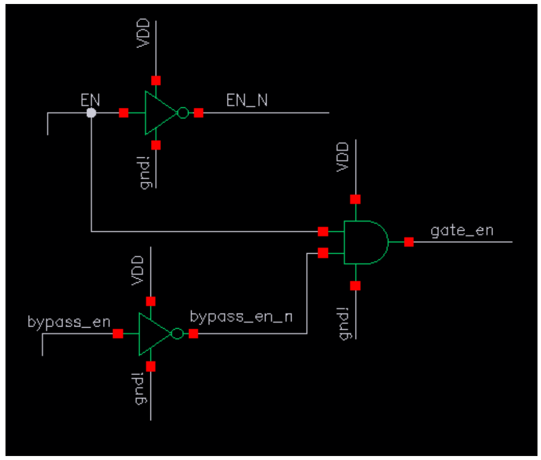

Figure 13. Logic of the Enable Signals

Table 7 summarizes the devices sizes in the LDO:

Component	**Type**	**Width (μm)**	**Length (μm)**	**Purpose**
M0	PMOS (pch_18_mac)	600	0.15	Pass transistor for LDO output regulation
M3	PMOS (pch_18_mac)	10	0.15	Pull-up to disable pass transistor gate
M4	PMOS (pch_18_mac)	25	0.15	Pull-up to output node (bypass mode)
M2	NMOS (nch_18_mac)	0.3	0.5	Pull-down to output node (disable mode)
R1	Resistor (rupolym_m)	0.4	28.5	Feedback resistor ($\sim 41.39\,k\Omega$)
R16	Resistor (rupolym_m)	0.53	9.5	Feedback resistor ($\sim 10.43\,k\Omega$)
Output Capacitance	Capacitor (dig. block/top-level)	–	–	Output filter (optional, 100–300 fF)
Series Resistor	Resistor (dig. block/top-level)	–	–	ESR (optional, 5–50 mΩ)

Table 7
Component Sizing Summary for LDO Circuit

3.5 Layout Implementation

The implementation of analog integrated circuit layout differs significantly from digital design methodologies. While digital layouts generally focus on standard cell placement, automated routing, and achieving timing closure across a large number of logic gates, analog layouts must instead emphasize precise matching, symmetry, parasitic control, and minimal sensitivity to process and environmental variations.

In analog blocks, even small mismatches or unintended parasitic effects can severely degrade performance parameters such as offset, gain, and common-mode rejection, which are often negligible in purely digital designs. As a result, the layout process for analog blocks is typically much more manual, with careful planning of device placement, routing symmetry, current mirror structures, and guard rings to ensure predictable and stable circuit behavior.

The following subsections will describe the complete layout flow of each sub-block in the project (the Bandgap Reference, OTA blocks, and LDO) with an emphasis on how design constraints, matching considerations, and routing strategies were addressed. This approach aims to guide the reader step by step through the entire physical realization of the system.

OTA Layout

The OTA used in this design is a two-stage architecture that incorporates a differential input stage. Due to the inherent sensitivity of differential pairs, it is essential to ensure a high level of symmetry and matching. From the experience gained in this project, a critical principle emerged: parasitics can never be completely eliminated, but their effect can be controlled by guaranteeing that every transistor requiring matching sees the exact same parasitic environment as its counterpart. This matching strategy is achieved through the following:

- **Dummy Transistors:** Placed around the matched devices to equalize the parasitic capacitances and resistances from all four directions. These dummy devices are dimensionally identical to the active transistors and have all terminals shorted together to avoid influencing the circuit operation by ensuring they never carry any signal.
- **Symmetric Routing:** All critical interconnects are routed as symmetrically as possible, since every wire introduces unavoidable capacitances and resistances that must be balanced between matched devices. The same applies to vias, which must be placed and sized in a symmetric fashion.

M3 and M4: These PMOS devices act as the active load for the differential pair and, through their internal feedback, are responsible for ensuring current equality between the two branches. They play a critical role in maintaining the symmetry of the differential stage. The layout, in Figure 14, demonstrates mirror-like symmetry and the use of dummy devices to balance parasitic effects.

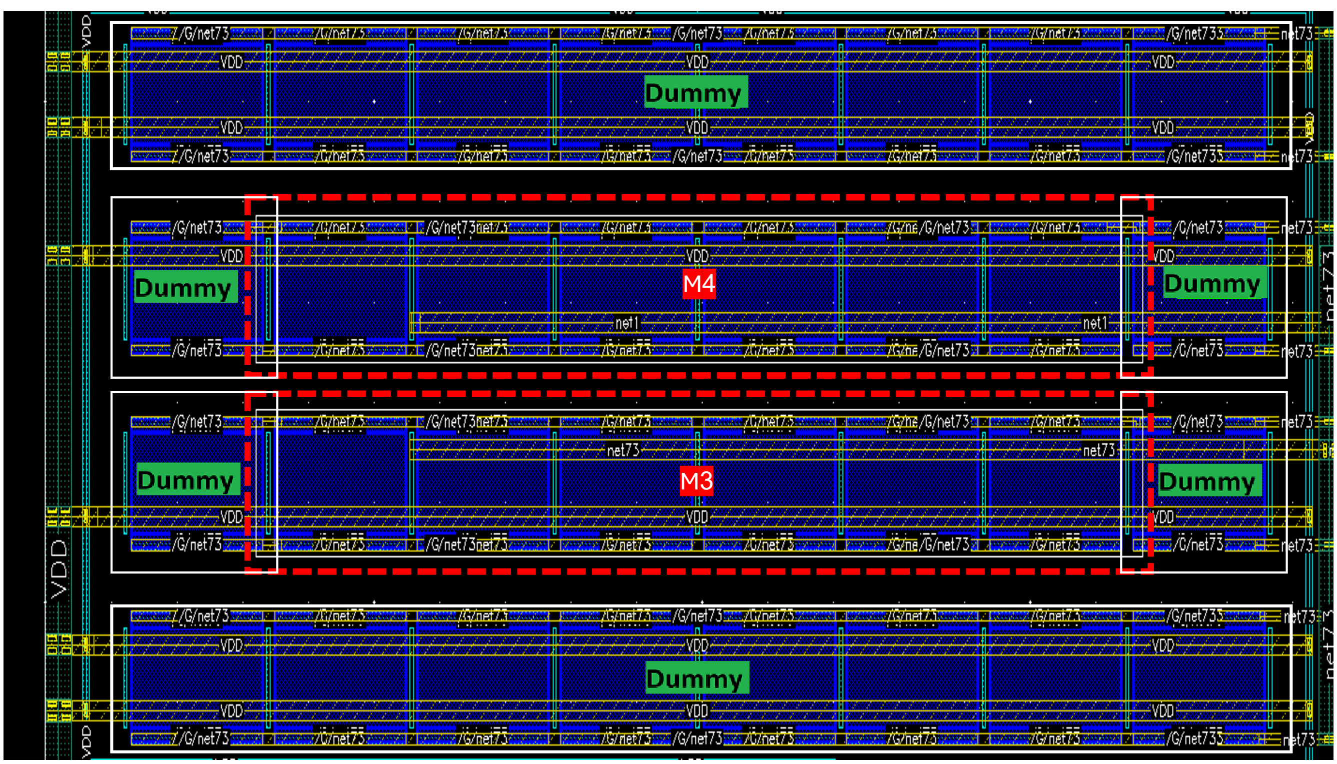

Figure 14. Layout View of M3 and M4 with Surrounding Dummies and Symmetric Routing

M1 and M2: These transistors implement the input differential pair of the OTA, a critical part of the amplifier that requires extremely high matching. To achieve this, a common-centroid layout technique was adopted. In this method, the two devices are split and arranged such that their "effective center" is identical, which greatly enhances symmetry and reducing systematic mismatch.

Figure 15 illustrates how the differential pair is split and interleaved, with an H-shaped connection scheme that further improves routing symmetry. This ensures that both transistors are affected equally by parasitic resistances and capacitances along all routing directions.

Beyond standard layout parasitics (metal capacitances, resistances, and via effects), the common-centroid strategy also helps counteract wafer-level doping gradients. Because the doping concentration across a silicon wafer is typically radially symmetric but varies from the center outward, placing both transistors to share the same centroid means that any doping gradient will affect them equally. Although doping profiles can be relatively uniform, minor variations still exist due to process imperfections, and the common-centroid helps mitigate these effects, leading to better current and offset matching.

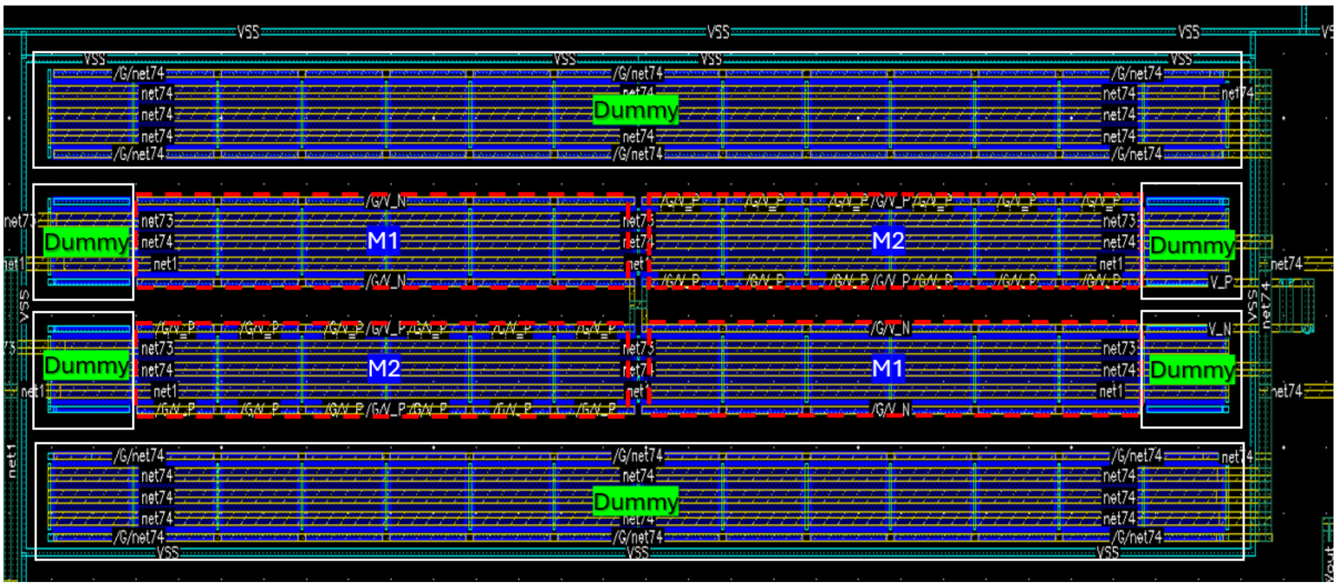

Figure 15. Layout View of the Differential Pair in a Common-Centroid Arrangement, showing the Symmetrical H-shaped Routing

M5, M7, and M8: These devices form the current mirror that defines the bias currents of the OTA. Although these transistors do not directly affect the differential stage matching, it is still beneficial to place them in a well-controlled, symmetrical layout. By matching their geometry and carefully routing their interconnects, the designed current ratios between the mirror branches are better preserved, which in turn guarantees that the amplifier operates with the intended bias currents and maintains its performance targets. Even small mismatches in these transistors could slightly alter the transconductance or reduce the headroom of the second stage, so implementing a consistent layout strategy for the current mirror is still an important consideration, despite it being less critical than for the differential pair itself.

The overall floorplanning of the OTA was carefully designed to follow the natural current flow from V_{DD} to V_{SS}. This means that the transistors were physically placed from left to right according to the natural path of the current. Such an arrangement also supports efficient routing and minimizes the total interconnect resistance and capacitance. In addition, placement decisions took into account silicon area constraints as well as the need to maintain symmetrical routing wherever possible, in order to achieve optimal matching and predictable performance.

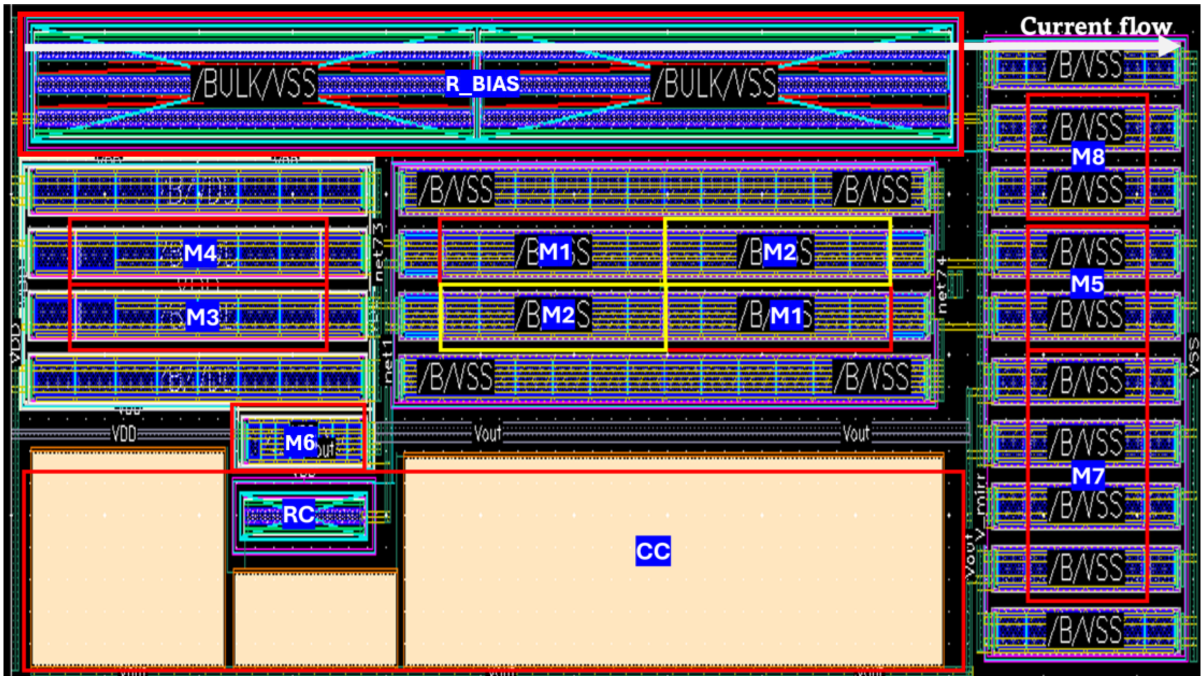

Figure 16. Entire Floorplan of the OTA Layout showing the Arrangement of Functional Blocks

Bandgap Reference Layout

The diodes within the Bandgap Reference were implemented using PNP bipolar transistors, configured by shorting the base and collector terminals to behave as diode-connected devices. Already in the schematic design phase, a size ratio of 1:8 was defined between the two diodes to achieve the required ΔV_{BE} behavior for the PTAT current generation.

In order to preserve excellent matching between these two devices, a common-centroid structure was chosen for their layout, similar to the techniques used in the OTA. As shown in the figure, this common-centroid placement ensures that both devices are subject to similar process gradients and systematic variations, thereby reducing mismatch and improving temperature stability.

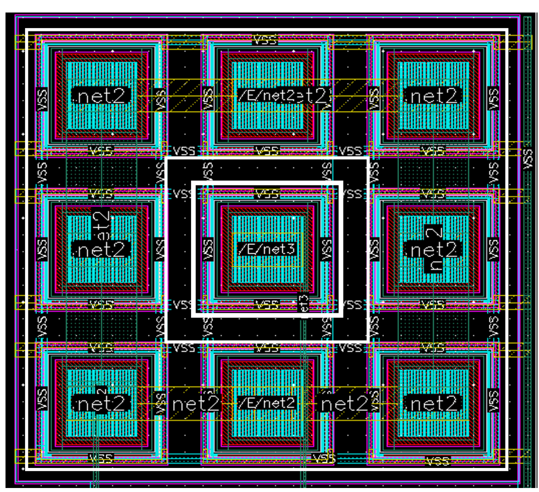

Figure 17. Layout View of the Two Diode-Connected PNP Transistors in a Common-Centroid Arrangement

OTA for LDO Layout Modification

Although the same OTA architecture was reused within the LDO (with identical electrical parameters), a separate layout instance was created to optimize silicon area and to achieve a more square-shaped overall block, which is often easier to integrate. This variant was slightly rearranged in floorplan, while maintaining identical device dimensions and matching rules. In addition, both OTAs share the same biasing network, allowing reuse of the operating point and saving area and current consumption.

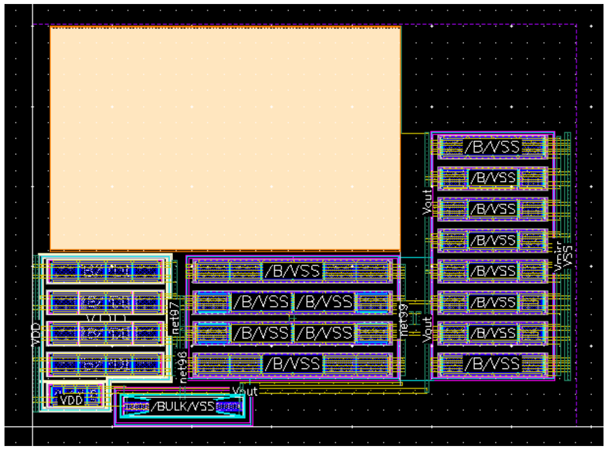

Figure 18. Modified OTA Layout Implemented for the LDO, Preserving Device Parameters but Rearranged for a Square Layout and Shared Biasing.

LDO Layout

At the heart of the LDO stands the PMOS pass transistor, whose role is to handle a wide range of load currents and supply the required current to stabilize the output voltage. As described at the beginning of this book, the maximum load current is specified as 55.5 mA. To achieve this performance, a wide transistor was chosen, with a total width of 600 μm.

In order to distribute such a large current efficiently across the transistor, it was divided into 80 fingers and implemented in 5 separate instances. This segmentation improves current spreading and reduces localized current crowding effects. In addition, the physical placement was chosen carefully: the V_{DD} supply enters from the lower left corner of the device, while the regulated output voltage is taken from the upper right corner. This orientation encourages the current to spread evenly across the transistor's width, minimizing hot spots.

The routing strategy followed a **metal stacking** approach, where the current flows from M2 up to M8 layers, allowing the metal hierarchy to carry higher current densities without exceeding electromigration limits. The metal layers in modern processes have differing thicknesses and current-handling capabilities, with higher-numbered metals being thicker. The wire widths and spacings were determined based on the current density specifications outlined in the process design kit (PDK). While the exact values are proprietary and cannot be disclosed, all routing complies with the PDK's reliability and electromigration guidelines.

Furthermore, the layout employed dedicated wide metal connections between critical regions of the pass transistor, further helping to share the current load evenly along both the device and its associated routing. Figure 19 shows the resulting layout and metal routing arrangement.

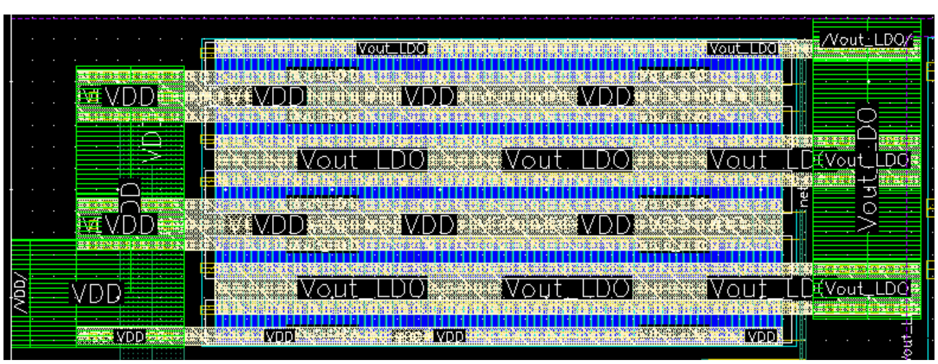

Figure 19. Layout View of the PMOS Pass Transistor showing Metal Stacking and Current Spreading Structures

Final Integration

The final integration of the analog block was carried out with a focus on achieving a compact layout, ensuring robust power distribution, and maintaining appropriate isolation between sensitive analog nodes and the power-handling sections. The total block area was measured to be approximately 4699.7 μm^2.

Within this area, the total drawn metal area across all routing layers (M1 to M9) amounted to 914.2 μm^2, giving an overall pre-fill metal density of about 19.45%. This metal density complies with standard DRC rules and supports robust current delivery while maintaining acceptable process uniformity during manufacturing. During the final tapeout flow, additional dummy metal fill will be inserted.

Special care was taken to align power supply rails and sensitive analog references in a consistent floorplan, following a left-to-right hierarchy from V_{DD} to V_{SS}, to ensure current naturally flows through the block with minimal parasitic IR drops and uniform thermal distribution.

The block's final outline is shown in Figure 20, demonstrating a clean, symmetrical arrangement ready for top-level integration.

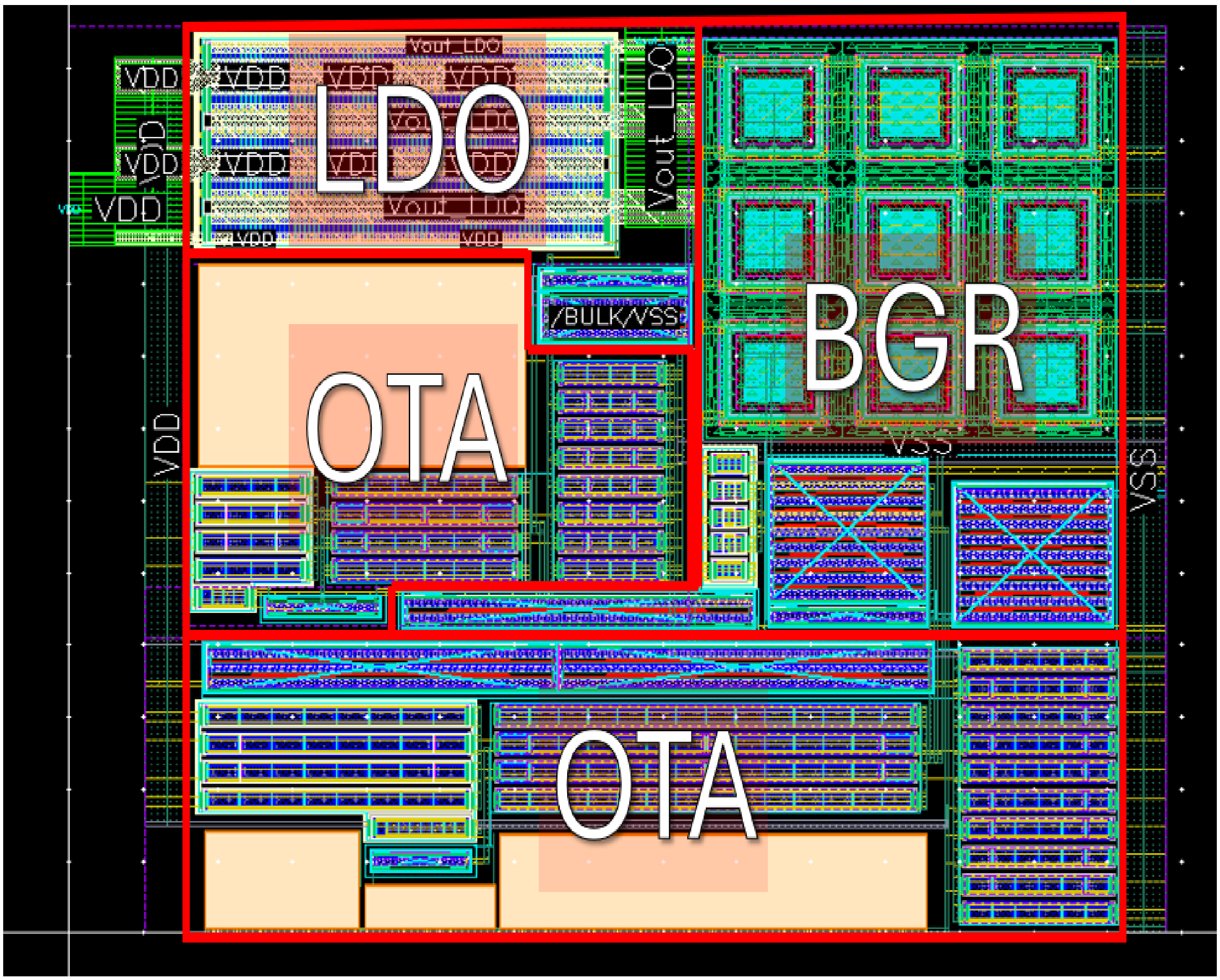

Figure 20. Final Integrated Layout of the Entire Analog Block including OTA, Bandgap Reference, and LDO

4 Simulations

This section presents the key simulation results for each of the main circuit blocks: the OTA, the bandgap reference (BGR), and the LDO. For each block, post-layout results are shown. When relevant, a comparison with the schematic results is made.

4.1 OTA

OTA DC Operating Point

The OTA was implemented in a two-stage topology with the post-layout DC operating point presented in Figure 21. The currents refer to each finger and not the entire device.

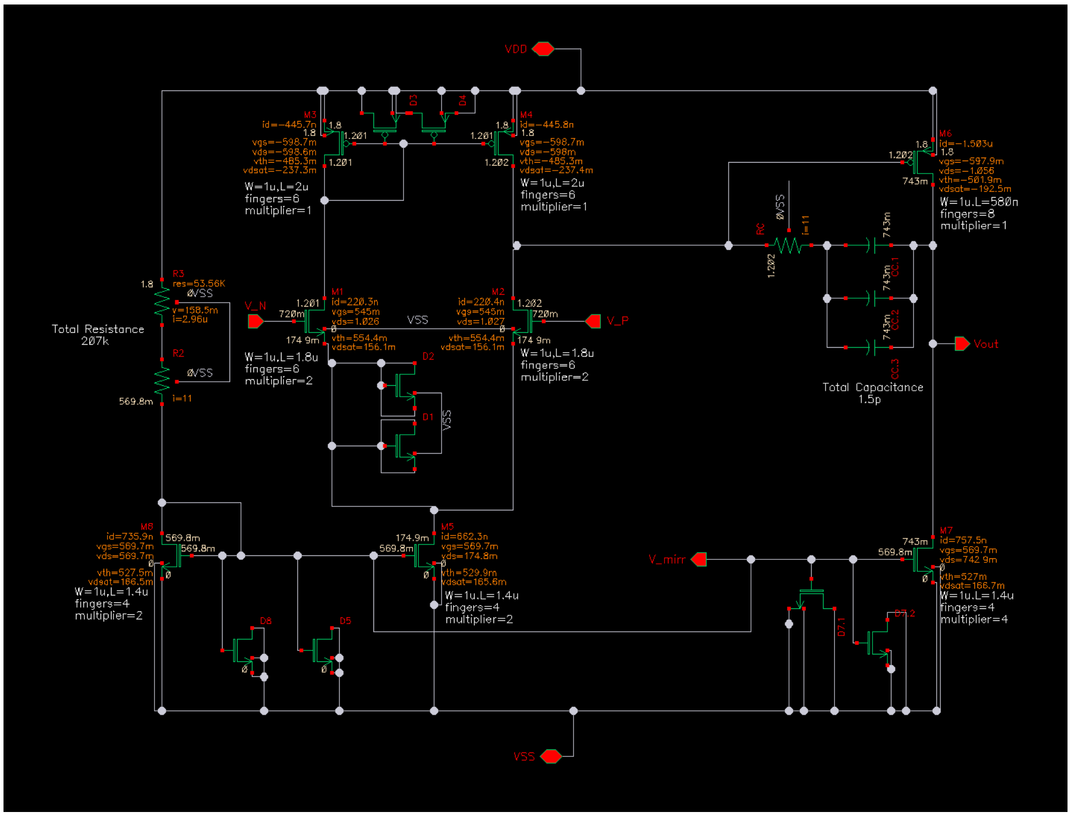

Figure 21. Post-layout DC operating point of the OTA.

The circuit includes the following main elements:

- **M1, M2:** Differential pair transistors forming the input stage.
- **M3, M4:** Active load for the first stage, helping to preserve symmetry in the differential path and support differential-to-single-ended conversion.
- **M6:** Common-source gain stage that increases the overall gain of the amplifier.
- **M5, M7, M8:** Current mirror structure, providing the bias current for both amplifier stages.
- **D1–D7:** Dummy transistors placed in the layout to ensure matching of the devices and preserve symmetry, as described in the implementation section.
- A compensation network composed of a capacitor and a resistor, which guarantees the stability of the OTA across multiple process, voltage, and temperature corners.

AC Response of the OTA

Figure 22 presents the open-loop AC response of the OTA. The OTA achieves a DC gain of approximately 77.5 dB, with a unity-gain frequency (UGF) around 5.6 MHz and a gain-bandwidth product (GBW) of about 5.9 MHz, which indicates a well-defined system with a single dominant pole. In this design, a deliberate trade-off was made between gain and bandwidth. Since the OTA primarily responds to slow variations in supply voltage, load current and temperature, which based on literature and similar designs generally occur below 100 Hz, the circuit effectively behaves as a quasi-DC amplifier. Therefore, a high DC gain was prioritized to minimize steady-state offset in the BGR and LDO as much as possible. Additionally, the near overlap of the UGF and GBW suggests good pole separation, enabled by the careful selection of the compensation capacitor and resistor. This supports robust phase margin and overall stability within the expected operation frequency range.

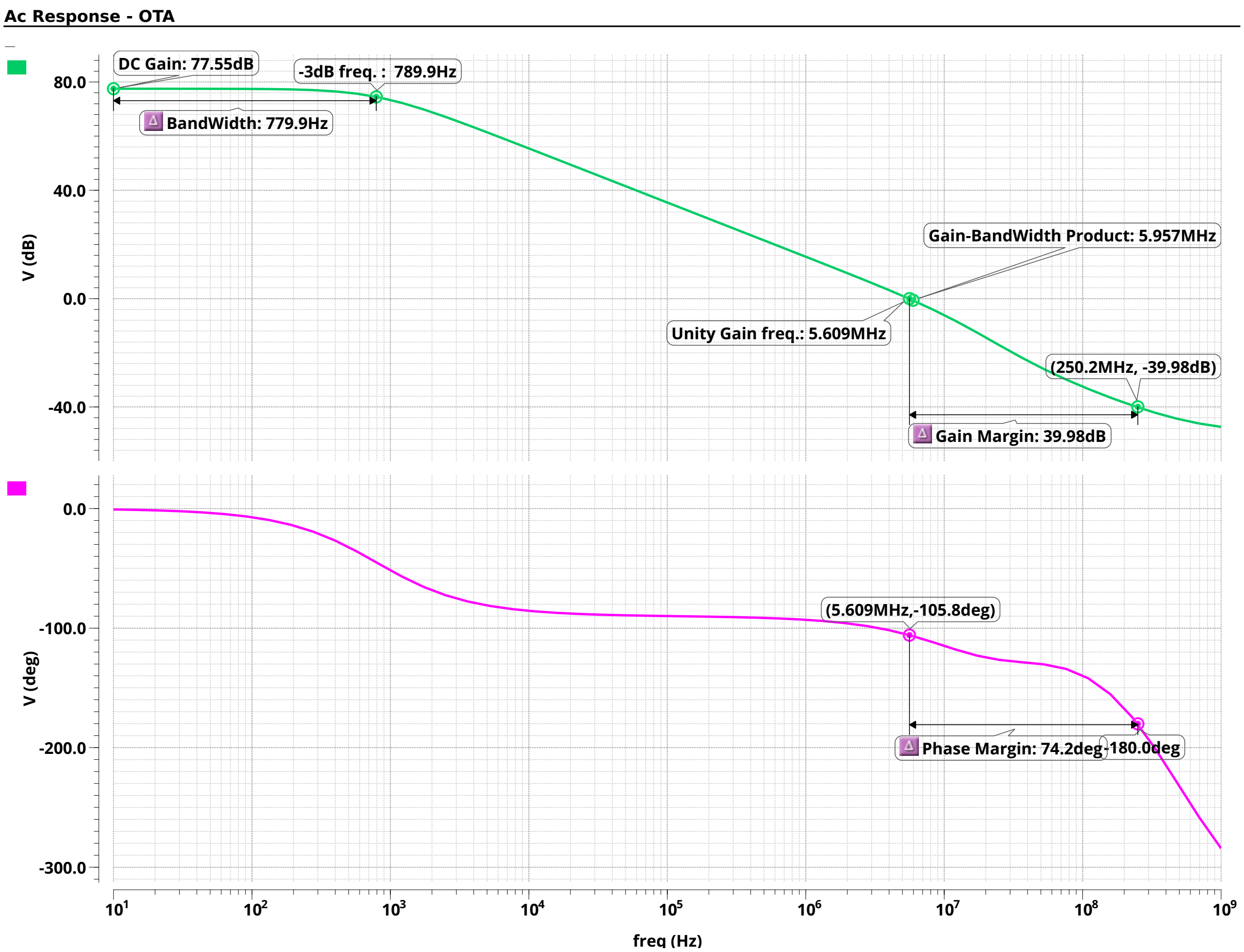

Figure 22. Open-loop AC response of the OTA, demonstrating a DC gain of 77.5 dB and a unity-gain frequency of 5.5 MHz.

Output Swing

Figure 23 shows the output voltage of the Operational Transconductance Amplifier (OTA) as a function of the differential input, obtained via a DC sweep simulation. In this test, the inverting input (V_-) is fixed at 720,mV, while the non-inverting input (V_+) is swept from 710,mV to 730,mV. The resulting curve highlights three distinct regions of operation:

- Negative differential input — ($V_+ < V_-$): The OTA output saturates near 0V.
- Positive differential input — ($V_+ > V_-$): The OTA output saturates near the supply voltage (1.8V).
- Small differential input — ($V_+ \approx V_-$): The OTA operates in its linear region with high gain.

This characterization is essential for verifying that the OTA meets the design requirements for its role in both the Bandgap Reference (BGR) and the Low-Dropout Regulator (LDO). In these circuits, the OTA operates in a closed-loop configuration, typically experiencing only small differential input voltages (on the order of a few millivolts). However, it must be capable of driving internal nodes across a wide output voltage range, especially to control the gate of the pass transistor in the LDO. Therefore, verifying that the OTA output can swing from near ground to near V_{DD} is critical for ensuring correct loop operation and high regulation accuracy under all process, voltage, and temperature conditions.

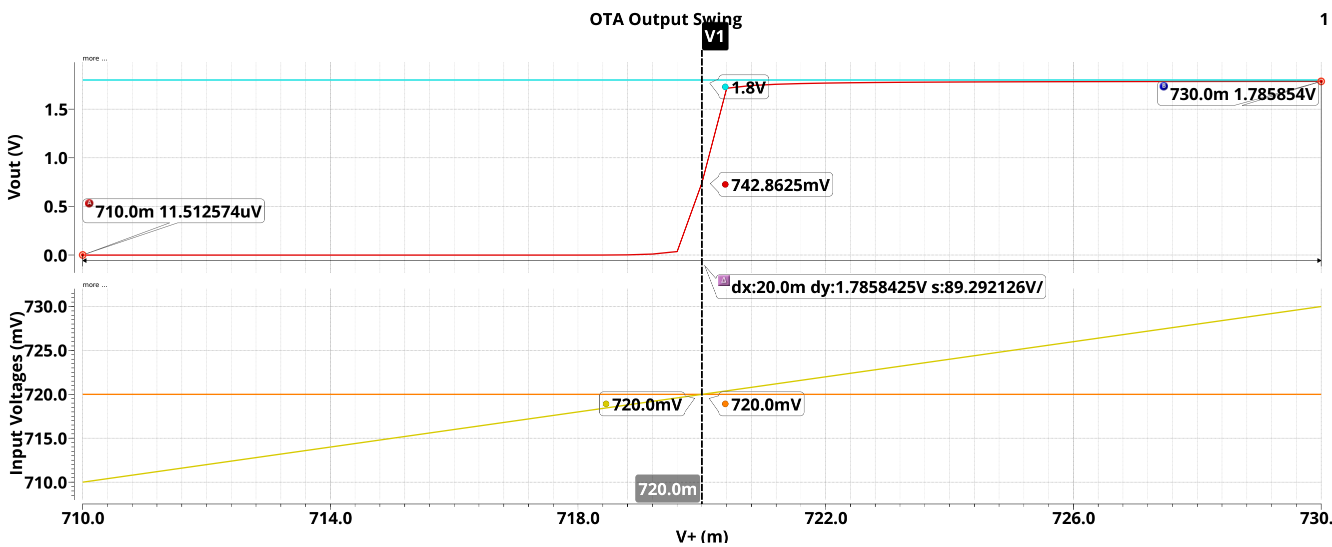

Figure 23. Simulated output swing of the OTA demonstrating nearly rail-to-rail behavior.

OTA Output Swing Under Linear Sine Input Operation

Figure 24 shows the OTA linear region response to a sine input. The measured minimum output voltage was approximately 73.9 mV, while the maximum reached 1.67 V, for a supply voltage of 1.8 V. This indicates that the amplifier operates effectively across nearly the entire supply range, with only a small margin to the rails. Such performance demonstrates careful design of the output stage and good selection of transistor sizing on the output stage to achieve rail-to-rail capability or close to it.

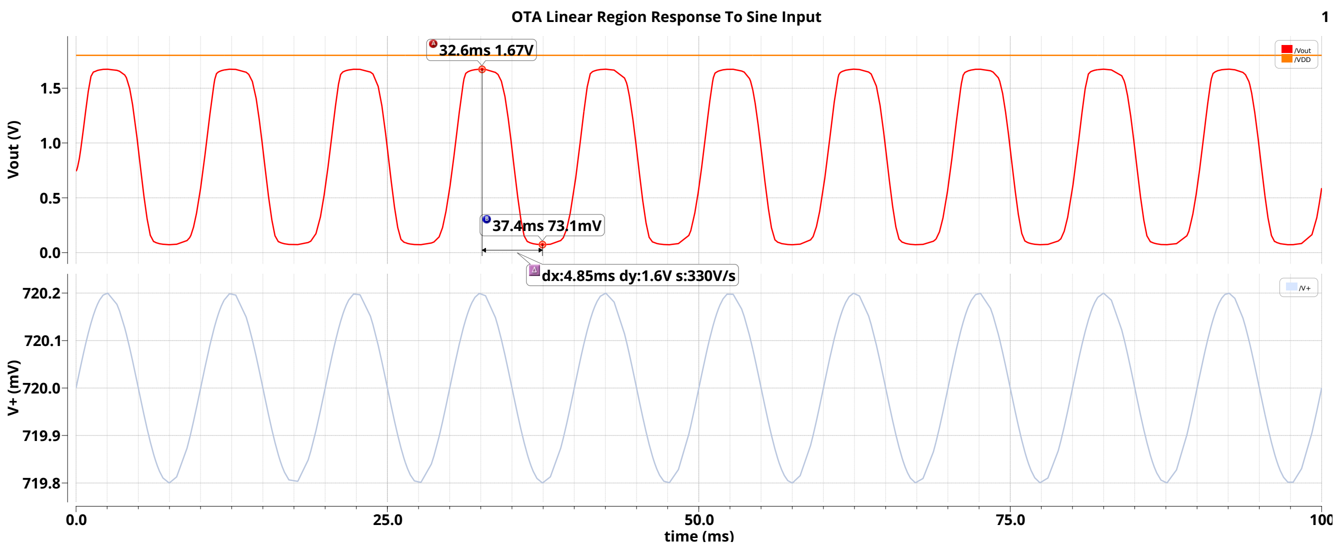

Figure 24. Simulated Output Swing of the OTA in the Linear Range

4.2 Bandgap Reference Circuit

BGR DC Operating Point

The Bandgap Reference (BGR) circuit shown in Figure 25 consists of the following elements:

- Two PNP BJTs, which are effectively configured as diodes by shorting their base to collector. A size ratio of 1:8 was chosen for these devices due to layout considerations and the use of a common-centroid technique, as discussed in the implementation chapter.
- An OTA amplifier, which forces equal voltages across the two diode-connected BJT branches to generate a well-defined PTAT voltage.
- Three PMOS transistors that form a current mirror.
- Resistors that weigh the PTAT (Proportional To Absolute Temperature) and CTAT (Complementary To Absolute Temperature) currents, to enable effective cancellation of temperature dependence.

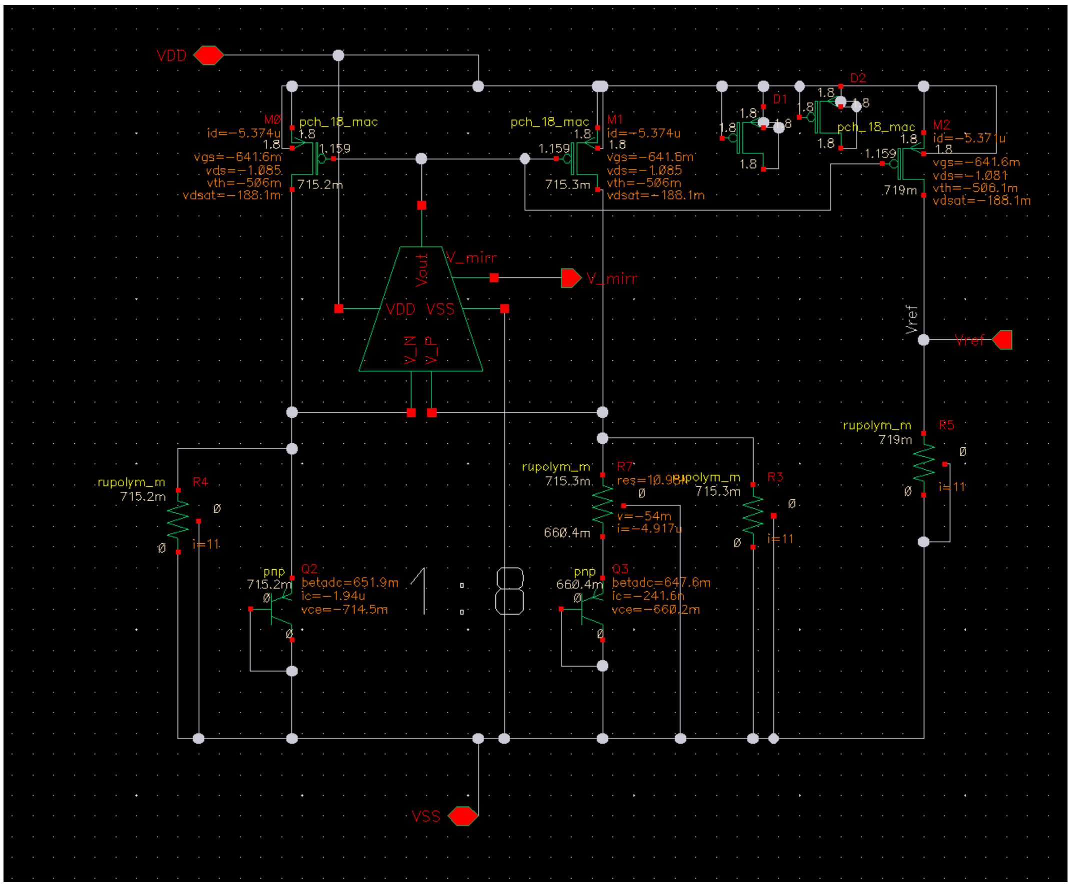

Figure 25. Bandgap reference schematic with annotated operating points.

Note: This is not a conventional BGR configuration, in which, a reference voltage of approximately 1.2 V is generated. Due to the project requirement for a final output voltage of 0.9 V, this modified topology was chosen. It provides more degrees of freedom, allowing reference voltages lower than the usual 1.2 V. Here, the achieved V_{ref} is approximately 0.72 V. For further details, please refer to the theoretical background chapter.

Bandgap Reference Principle

The fundamental principle of the Bandgap Reference relies on a precise scaling of two temperature-dependent components: a voltage that increases proportionally to absolute temperature (PTAT), and a voltage that decreases as a function of temperature (CTAT). By carefully combining these two contributions, a reference voltage V_{ref} is obtained that exhibits very low sensitivity to temperature variations, ideally achieving a zero temperature coefficient over a targeted temperature range.

Figure 26 illustrates the CTAT and PTAT components, which exhibit equal magnitudes but opposite temperature coefficients. As a result, their sum remains nearly constant across a wide temperature range, which demonstrates effective temperature compensation.

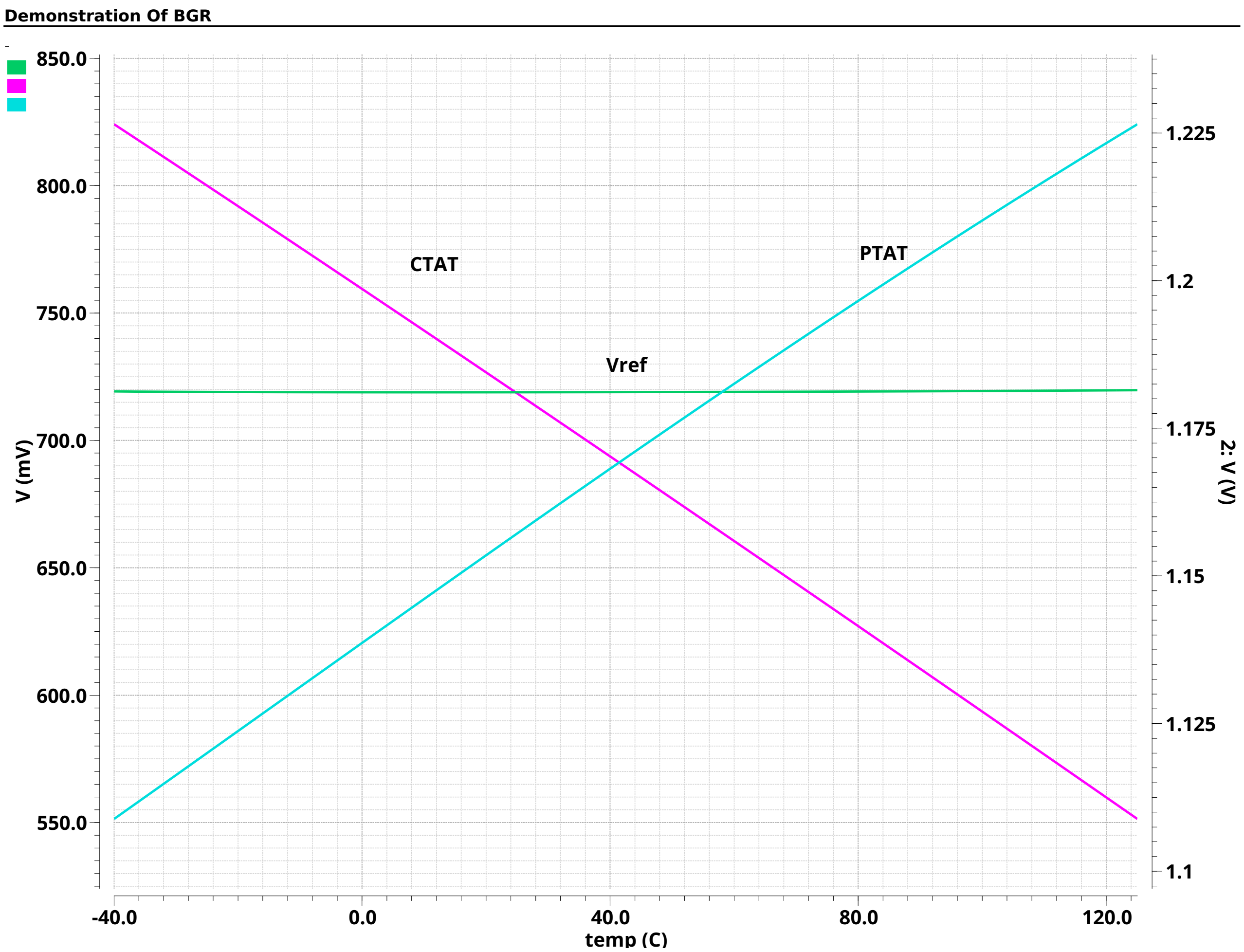

Figure 26. Demonstration of the BGR concept: summing a PTAT voltage (increasing with temperature) and a CTAT voltage (decreasing with temperature) to obtain a reference voltage with minimal temperature dependence.

Bandgap Reference Loop Gain

The loop gain plot of the Bandgap Reference (BGR) circuit reveals several important characteristics regarding its stability and frequency response.

The figure clearly indicates that the BGR loop is stable, as evidenced by a large phase margin at the unity-gain frequency. A phase inversion is observed at low frequencies, which is attributed to the common-source (CS) configuration of the transistors in the current mirror. This stages introduces a 180° phase shift into the loop. At low frequencies, the loop gain is dominated by the high DC gain of the OTA, which ensures effective regulation and temperature compensation.

As frequency increases, the gain begins to roll off due to the dominant pole of the OTA. However, at higher frequencies, the loop gain flattens and reaches a plateau. This plateau is primarily caused by a pole-zero cancellation.

The output pole of the OTA lies at a very high frequency due to the small output capacitance, a result of the minimal sizing of the current mirror transistors. Additionally, the compensation network introduces a left-half-plane (LHP) zero via the series resistor and compensation capacitor. This zero effectively cancels the phase and gain degradation from the high-frequency output pole, resulting in a flattened gain response.

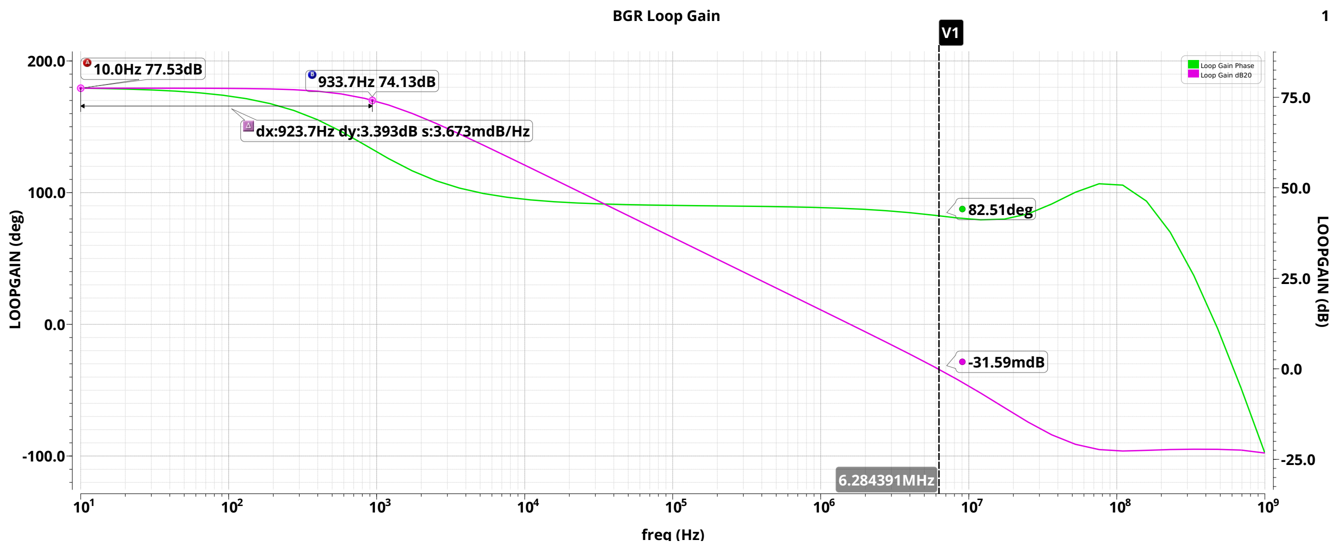

Figure 27. The Feedback Loop Gain of the BGR

BGR Temperature Coefficient

In order to evaluate the quality of temperature dependence cancellation in the Bandgap Reference, a common performance metric is the temperature coefficient (TC), as discussed in the theoretical background chapter. Figure 28 shows the simulated TC of V_{ref} before and after the physical layout of the BGR block.

The measured temperature coefficient before layout was approximately 4.6 ppm/°C, and after layout increased to about 7 ppm/°C. Additionally, it can be seen that the mean reference voltage decreased by about 2 mV after layout extraction. This negligble drop is likely caused by parasitic resistances at the output branch path metal wires.

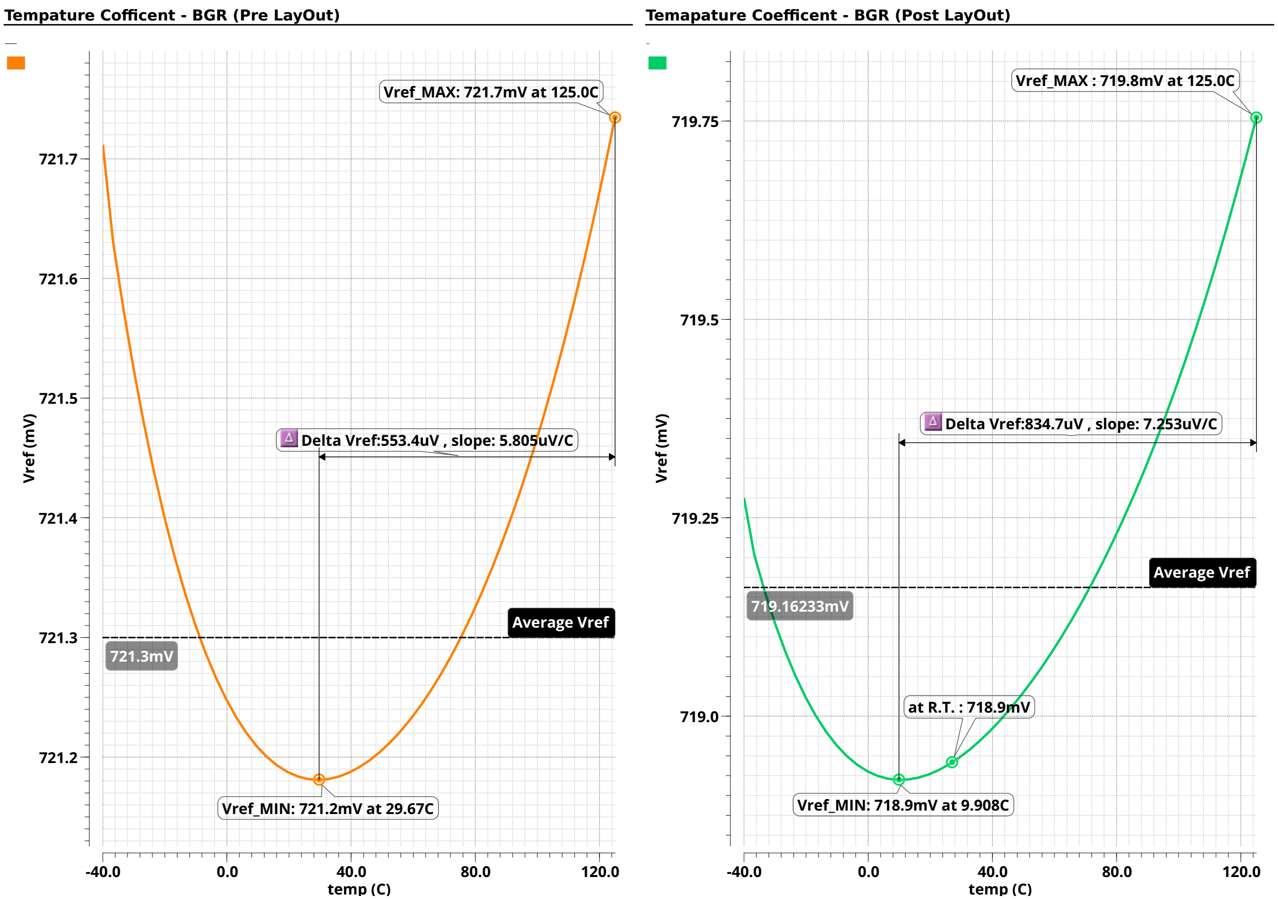

Figure 28. Simulated temperature coefficient of V_{ref} before and after layout.

BGR Line Regulation

Although the primary objective of the BGR block is to minimize temperature dependence, it is equally important to evaluate its performance under varying supply voltages. This is crucial because the BGR serves as the reference for the LDO, which requires a supply-independent reference voltage to maintain accurate regulation. Figure 29 shows the simulated line regulation of V_{ref} before and after layout extraction.

Interestingly, a slight improvement was observed after extraction, with the line regulation improving from approximately 945.9 μV/V before layout to 819.4 μV/V after layout. The reason for this improvement is not fully clear and might relate to parasitic resistances or capacitances modifying the biasing conditions. However, since this measurement reflects only the behavior of the standalone BGR sub-block, it should not be directly assumed to represent the final behavior of the entire LDO block.

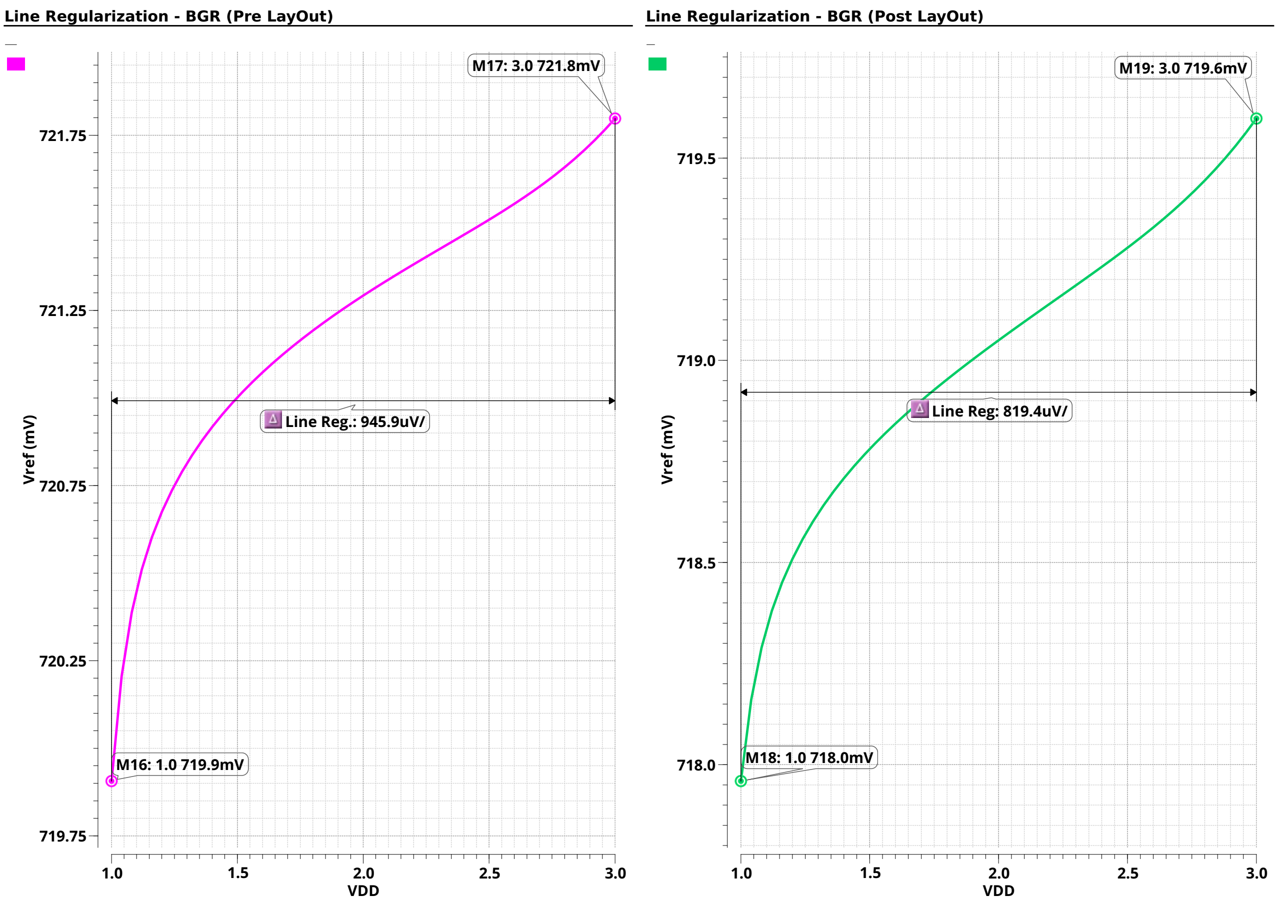

Figure 29. Simulated line regulation of V_{ref} before and after layout extraction.

BGR Power Supply Rejection Ratio (PSRR)

The PSRR performance of the Bandgap Reference was also evaluated using an AC analysis. As described in the theoretical background, this metric reflects the circuit's ability to reject supply ripple and noise. The simulation results show that the PSRR is better than −60 dB up to 10 kHz, which ensures a clean reference for the LDO under low-frequency supply variations. Beyond this point, the PSRR begins to degrade, reaching approximately −20 dB around 1 MHz. While this reduction may seem significant, it is acceptable in this design, as local decoupling at the LDO output helps attenuate high-frequency noise.

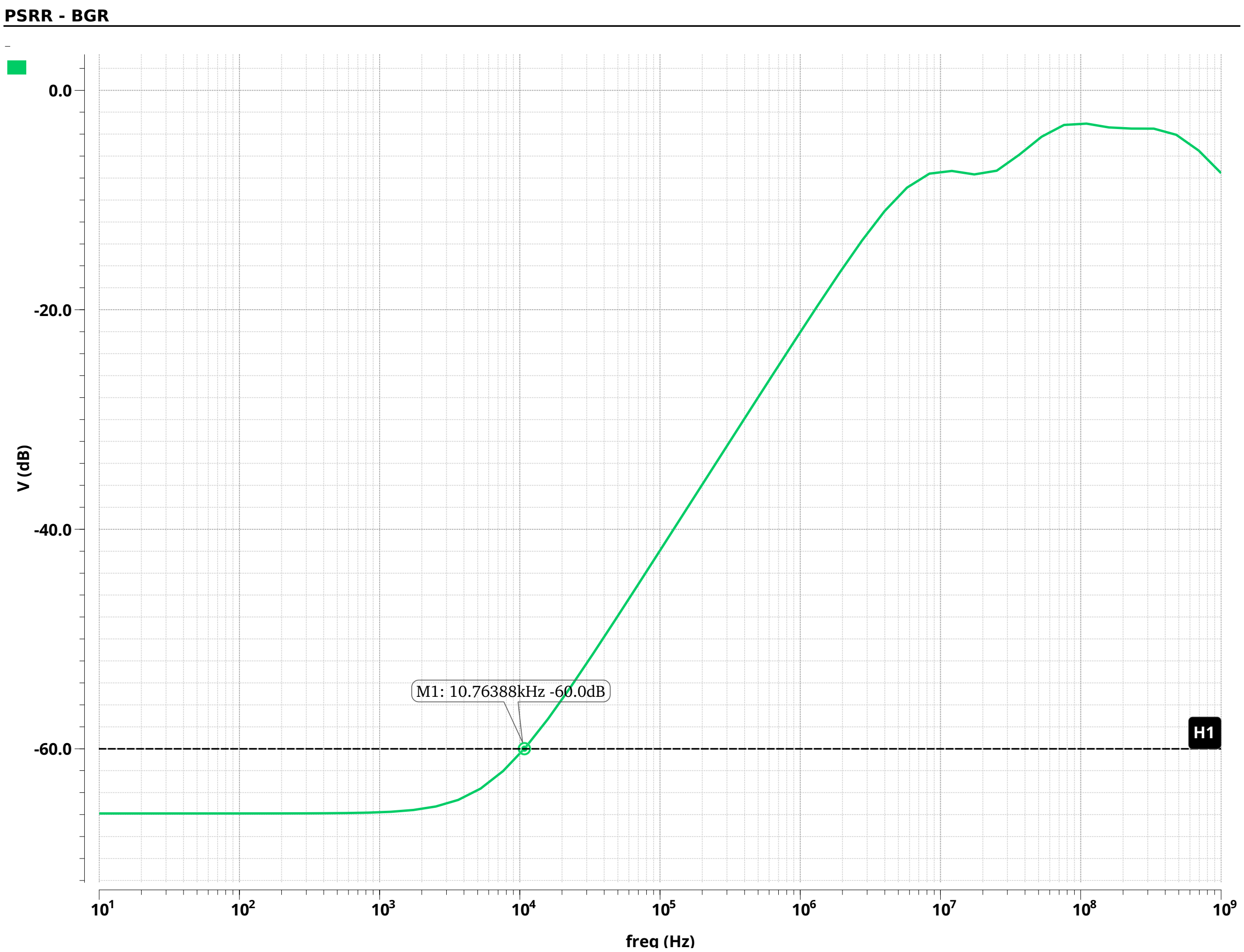

Figure 30. Simulated PSRR of the Bandgap Reference block.

BGR Start-Up Transient Response

Figure 31 presents the transient response of the bandgap reference (BGR) output voltage (V_{REF}) during power-up, where the supply voltage (V_{DD}) ramps from 0 V to 1.8 V over 1 µs. This rise time reflects a typical scenario, as integrated circuits powered by batteries usually experience supply ramp times on the order of microseconds. The transient simulation is performed at three temperature corners: −40°C (green), 27°C (yellow), and 125°C (red).

The plot shows that V_{REF} successfully settles to its designed target voltage across all temperature corners, confirming robust startup behavior of the BGR under realistic supply ramp conditions and temperature variations.

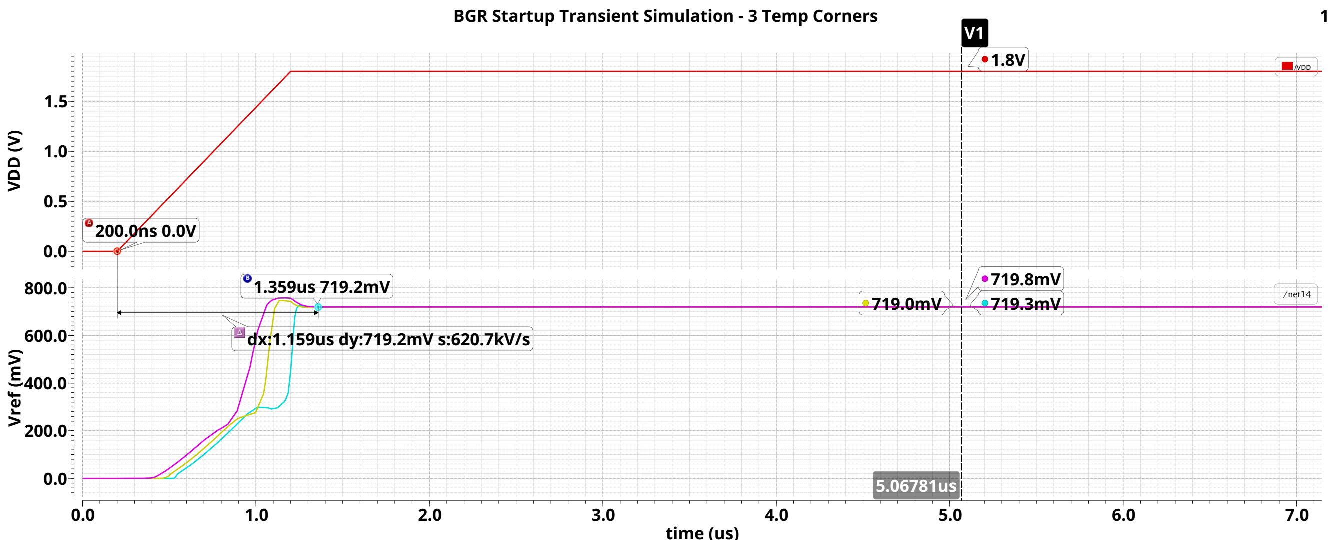

Figure 31. Startup Behavior of the BGR

4.3 LDO

LDO DC Operating Point

The Low Dropout Regulator (LDO) contains the following building blocks:

- **Bandgap Reference (BGR)** — provides a supply voltage and temperature independent reference voltage to define the regulated output.
- **OTA (Error Amplifier)** — compares the scaled output voltage to the reference and generates a control signal for the pass transistor. Note that this OTA was slightly modified from the BGR OTA due to layout and floorplanning considerations, as explained in the implementation chapter.
- **PMOS Pass Transistor** — regulates the current flow to the load under the control of the OTA, and ensures a stable output voltage.
- **Resistive Feedback Network** — scales down the output voltage to feed back into the OTA for closed-loop regulation.

These elements together ensure that the LDO maintains a stable, well-regulated output under different load and supply conditions.

Note: The current shown through the pass transistor in Figure 32 represents the current in a single finger. Since the pass device consists of 400 fingers in total, the actual total current is approximately 60mA.

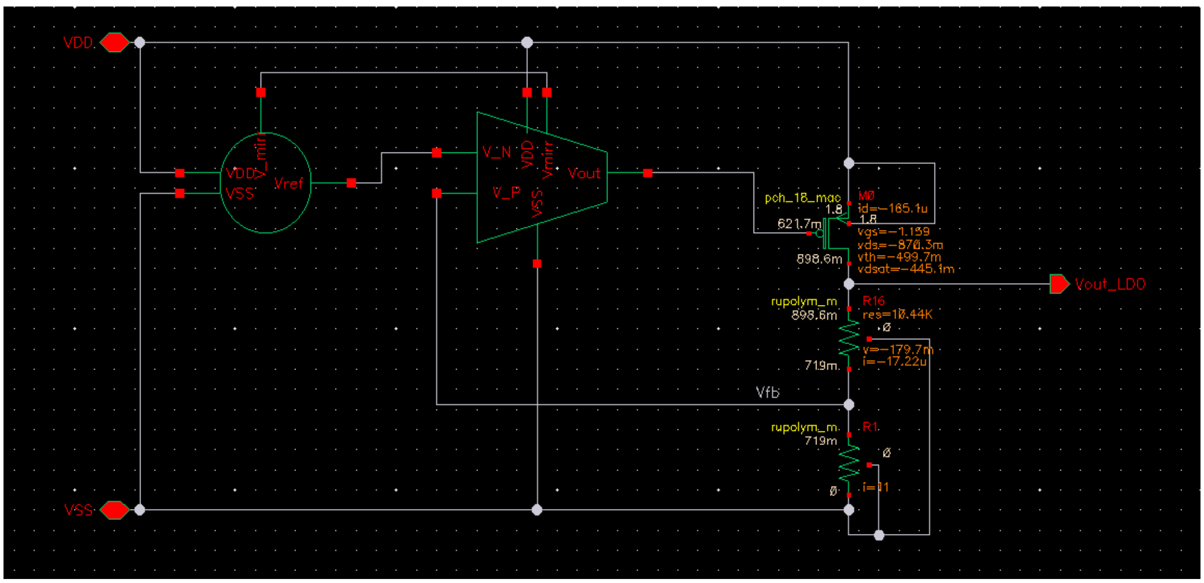

Figure 32. LDO schematic with annotated DC operating points

LDO Feedback Loop Operating Principle

Figure 33 illustrates the fundamental operating principle of the LDO. As the load current increases, the effective load resistance decreases, leading to a drop in the output voltage. To maintain regulation, the pass transistor must deliver more current. This is achieved through the OTA, which senses the output voltage drop and responds through negative feedback by lowering the gate voltage of the pass transistor. As a result, the transistor's gate-source voltage (V_{SG}) increases, which increases its drain current and restoring the output voltage. This mechanism highlights the importance of a wide output swing in the error amplifier, since large load current variations necessitate significant voltage swings at the OTA output to maintain tight regulation.

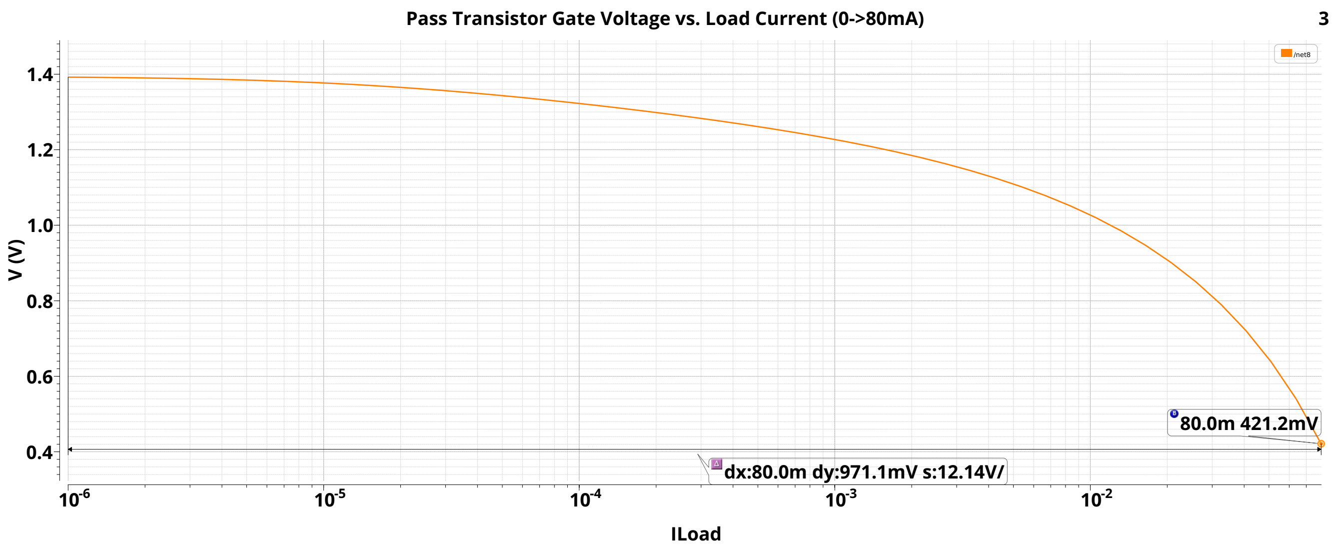

Figure 33. Gate voltage Modulation of the Pass Transistor in Response to Load Current Changes

LDO Temperature Coefficient

The temperature coefficient (TC) of the LDO output voltage was simulated to evaluate the circuit's robustness across temperature variations. As illustrated in Figure 34, the pre-layout temperature coefficient was approximately 5.1 ppm/°C, while after layout it increased to about 6.4 ppm/°C. This closely resembles the TC of the reference voltage, which determines the TC of the LDO's output, since it is regulated to track the reference voltage. If the reference voltage drifts with temperature, the LDO output follows that drift.

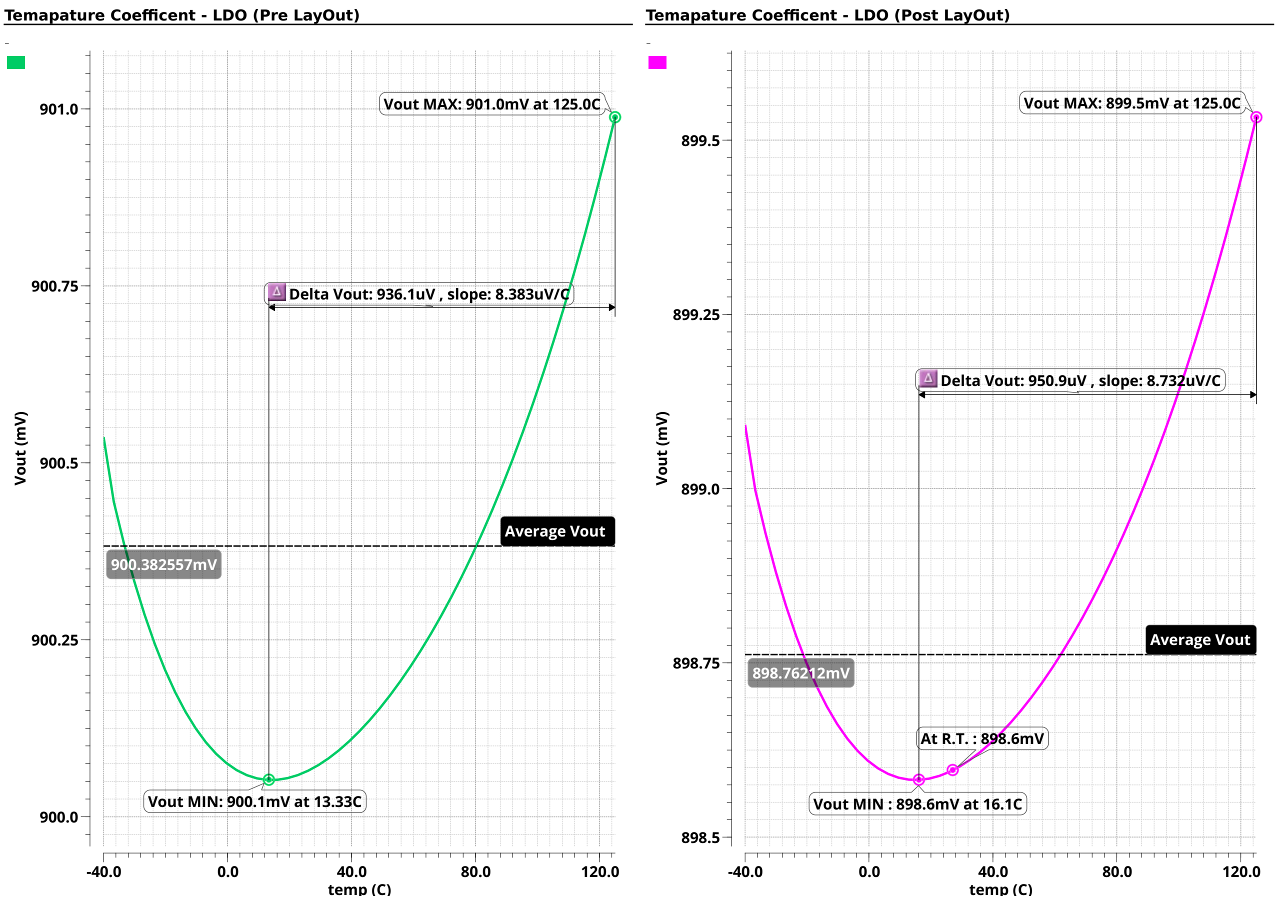

Figure 34. Simulated temperature coefficient of the LDO before and after layout.

LDO Line Regulation

Line regulation characterizes how the LDO output responds to changes in the supply voltage. Figure 35 shows that the simulated line regulation before the layout was approximately 774.4 μV/V, which increased slightly to about 893.8 μV/V after the layout. High OTA gain combined with excellent line regulation of the reference voltage were key factors in obtaining optimal line regulation performance. Beyond the change in the regulation value itself, it can also be observed that the operating voltage range of the LDO has been reduced after the layout. Before the layout, the circuit was able to maintain regulation up to a supply voltage of 1.08 V, while after the layout, the minimum operating supply voltage increased to approximately 1.3 V. This shrinkage of the input voltage range is mainly attributed to parasitic resistances and voltage drops introduced by the routing and layout structures.

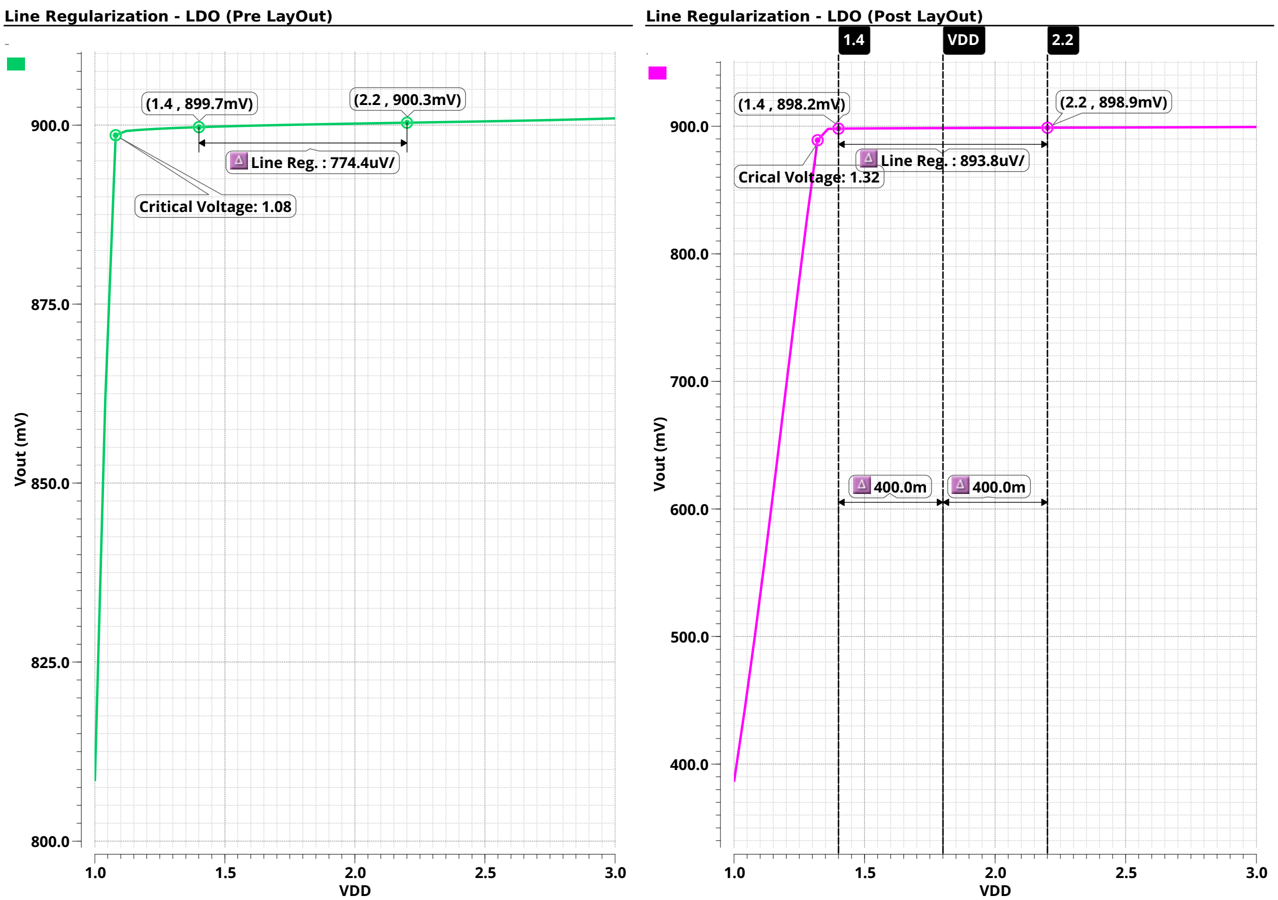

Figure 35. Simulated line regulation of the LDO before and after layout.

LDO Load Regulation

The LDO load regulation was evaluated to verify its ability to maintain a constant output voltage for different load currents. As shown in Figure 36, the simulated regulation before layout extraction was approximately 2.25 mV/A, while after extraction it degraded to approximately 32.7 mV/A. This considerable deterioration corresponds to an effective series resistance of approximately 0.0327 Ω in the output path, which is due to parasitic resistances of the metal interconnects and layout parasitics. This added resistance results in increased voltage drops under load variations, which degrades load regulation. The addition of metal stacking proved vital, reducing the parasitic resistance by up to tenfold. The resulting resistance of approximately 0.0327 Ω was sufficiently low, and further attempts to reduce it were deemed unnecessary.

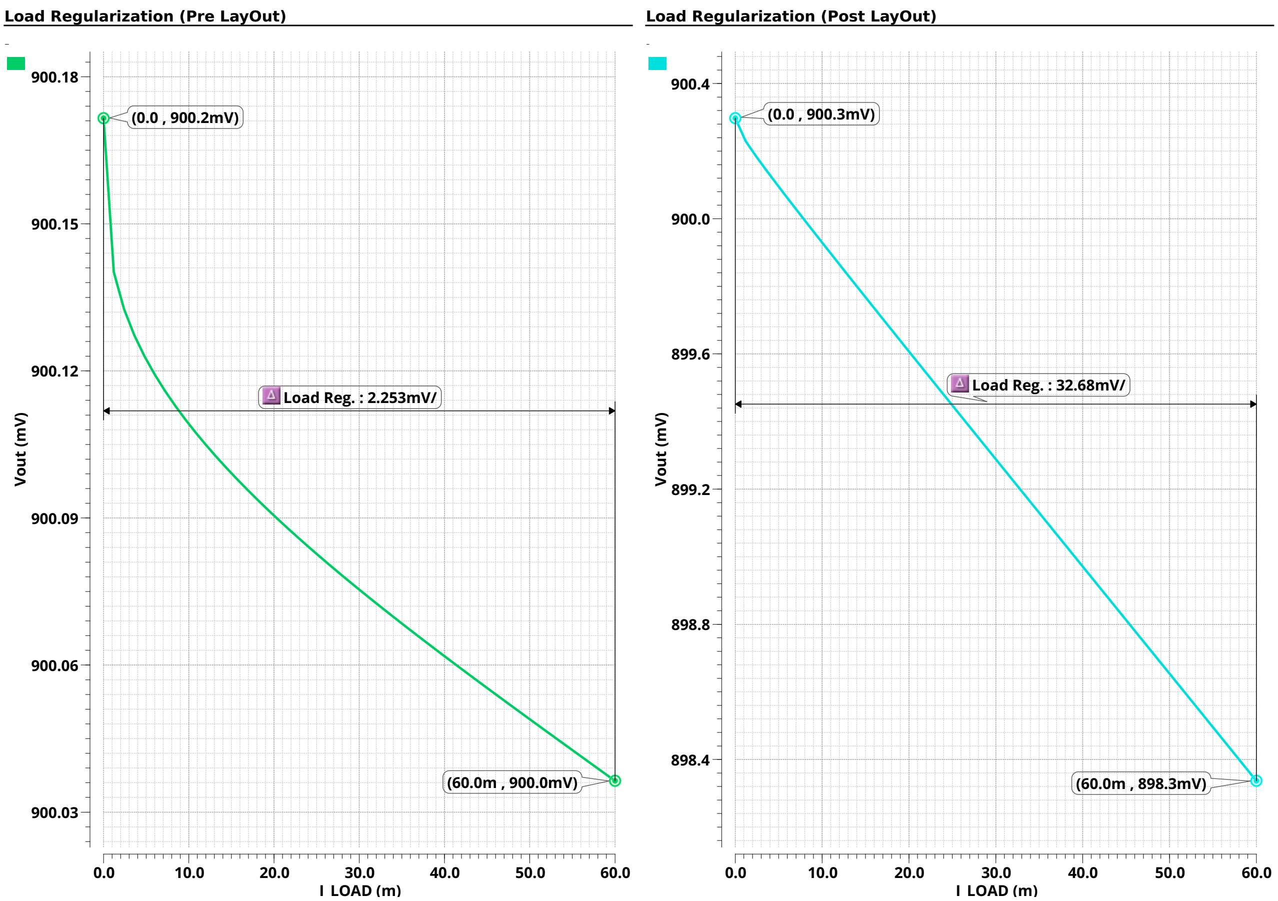

Figure 36. Simulated load regulation of the LDO before and after layout, showing the significant impact of parasitic routing resistance.

LDO Loop Gain

The LDO loop gain was simulated to evaluate stability and phase margin. Figure 37 shows the open-loop gain response, with a unity-gain bandwidth at approximately 3.73 MHz and a phase margin of 57.8°, which ensures adequate stability for the designed operating range. The dominant pole originates from the internal OTA, causing the gain to roll off at a rate of -20 dB/decade up to the gain-bandwidth product (GBW). Beyond this frequency, the influence of the non-dominant gate pole and the OTA zero become apparent. The LDO output pole contribution appears slightly after, causing the gain to roll of at a rate of -40 dB/decade.

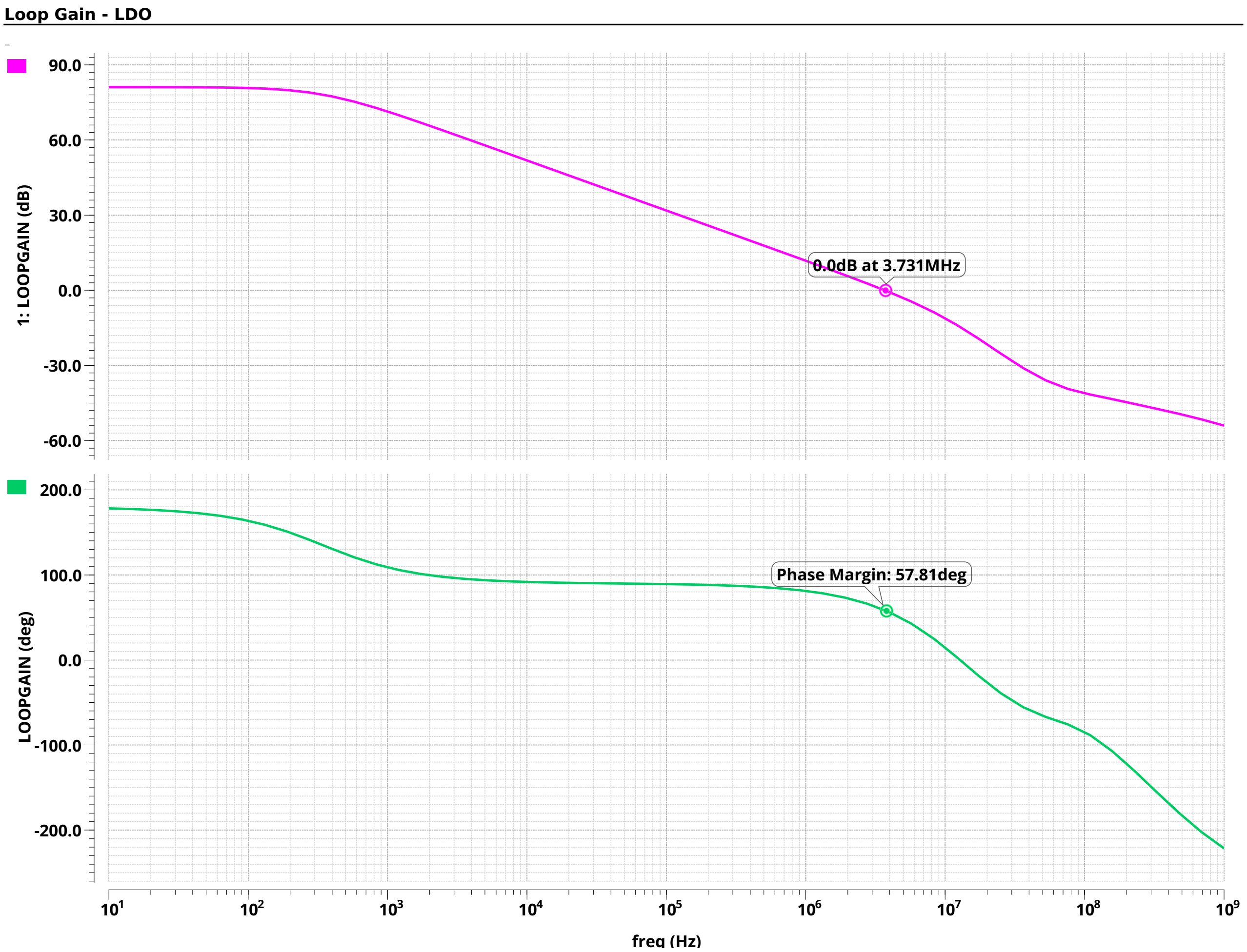

Figure 37. Simulated Loop Gain and Phase Margin of the LDO ($C_L = 200fF$)

LDO Transient Response to a Supply Voltage Disturbance

Figure 38 shows that the LDO exhibits stable behavior with minimal overshoot, consistent with its 60° phase margin. The output settles within the ±1% tolerance band approximately 500ns after the disturbance, indicating fast and well-damped transient response.

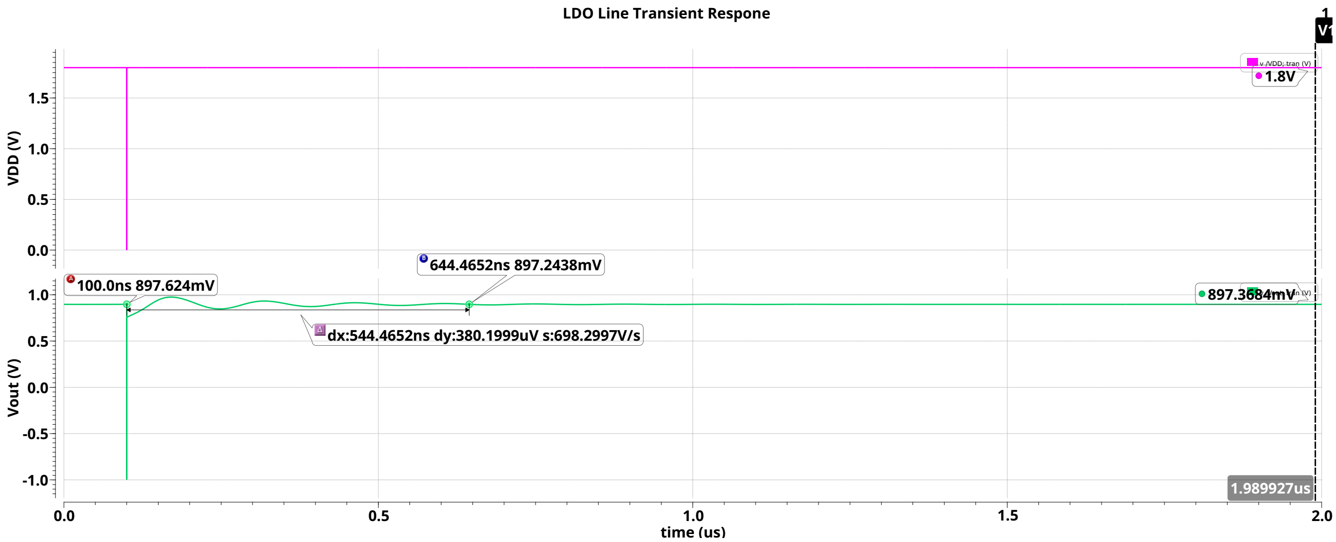

Figure 38. Post-Layout LDO Response to a 10ps Supply Voltage Impulse (1.8V->0V->1.8V), $I_{Load} = 55.5mA$, $T = 27\,°C$, No Output Capacitor

Since DPLL blocks on the chip have decoupling capacitors (on the order of 10s of fF), the effective output capacitance is typically in the 100-200fF range. This capacitance plays a crucial role in buffering against large and fast supply voltage fluctuations. Without it, the LDO output voltage tracks input dips directly, for example, a 1.8 V drop in VDD can cause an immediate 1.8 V drop at the output as seen in the figure above. The LDO's internal regulation loop is too slow to react to such a short impulse, as its bandwidth is the low MHz range. With a 200 fF output capacitor, however, charge is temporarily supplied to the load, significantly reducing the instantaneous voltage drop. While this improves line transient robustness, it also slightly slows down the LDO's response. Increasing the output capacitance further would reduce the instantaneous voltage dip even more, but would also slow down the overall settling time due to the larger time constant introduced. Figure 39 presents the response to a 10 ps impulse in the supply voltage with a 200fF effective output capacitance:

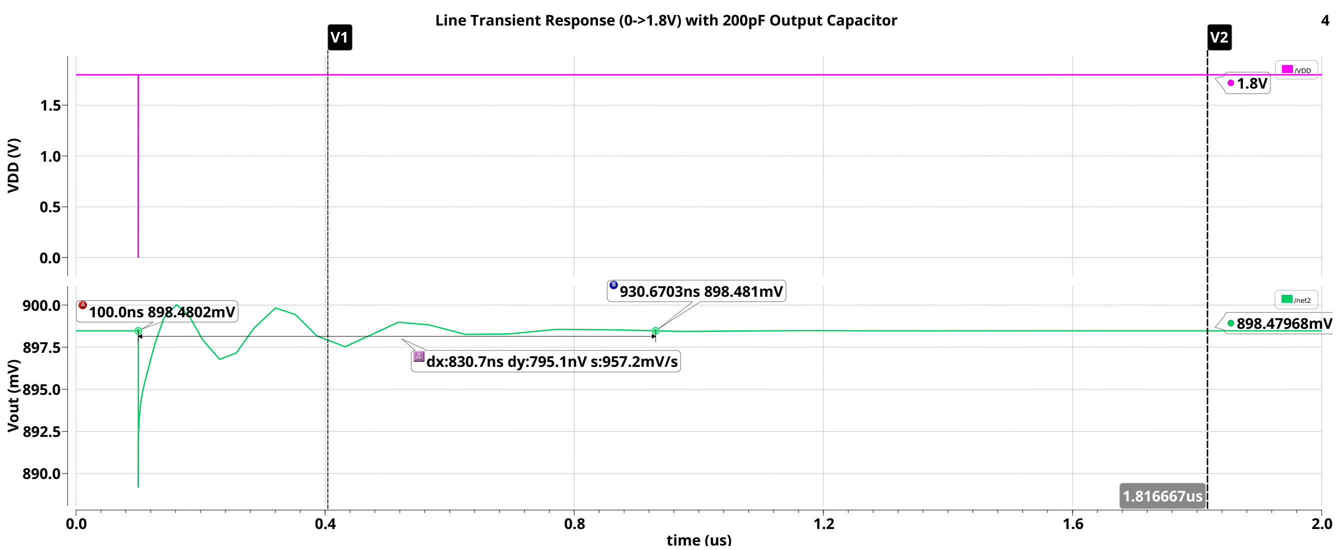

Figure 39. Post-Layout LDO Response to a 10ps Supply Voltage Impulse (1.8V->0V->1.8V), $I_{Load} = 55.5mA$, $T = 27\,°C$, 200fF Output Capacitor

This highlights the critical role of decoupling capacitors in filtering voltage dips and spikes for the DPLL blocks, particularly for the DCO, which is highly sensitive to supply variations due to its frequency dependence on V_{DD}. This test scenario represents a worst-case condition and is not expected during normal operation, but it is still essential to verify that the LDO remains stable and maintains regulation even under such extreme conditions.

LDO Transient Response to a Load Current Disturbance

Figure 40 shows the LDO's response to a 10 ps transient disturbance in the load current, dropping from its nominal value of 55 mA to 0 mA and then returning. The output voltage settles within 1 μs, and exhibits minimal overshoot and undershoot. This indicates a well-damped, stable transient response.

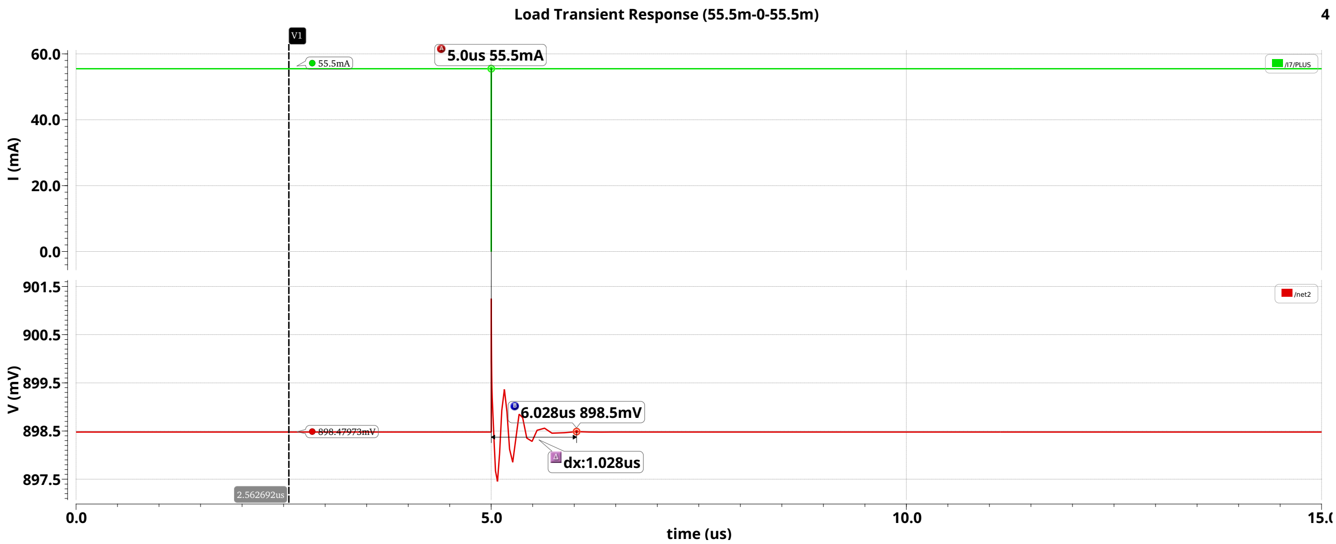

Figure 40. Post-Layout LDO Load Transient Impulse Response (55.5mA->0->55.5mA), $V_{DD} = 1.8V$, $T = 27\,°\mathrm{C}$, 200fF Output Capacitance

Figure 41 illustrates the LDO's response to a sharp load current disturbance, transitioning from 0 mA to 55 mA and back to 0 mA within a 10 ps window. In this scenario, the output exhibits noticeable ringing and slower settling compared to the previous step-down case. This behavior is attributed to the injection of rapid large charge into the output capacitor and any parasitic inductances in the layout. The resulting underdamped response leads to oscillations in the output voltage.

To mitigate this, a small series resistor (e.g., 5 mΩ) was added to the output path. This resistor passively dissipates excess energy and improves damping, thereby enhancing stability. However, increasing the resistance further, for instance, to 5 Ω, can accelerate settling by improving damping, but it also introduces a larger instantaneous voltage dip during fast load transitions. In this case, such a resistor would cause the output voltage to momentarily drop to approximately 0.4 V, severely degrading load regulation. Thus, this trade-off must be carefully considered when selecting the output resistor size.

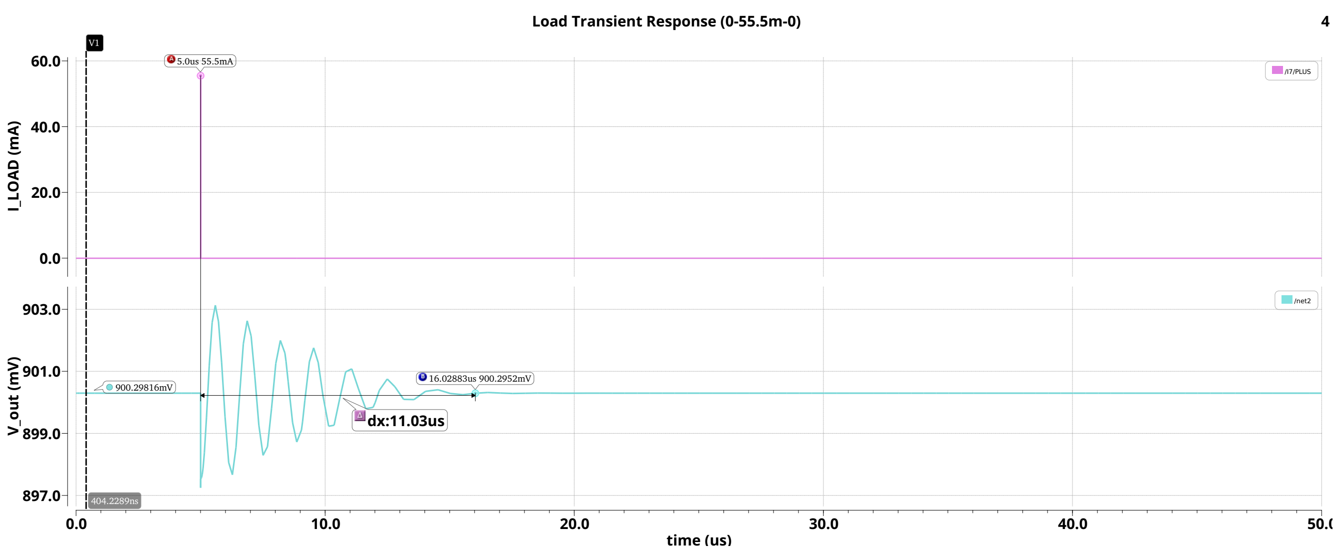

Figure 41. Post-Layout LDO Load Transient Impulse Response (0->55.5mA->0), $V_{DD} = 1.8V$, $T = 27\,°\mathrm{C}$, 200pF Output Capacitance With Series $5m\Omega$ Resistor

PSRR of the LDO

As expected, the low-frequency PSRR aligns with that of the BGR, remaining better than −60 dB. At higher frequencies, a noticeable peaking occurs, which is primarily due to the output capacitance interacting with parasitic inductance in the layout. Since the OTA gain is already low in this frequency range, the filtering effectiveness drops, and this LC resonance leads to a peak in the PSRR response.

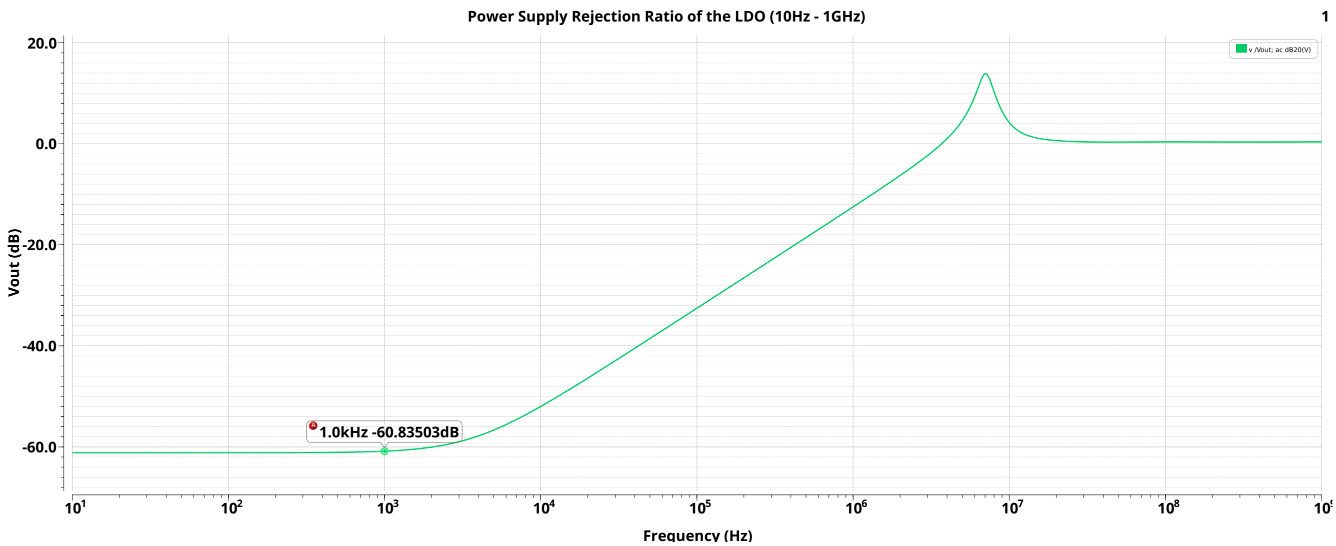

Figure 42. Post-Layout Power Supply Rejection Ratio (PSRR) of the LDO (10Hz–1GHz), $V_{DD} = 1.8V$, $I_{Load} = 55.5mA$

Output Noise of the LDO

Figure 43 shows the noise spectral density of the LDO output. At low frequencies, the noise density begins around 25 μV/$\sqrt{\text{Hz}}$ and decreases with frequency. At 100 kHz, it reaches approximately 672 nV/$\sqrt{\text{Hz}}$. This level of output noise is considered moderate and acceptable for DPLL blocks, where the supply sensitivity is most critical within the loop bandwidth of the DPLL, typically below 100 kHz. Beyond this range, the influence of supply noise on phase noise and jitter is significantly reduced.

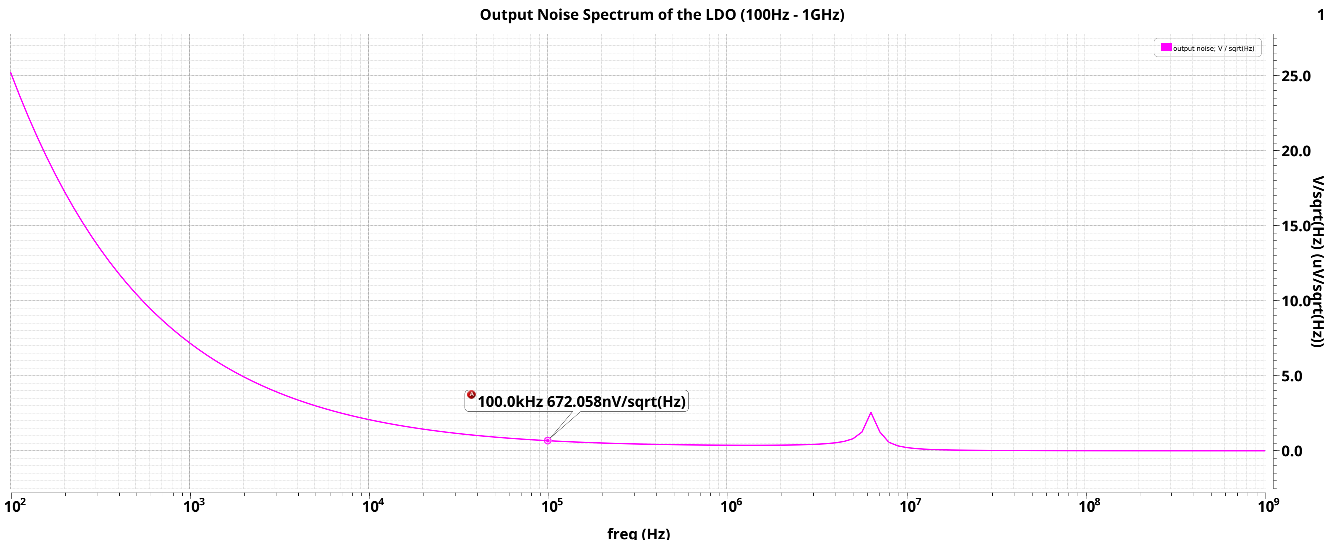

Figure 43. Post-Layout Output Noise Spectrum of the LDO (100Hz–1GHz), $V_{DD} = 1.8V$, $I_{Load} = 55.5mA$

Quiescent Current and Power Efficiency

The total current drawn from the supply was measured to be 55.58 mA, of which 55.5 mA was delivered to the load. This implies a quiescent current of approximately 80 μA. The corresponding current efficiency is:

$$\eta_{current} = \frac{I_{Load}}{I_{Total}} = \frac{55.5m}{55.58m} = 0.9985 = 99.85\%$$

This means that the periphery circuits of the LDO consume very little power compared to the load. However, since the output voltage V_{DO} is intentionally regulated to half the supply (0.9 V), the power efficiency is:

$$\eta = \frac{(V_{\text{DD}} - V_{\text{DO}}) \times I_{\text{Load}}}{V_{\text{DD}} \times (I_{\text{Load}} + I_Q)} = \frac{0.9 \times 55.5\,\text{mA}}{1.8 \times 55.58\,\text{mA}} \approx 0.4993 = 49.93\%$$

As expected, due to the deliberately large V_{DS} (dropout voltage) defined in the specification, approximately half of the input power is dissipated as heat, resulting in a poor power efficiency during operation. Therefore, designers usually focus on minimizing V_{DO} to improve power efficiency.

Line and Load Regulation Across All Corners

In order to ensure correct operation of the LDO across all PVT (Process, Voltage, Temperature) corners, line and load regulation metrics must be evaluated under various conditions. Specifically, simulations were performed at five process corners: TT, FF, SS, FS, SF, all at room temperature (27°C), as well as at extreme temperatures with two representative corners: FF at 125°C and TT at -40°C to capture best and worst-case thermal conditions.

- **TT (Typical-Typical)**: Both NMOS and PMOS transistors exhibit nominal electrical characteristics, representing typical behavior. Shown by the red curve.
- **FF (Fast-Fast)**: Both NMOS and PMOS devices are faster than nominal due to process variations such as higher mobility and lower threshold voltage. In analog circuits, this can result in higher transconductance and faster settling. Represented by the orange curve at room temperature and the white curve at 125°C.
- **SS (Slow-Slow)**: NMOS and PMOS are both slower than nominal, with higher threshold voltages and lower mobility. This often reduces bias currents and amplifier gain, making it a worst-case for analog performance. Represented by the yellow curve at room temperature and the blue curve at -40°C.
- **FS (Fast-Slow)**: NMOS devices are faster than typical while PMOS devices are slower. This asymmetry may cause mismatch in current mirrors and unbalanced differential pairs. Shown by the pink curve.
- **SF (Slow-Fast)**: PMOS devices are faster, NMOS devices are slower. Like FS, this represents a mismatch corner, and possibly degrading analog biasing or offset. Represented by the green curve.

As shown in Figure 44, the LDO maintains proper line regulation across all tested corners and temperatures.

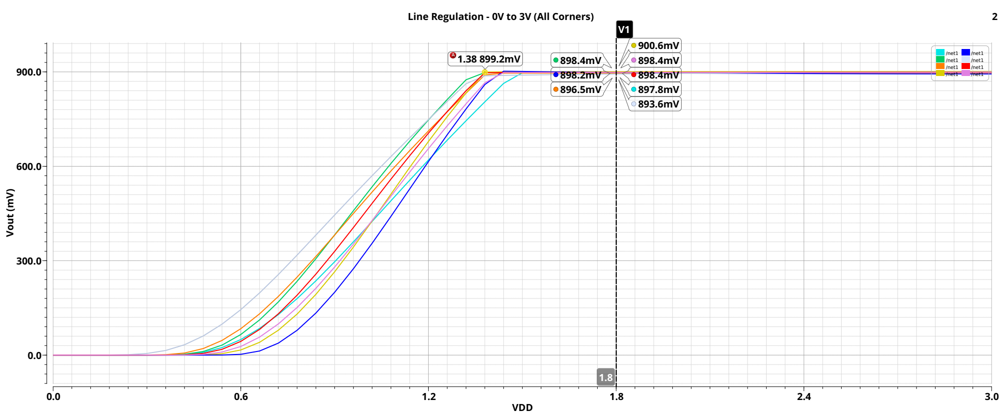

Figure 44. Post-Layout Line Regulation of the LDO for All Corners, $I_{Load} = 55.5mA$

The load regulation of the LDO was also evaluated across the same set of PVT corners to ensure reliable output voltage under varying load conditions. The results are shown in Figure 45, using the same color scheme as described previously for each process corner. As observed, the LDO maintains stable output voltage across all simulated corners and temperatures.

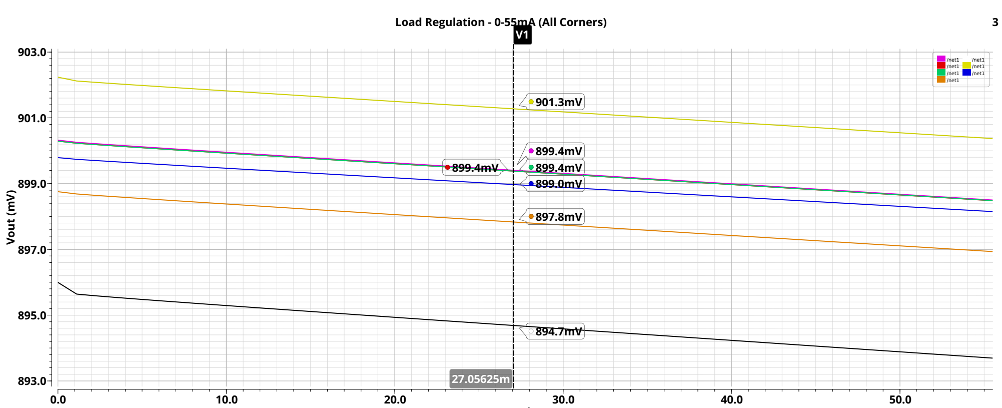

Figure 45. Post-Layout Line Regulation of the LDO for All Corners, $V_{DD} = 1.8V$

LDO Operating Modes Based on Control Signals

Figure 46 shows the DC operating point of the LDO when `EN = 0` and `bypass_en = 0`. In this disabled state, the output is pulled to 0 V by an NMOS pulldown device, while the PMOS pullup and the pass transistor remain in cutoff. Additionally, the gate of the pass transistor is shorted to the supply through a gate pull-up device to eliminate leakage and reduce static power.

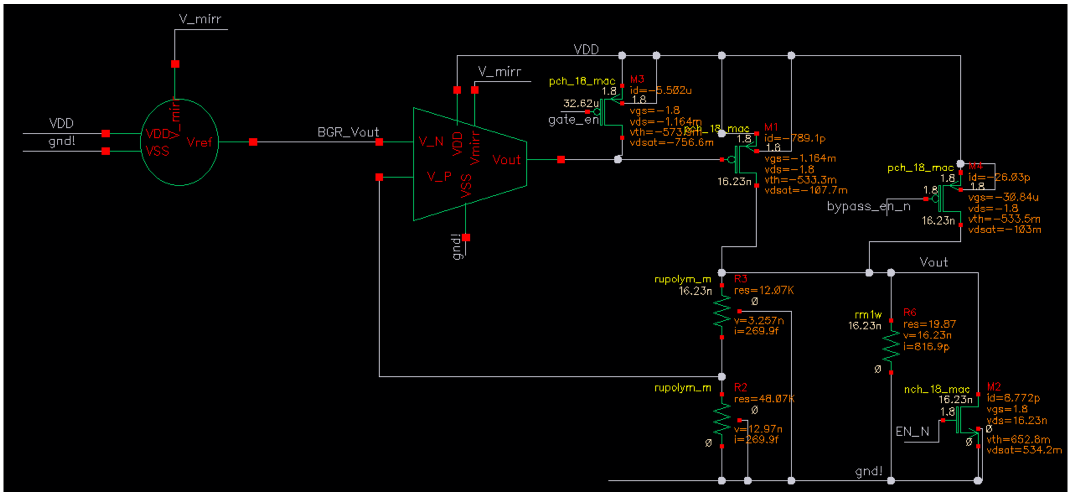

Figure 46. LDO DC operating point when disabled (`EN = 0`, `bypass_en = 0`).

Figure 47 illustrates the nominal operation of the LDO with `EN = 1` and `bypass_en = 0`. In this mode, the regulator is enabled and produces the target output voltage of 0.9 V. The output pullup and pulldown transistors are off, and the pass transistor is control by the OTA control loop, as the gate pull-up is disconnected.

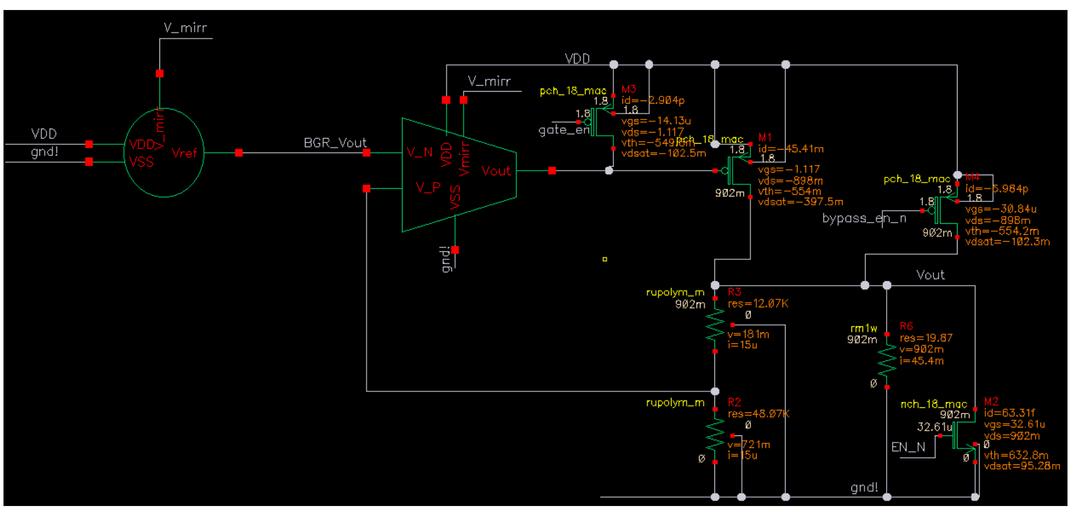

Figure 47. LDO DC operating point under normal operation (`EN = 1`, `bypass_en = 0`).

Figure 48 shows the bypass mode, where `EN = 1` and `bypass_en = 1`. In this configuration, the LDO output is directly connected to the supply voltage (1.8 V) through an active PMOS pullup device, bypassing the internal regulation path. The pass transistor is disabled by shorting its gate via the gate pull-up, and the NMOS pulldown remains off.

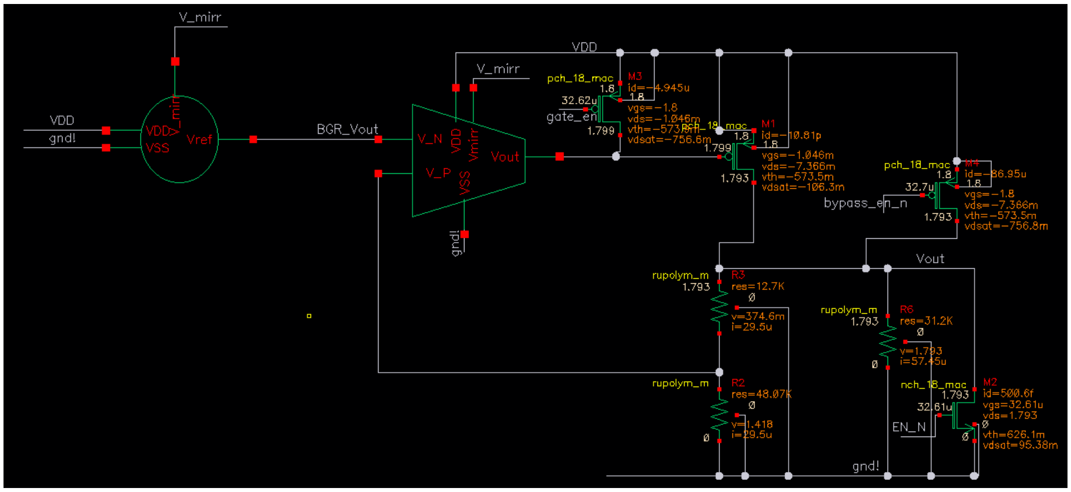

Figure 48. LDO DC operating point in bypass mode (`EN = 1`, `bypass_en = 1`).

5 Results Analysis

This section summarizes the performance results of the designed LDO regulator and its sub-blocks, comparing pre-layout and post-layout simulations. Metrics include nominal output voltage, line regulation, load regulation, temperature coefficient, and PSRR, evaluated under the specified supply voltages and load conditions.

The Following tables compile these results to provide a concise overview of overall system behavior as well as the behavior of each sub-block,

Table 8 summarizes the OTA results pre- and post-layout.

Parameter	Spec	Pre-Layout	Post-Layout	Change (%)	Improvement over Spec
DC Gain (dB)	>70	77.6	77.5	−0.13 %	↑ 7.5 dB
BW (Hz)	N/A	816.7	789.9	−3.3 %	N/A
UGF (Hz)	≥5 M	6.27 M	5.6 M	−10.7 %	↑ 1.12×
GBW (Hz)	≥5 M	6.4 M	5.9 M	−7.8 %	↑ 1.18×
Phase Margin (deg)	>60	76.5	74.2	−3.0 %	↑ 14.2°

Table 8
OTA Performance Comparison Before and After Layout

Table 9 summarizes the BGR results pre- and post-layout.

Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement Over Spec
Temperature Coefficient (ppm/°C)	<60	4.6	7.03	−40–125 °C	+52.8 %	↑ 8.5×
Line Regulation (V/V)	<0.01	0.00058	0.00054	1.4–2.2 V	−6.9 %	↑ 18×

Table 9
BGR Performance Comparison Before and After Layout

Table 10 summarizes the LDO results pre- and post-layout.

Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement from Spec
Temperature Coefficient (ppm/°C)	< 60	5.1	6.4	−40–125 °C	+25.5 %	↑ 9.4×
Line Regulation (V/V)	< 0.01	0.0009	0.00089	1.4–2.2 V	−1.1 %	↑ 11.2×
Load Regulation (mV/mA)	< 0.1	0.0024	0.033	0–60 mA	+1275 %	↑ 3×
PSRR @ 1kHz	≤ −40 dB	-61 dB	-60.8 dB	N/A	−0.4 %	↑ 11×
Quiescent Current	≤ 1 mA	77 µA	80 µA	N/A	+3.75 %	↑ 12.5×

Table 10
LDO Performance Comparison Before and After Layout

Although performance variations can be observed after layout extraction in all measured parameters, these deviations remain within the acceptable design specifications. For example, the OTA bandwidth showed a modest decrease of approximately 3%, attributed to additional parasitic capacitances introduced by the layout, which is considered a minor effect in practical terms. Regarding the LDO line regulation, while values were simulated across a supply range of 1.4–2.2 V, the regulator has actually demonstrated correct operation at higher supply voltages as well, including up to 3 V, with stable behavior.

The most significant post-layout change was observed in the load regulation, which increased substantially. As explained in the implementation section, this is due to the inherent difficulty of maintaining precise regulation across high load currents, particularly given the unavoidable parasitic resistance of the pass transistor's metal routing, leading to IR drops no matter how well the layout is optimized.

Finally, although the LDO was developed primarily for a DPLL system, its performance and specifications would make it suitable for a variety of other analog or mixed-signal applications as well. Overall, the design remains compliant with its intended requirements and demonstrates robust operation across a wide range of conditions.

The following table compares the performance of the post-layout LDO with that of similar designs reported in the literature.

Parameter	**This Work**	[7]	[9]	[8]	[13]	[3]	[10]
Process	28nm	28nm	40nm	600nm	180nm	130nm	180nm
Layout Area (mm^2)	0.0044	0.0086	N/A	0.307	0.031	0.008	0.109
Supply Voltage V_{DD} (V)	1.38–3	0.9	1.1	1.5-4.5	1.2-1.8	1-1.4	1.31-3.3
Output Voltage V_{OUT} (V)	0.9	0.85	0.9	1.3	0.8-1.6	0.8	1.2
Load Current I_{Load} (mA)	55.5	20	100	100	100	25	50
Quiescent Current I_Q (μA)	80	33	30	N/A	10.2	112	5.5
Line Regulation (mV/V)	0.89	17.5	0.2	2.1	10	2.25	4.13
Load Regulation (mV/mA)	0.033	0.26	0.25	0.065	0.081	0.173	0.0056
Temperature Coefficient (ppm/°C)	6.4	N/A	N/A	38	N/A	N/A	87.5
PSRR (dB @1kHz)	-60.8	-24 @1MHz	-70 @10kHz	-60	N/A	-60	-36.3
Output Noise (nV/$\sqrt{Hz}$ @100kHz)	672	N/A	N/A	380	N/A	N/A	N/A
Settling Time - Worst Case (μs)	0.7	N/A	1.2	2	0.22	0.19	N/A
$C_L[pF]$	≈ 0.2	36	100	N/A	100	25	50
FOM [ps]	0.008	0.52	0.007	N/A	0.02	N/A	0.033

Table 11
Comparison of this LDO with LDO Implementations from the Literature

All values reported are post-layout, including full parasitic extraction, thereby ensuring the performance metrics reflect realistic behavior and confirming the suitability of the proposed design for practical DPLL integration.

Designed specifically for integration within a Digital Phase-Locked Loop (DPLL) system, this LDO is optimized for compactness, stability, and high load and line regulation. All essential for noise-sensitive mixed-signal environments. Additionally, contrary to other LDO implementations, this implementation can work with a very low output capacitances which come from the decoupling of the DPLL blocks.

The LDO achieves a layout area of only 0.0044 mm^2, making it one of the most compact among the compared works, which is particularly beneficial for dense SoC integration. Despite the small layout, it supports a load current of 55.5 mA, and meets the stringent load current specification of the DPLL.

It also exhibits strong regulation behavior, with a line regulation of 0.89 mV/V and load regulation of 0.033 mV/mA, while maintaining a temperature coefficient of 6.4 ppm/°C, which indicates excellent thermal stability. The quiescent current of 80μA is sufficiently low for low-power operation yet high enough to support fast transient response.

A key advantage is the PSRR of −60.8 dB at 1 kHz, which is especially important in DPLL applications where suppression of supply noise is critical for maintaining phase noise performance.

Compared to prior works, this LDO demonstrates highly competitive or superior results across all major metrics.

6 Conclusions and Further Work

Conclusions:

- The project successfully implemented a 0.9 V low-dropout (LDO) voltage regulator in 28 nm CMOS technology, which is designed for integration within a DPLL system.
- The LDO meets key performance specifications, including line regulation, load regulation, temperature stability, PSRR, and transient response, thereby validating both the circuit design and layout methodology.
- Layout-driven design considerations were critical in maintaining performance, with only minor degradations observed after post-layout extraction. This highlights the importance of early layout floorplanning and the application of industry-standard layout techniques.
- In general, the implemented LDO exhibits robust and reliable operation under the specified supply and load conditions across all PVT corners.

Further Work:

- **Error Amplifier Gain Improvement:** The EA gain can be improved beyond 80 dB using advanced topologies such as folded cascode with gain-boosting stages or telescopic cascode topologies, which can achieve gains of 100–120 dB. This can significantly improve line and load regulation, as well as power supply rejection ratio (PSRR). However, the telescopic cascode has limited output swing, which makes it more suitable for use as the error amplifier in the bandgap reference (BGR).
- **Error Amplifier Bandwidth Improvement:** The OTA bandwidth can be increased while maintaining a similar gain by adjusting the two-stage OTA parameters according to the desired performance. Specifically, the gain-bandwidth product (GBW), and consequently the bandwidth (BW), can be increased by boosting the transconductance (g_m) of the first stage. This can be achieved by increasing the OTA's bias current. In strong inversion, a tenfold increase in bias current roughly results in a tenfold increase in GBW, leading to a significant bandwidth improvement. However, to maintain proper pole splitting and stability, the second stage current should also be increased proportionally. Thus, this bandwidth enhancement comes at the cost of higher power consumption. This trade-off might be worth it at times since increasing the error amplifer's GBW improves power supply rejection ratio (PSRR) at higher frequencies by extending the frequency range over which the feedback loop effectively suppresses supply noise. However, in a 2-stage OTA increasing the bias current also reduces the output resistance (r_{ds}), which can eventually degrade the gain. If pushed too far, at some point, the gain loss due to lower r_{ds} outweighs the benefit from higher g_m. Therefore, a topology change to a cascode stage can effectively address this gain-bandwidth trade-off by stacking transistors to significantly increase output resistance by approximately a factor of $g_m r_{ds}$, which allows for high gain to be maintained even at elevated bias currents. Therefore, cascode stages offer a more favorable trade-off for LDOs where both gain and speed are critical.
- **BGR Improvements:** Second-order curvature-correction techniques can be implemented in the BGR to further reduce the temperature coefficient (TC), especially at higher temperatures. The TC and PSRR of the reference voltage can also be improved by using a cascode current mirror, which increases output impedance and can help stabilize the reference voltage against supply and process variations.
- **Pass Transistor Sizing and Buffering:** Load regulation can be improved by increasing the width of the pass transistor, which improves current drive capability and increases g_m. However, this increases the gate capacitance and shifts the pole there to lower frequencies, which can potentially cause stability issues. To mitigate this, a buffer stage can be added between the OTA output and the gate of the pass transistor. This buffer isolates the OTA from the capacitive load, and thus can improve bandwidth and contribute additional gain and improve PSRR. The trade-offs include increased power consumption, area, and added design complexity because of an additional pole.
- **LDO Function Extension:** The current design includes an enable signal to activate the LDO and produce 0.9V and another to bypass the LDO, to deliver 1.8 V directly to the DPLL blocks. Additional control signals can be introduced to enhance LDO functionality. These can be a tunable resistor ladder network in the feedback path for output voltage trimming using control bits, a high-impedance output mode, or a low-power operational mode for energy efficiency.
- **Noise Reduction:** Output noise can be reduced by utilizing lower-noise OTA topologies such as folded cascode instead of a two-stage OTA. Additionally, the intentional addition of an output capacitor in series with a small resistor can help filter high-frequency noise and suppress voltage spikes, at the price of degrading the transient response speed.
- **Layout and Routing for Load Regulation:** Load regulation can be further enhanced through improved current routing, advanced current-splitting strategies, or better metal stack utilization. These layout optimizations might help reduce IR drop and improve the LDO's ability to handle high load currents efficiently.

7 Project Documentation

The complete project repository, including design files, deliverables, and figures, is available on GitHub at the following link:

`https://github.com/ArsenArutyunov/DPLL-LDO`

The repository is organized into the following main directories:

- **Weekly Presentation/** This folder documents the project's weekly progress. It is divided into two subfolders:
  - **Semester A/** – Contains presentation materials from the first semester.
  - **Semester B/** – Contains presentation materials from the second semester.
- **Deliverables/** This folder includes all formal submission items:
  - The final project book.
  - The final presentation.
  - The project poster.
  - The mid-project presentation.
  - The initial workplan.
- **Figures/** This folder contains visual materials, such as:
  - Circuit diagrams.
  - Simulation graphs.
  - Layout screenshots.
  - Devices Sizes Tables.
  - Results Tables.
- **Design/** This directory contains the full Cadence design environment for the Low Dropout Regulator (LDO), including schematics, layouts, testbenches, and simulation setups.

Acknowledgements

Special thanks to Pini Schnaider (Intel) for his additional guidance and valuable insights during the layout implementation phase.

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